Mapping Recaptures

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Mapping Genotype Recaptures in Central GA

This markdown will visualize the spatial distribution of bears with *spatial recaptures* from hair samples collected in central Georgia (2023 and 2024). These samples were genotyped via capture-seq (Rapid Genomics) or GT-seq (GTSeek). Capture-seq data were filtered to avoid tightly linked loci (no SNPs within 500kb with $r^2 > 0.8$) and focus on informative markers (MAF > 0.2) and combined with GT-seq data (which have a broader range of MAF), resulting in a total of 1325 SNP loci (see `4_combineGTSeek` for details on combining the data from the two marker panels). Genotypes were subjected to a clone analysis in `COLONY` (Wang 2016) and KING kinship coefficients (Manichaikul et al. 2010) were calculated in `PLINK2` (Chang et al. 2015). Inferences from the two approaches were combined following methods outlined in `10_recapFinal`.

Here, we'll plot the recaptures for bears as a way to assess a couple of things:

1. Confidence in recapture data - are samples spaced too far apart? Are we confident in the combined inferences from `COLONY` and `PLINK`?
2. Evidence for moving activity centers - do we have sufficient data to fit a model where activity centers are allowed to move between years? Does this appear to be happening in the dataset?

Load data

We need to load the recapture objects and the locations of the hair snares to visualize the distribution of recaptures...

```
#Load the sampleMetadata file
smd <- read.csv("../sampleinfo/sampleMetaData_cloneID.csv", header = TRUE)

#Load capture history objects
load("../scr/CGA_CH.Rda")
```

```r
#Load hair snare location data
load("../sampleinfo/snareLocations.Rda")

#Load the state space
library(raster)
```

```
## Loading required package: sp
```

```r
S360 <- raster("../input/state-space360.tif")

#Source functions to calculate some basic summary stats
source("../input/sumstats_4d.R")
```

Females

In the plots that follow, all detections for individuals with spatial recaptures are plotted on the same map. Primary sampling occasions are shown in different colors, while secondary sampling occasions are plotted with different symbols. In cases where the same individual is detected at multiple hair snares *in the same week*, points are connected with a dotted line (colored to indicate the year of the movement). Each plot also gives the total number of detections for a bear, and the number of hair snares J, weeks K, and years T where the individual was detected.

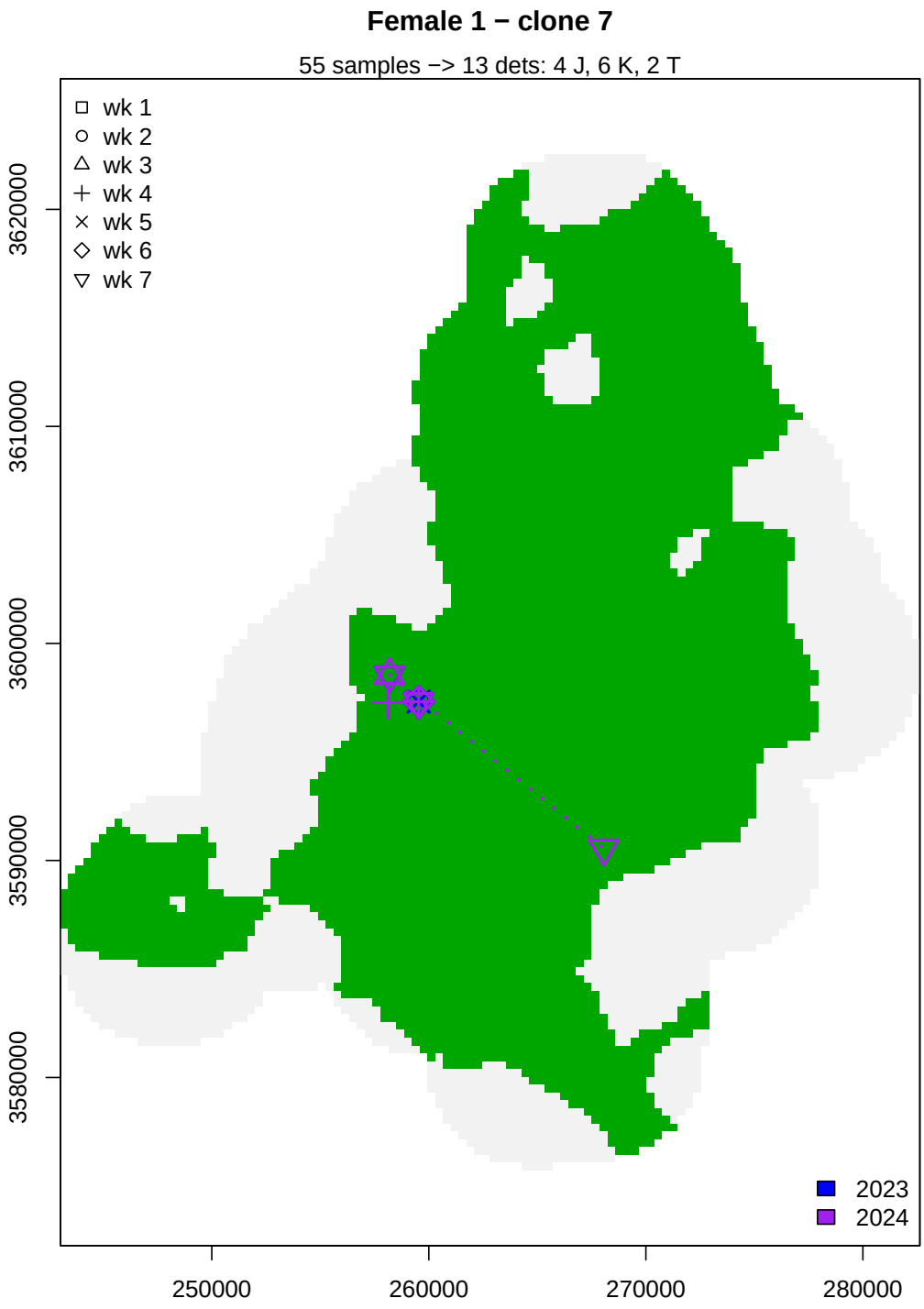
First, for females...

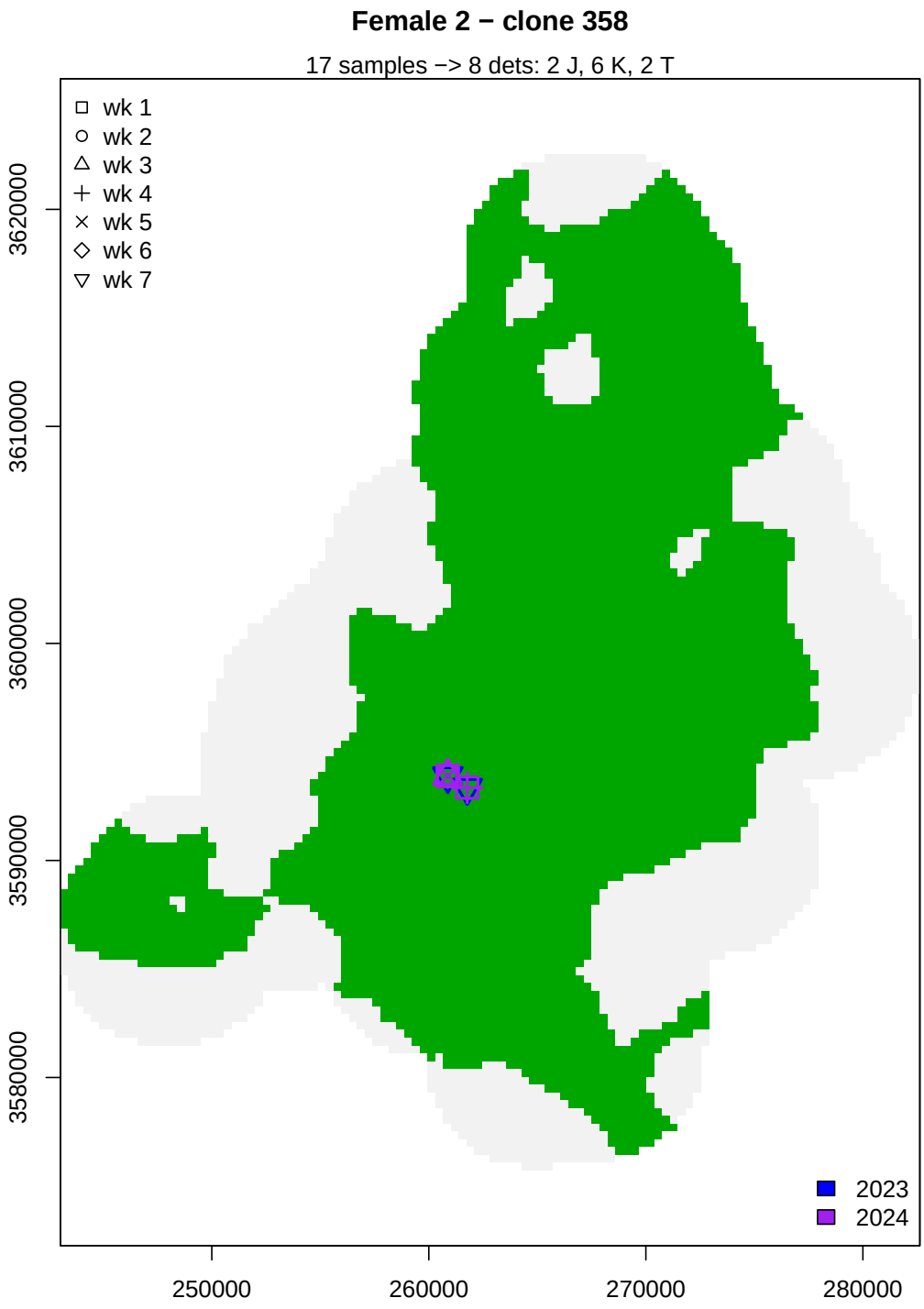
```r
#Females
caphist <- Fch$y
Fclones <- Fch$clone

sumstats_4d(caphist)
```

```
##        ss.totN     ss.totcaps     ss.avgcaps ss.indwsprecap    ss.avgtraps
##      98.000000     272.000000       2.775510      30.000000       1.540816
##      ss.avgyrs
##       1.255102
```

```r
nind <- dim(caphist)[1]

yr.col <- c("blue","purple")
wk.symbol <- 0:6

for(i in 1:nind) {
  i.ch <- caphist[i,,,]

  nTraps <- sum(rowSums(i.ch) > 0)
  if(nTraps > 1) {
    plot.me <- as.data.frame(which(i.ch == 1, arr.ind = TRUE))
    plot(S360, legend = FALSE, main = paste("Female",i,"- clone", Fclones[i]))
    mtext(paste(length(which(smd$finalClone == Fclones[i])), "samples ->",
                nrow(plot.me), "dets:",
                nTraps, "J,",
                length(unique(plot.me$dim2)), "K,",
```

```
                length(unique(plot.me$dim3)), "T"))
    points(x=locs[plot.me$dim1,1], locs[plot.me$dim1,2],
           pch = wk.symbol[plot.me$dim2], col = yr.col[plot.me$dim3],
           cex = 2, lwd = 2)

    for(r in 2:nrow(plot.me)) {
      if(plot.me$dim3[r]==plot.me$dim3[r-1] &
         plot.me$dim2[r]==plot.me$dim2[r-1]) {
        lines(x=c(locs[plot.me$dim1[r-1],1], locs[plot.me$dim1[r],1]),
              y=c(locs[plot.me$dim1[r-1],2], locs[plot.me$dim1[r],2]),
              lty="dotted", col = yr.col[plot.me$dim3[r]],
              lwd = 2.5)
      }
    }

    legend("topleft", bty = "n", pch = 0:6, legend = paste("wk",c(1:7)))
    legend("bottomright", bty = "n", fill = yr.col, legend = c("2023","2024"))

  }
}
```


Female 1 − clone 7
55 samples −> 13 dets: 4 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024


Female 2 − clone 358
17 samples −> 8 dets: 2 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
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3620000
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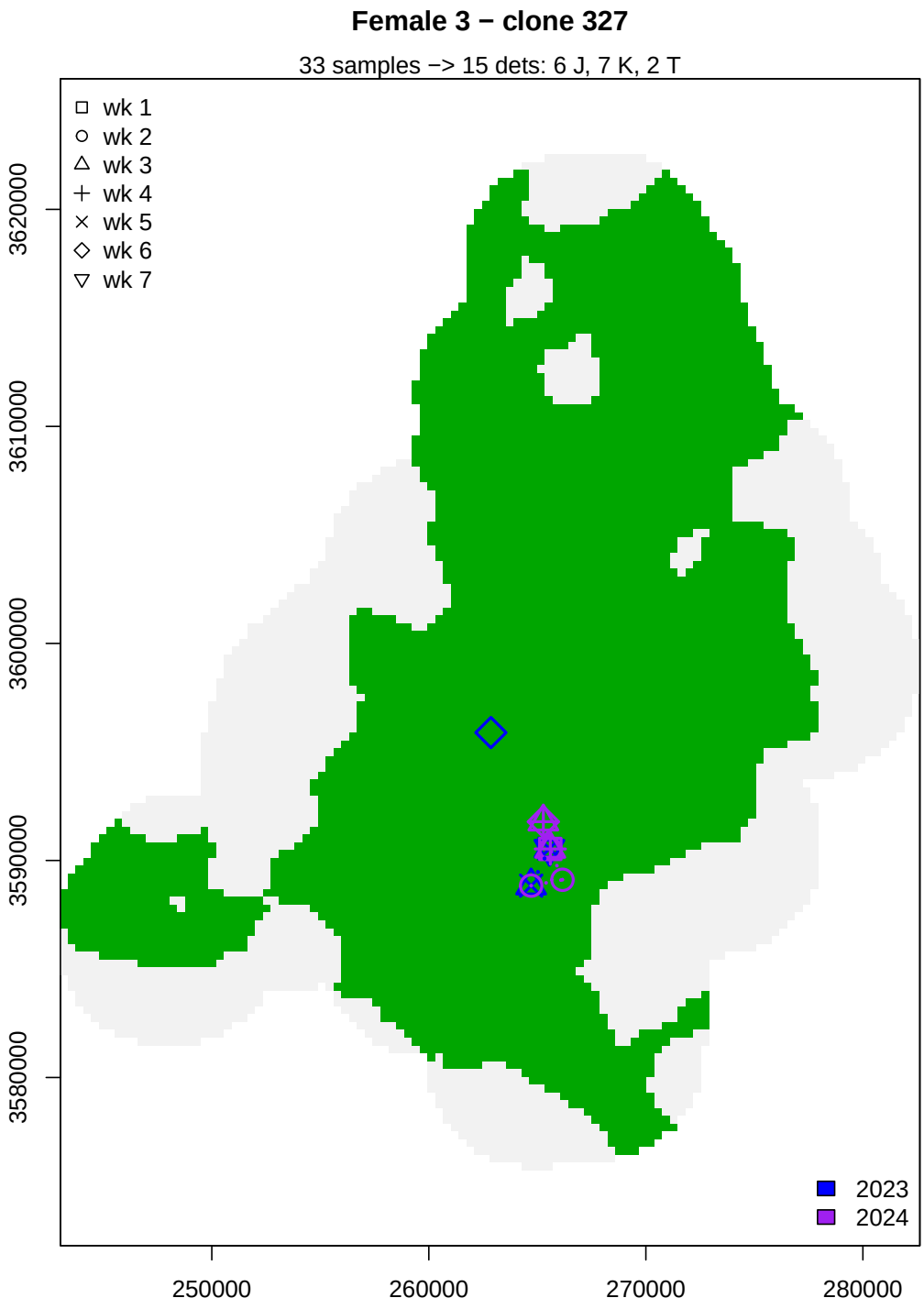
Female 3 − clone 327
33 samples −> 15 dets: 6 J, 7 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
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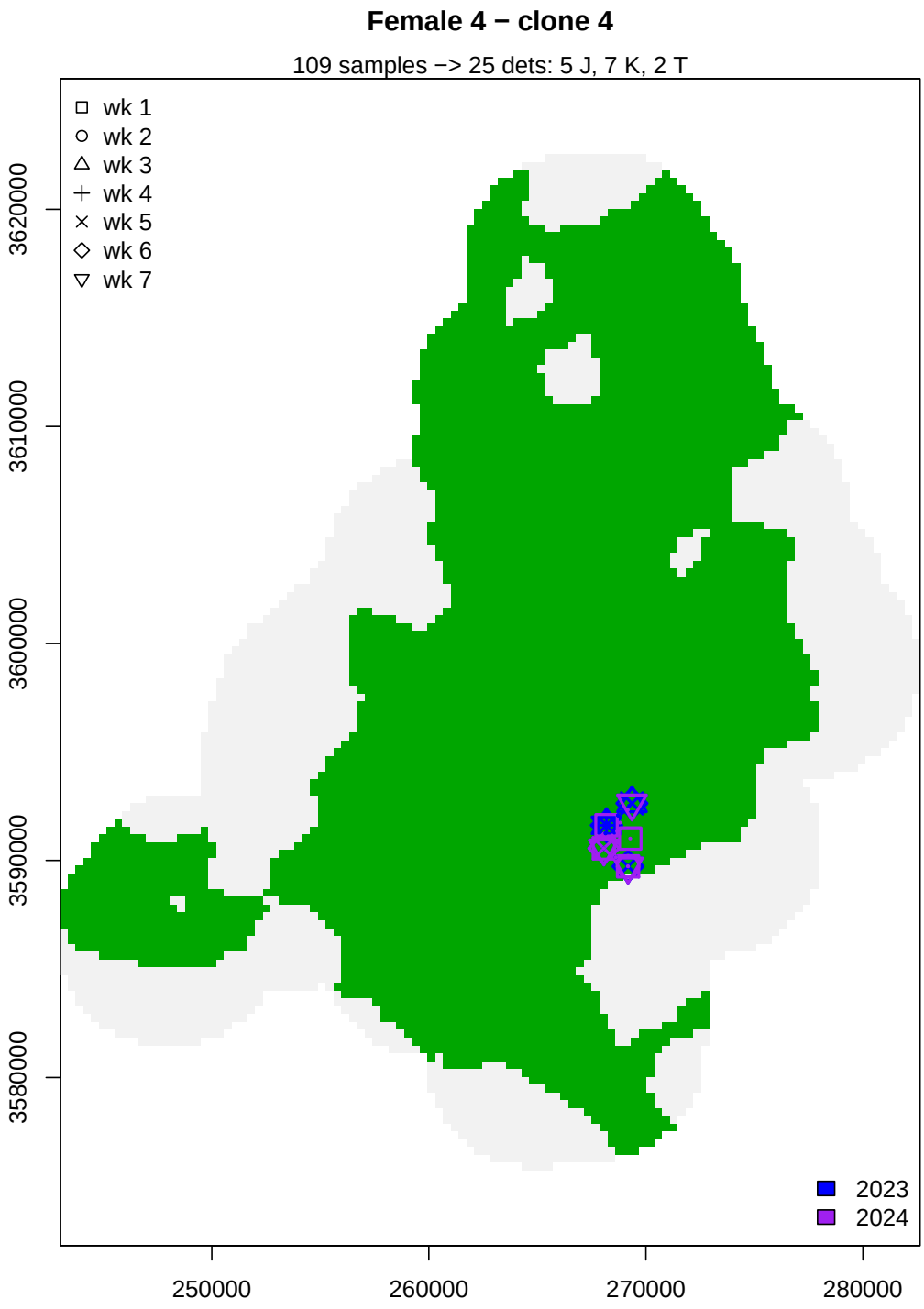
260000
270000
280000
2023
2024


Female 4 − clone 4
109 samples −> 25 dets: 5 J, 7 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
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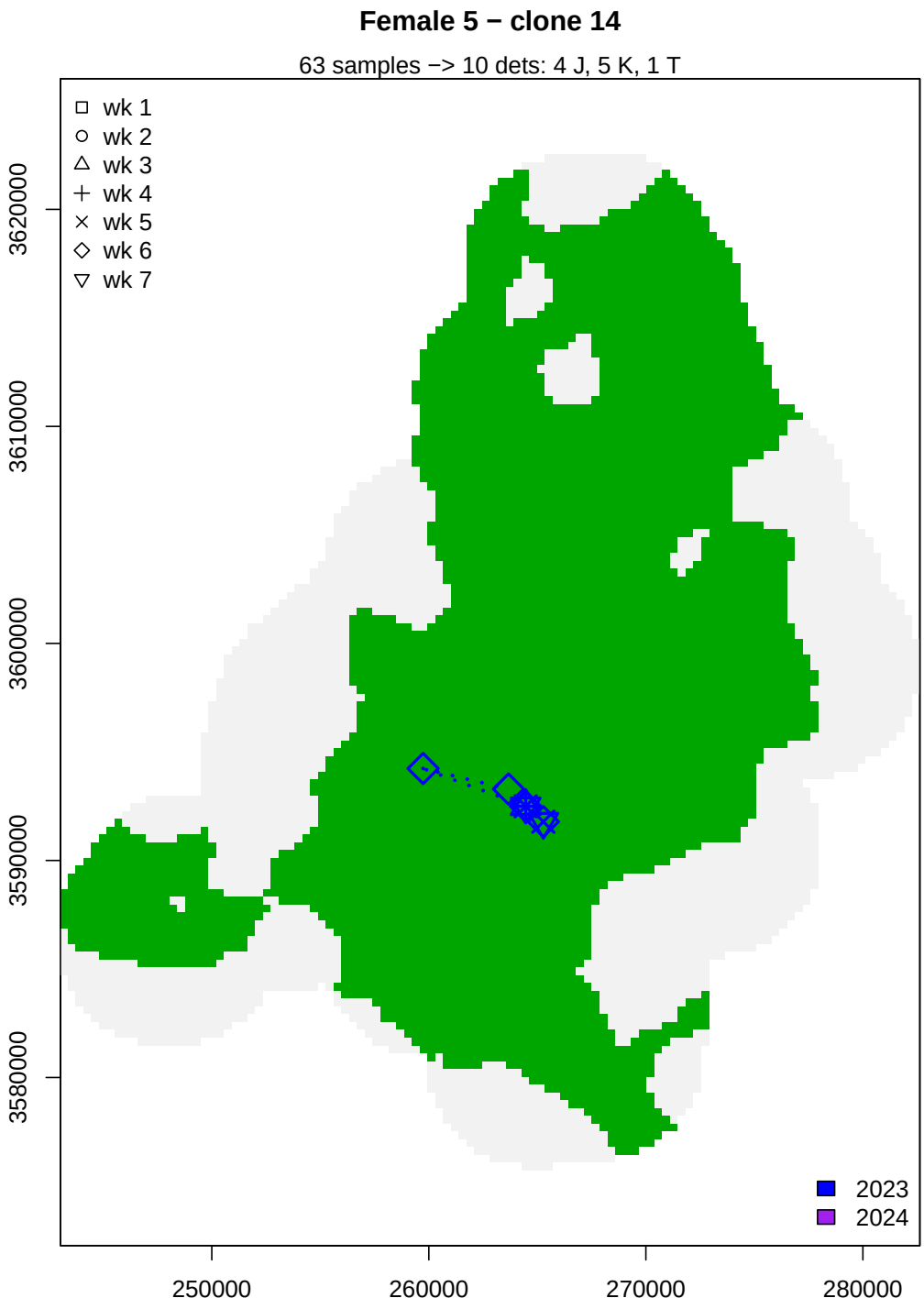
250000
260000
270000
280000
2023
2024


Female 5 − clone 14
63 samples −> 10 dets: 4 J, 5 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
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270000
280000
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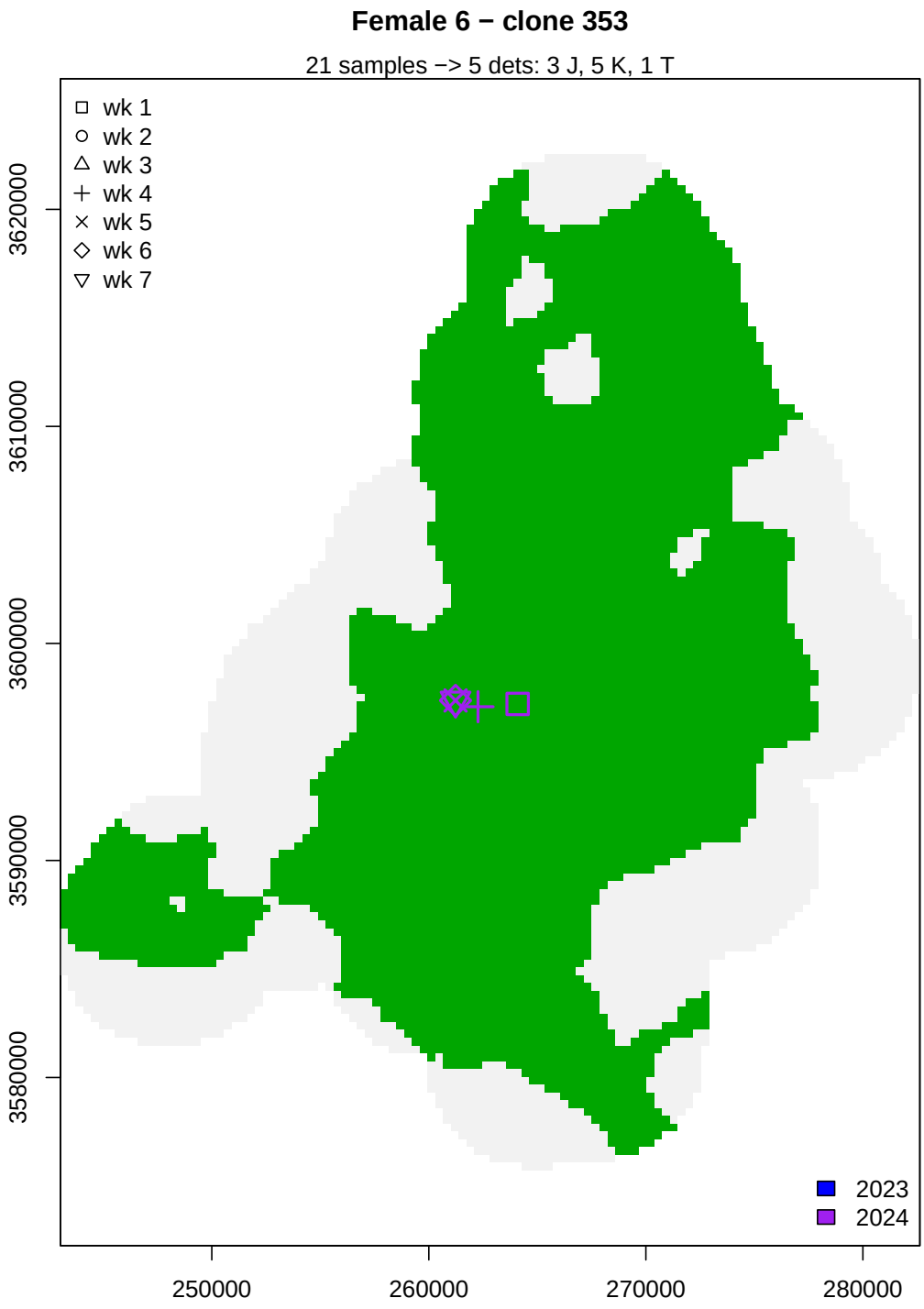
2024


Female 6 − clone 353
21 samples −> 5 dets: 3 J, 5 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
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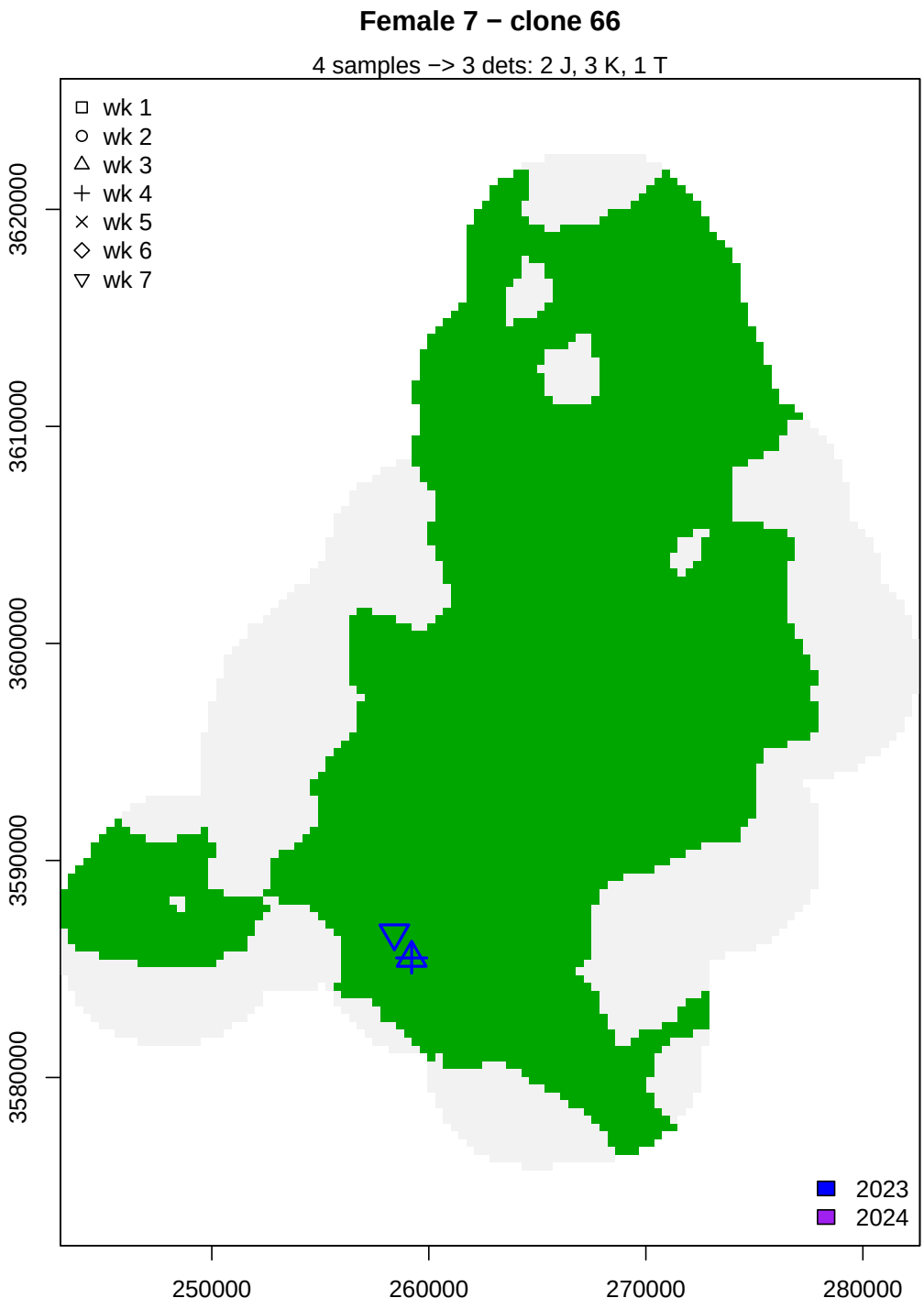
Female 7 − clone 66
4 samples −> 3 dets: 2 J, 3 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
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3590000
3580000
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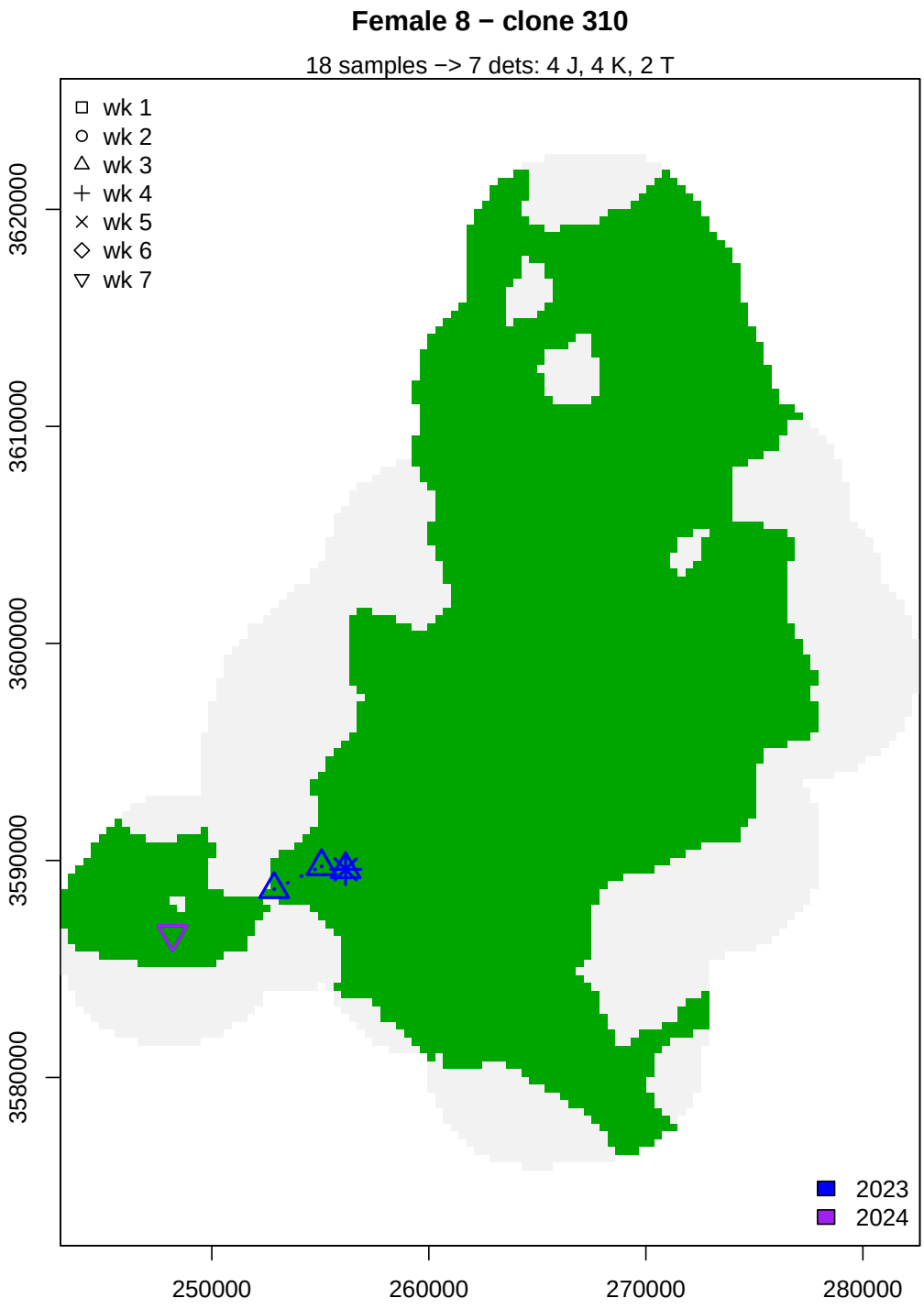
260000
270000
280000
2023
2024


Female 8 − clone 310
18 samples −> 7 dets: 4 J, 4 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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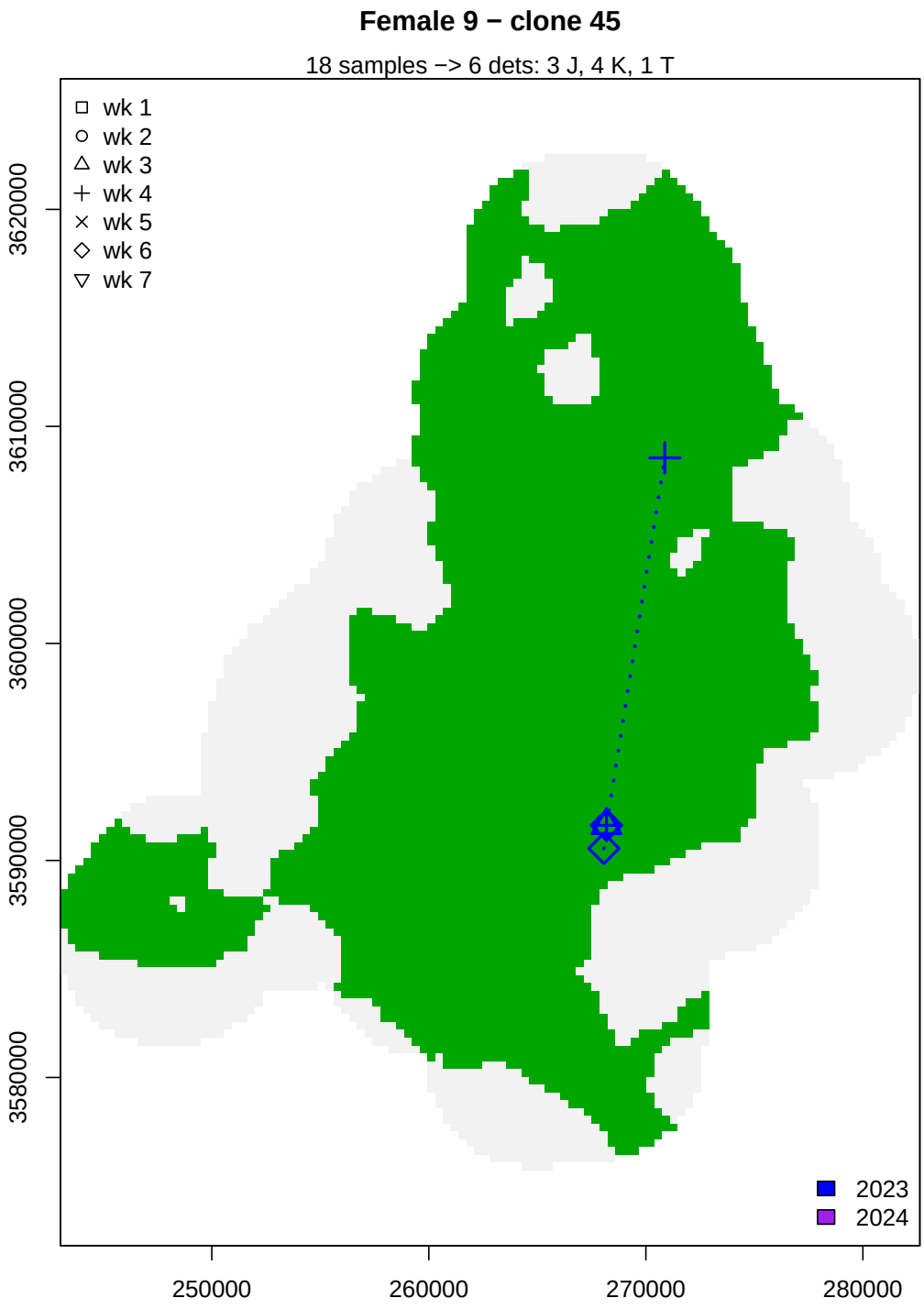
Female 9 − clone 45
18 samples −> 6 dets: 3 J, 4 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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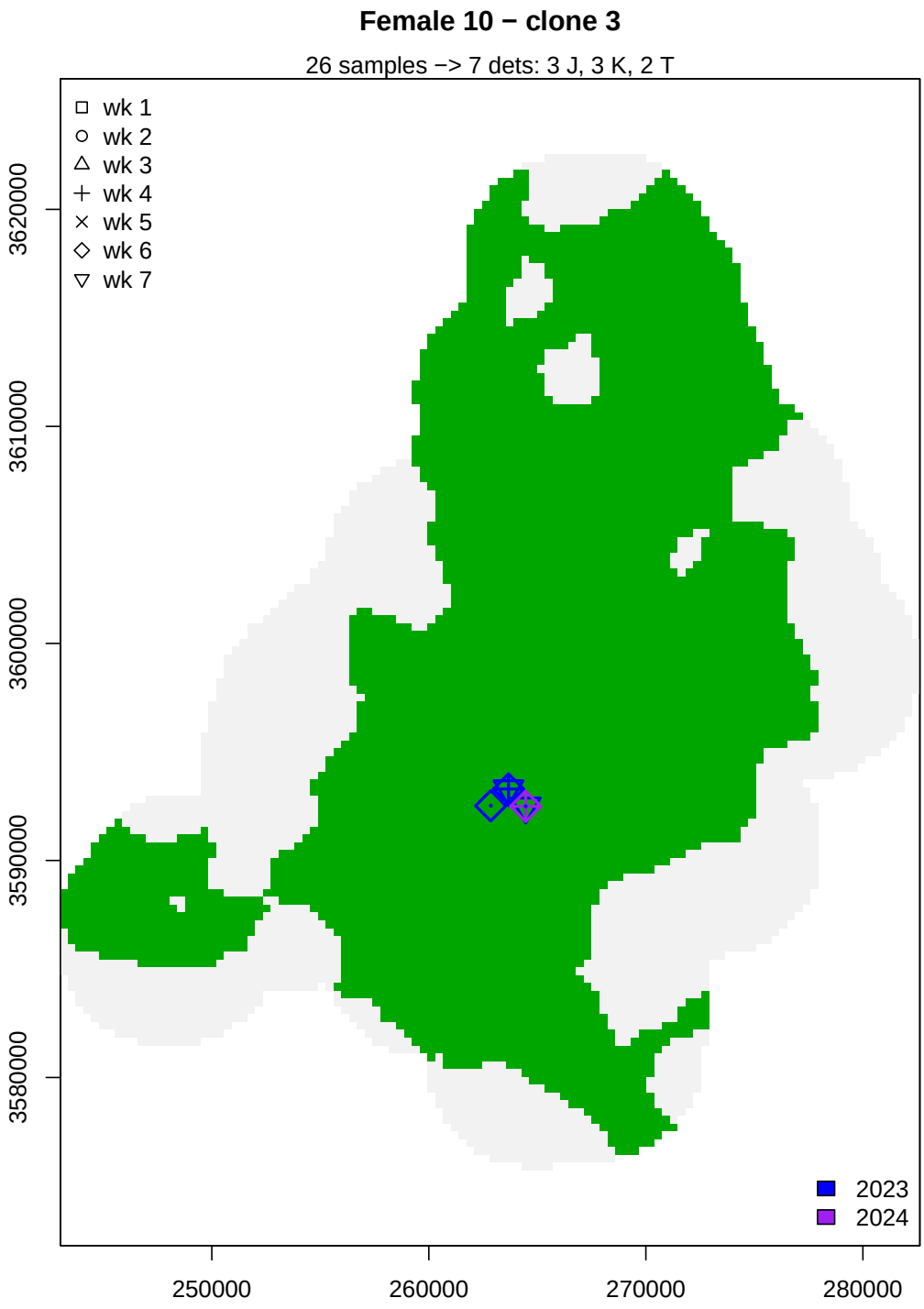
2024


Female 10 − clone 3
26 samples −> 7 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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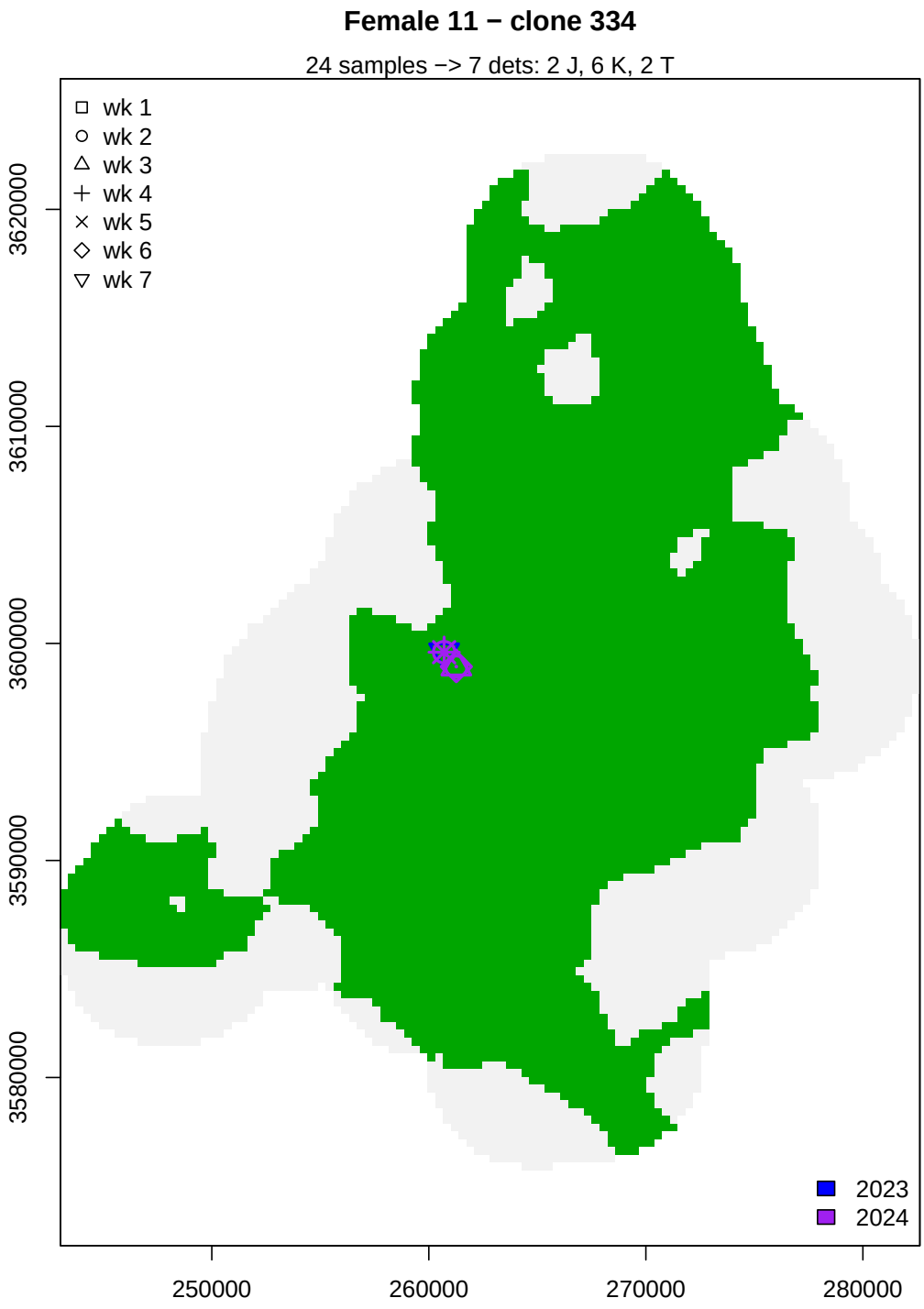
Female 11 − clone 334
24 samples −> 7 dets: 2 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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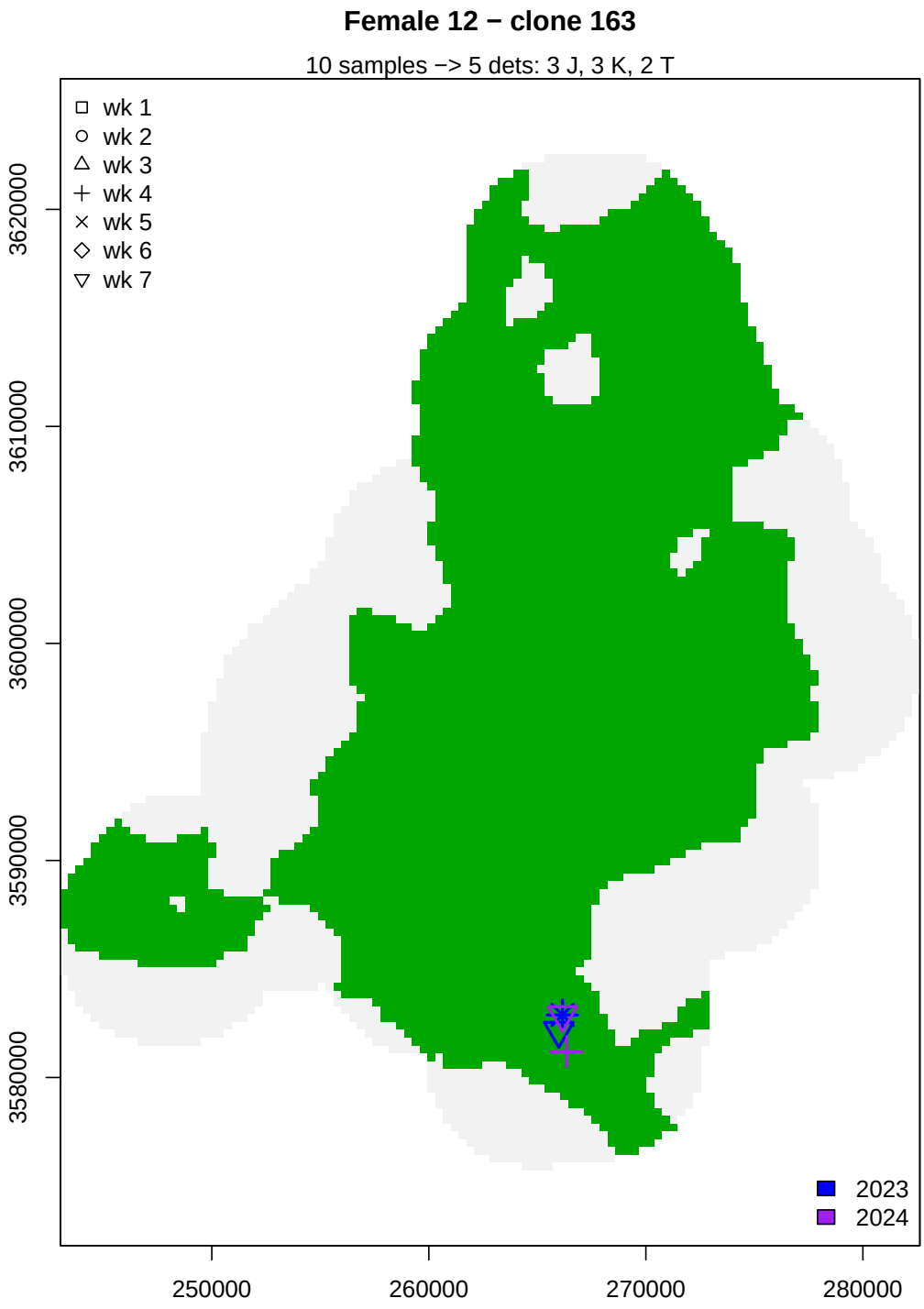
Female 12 − clone 163
10 samples −> 5 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
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wk 7
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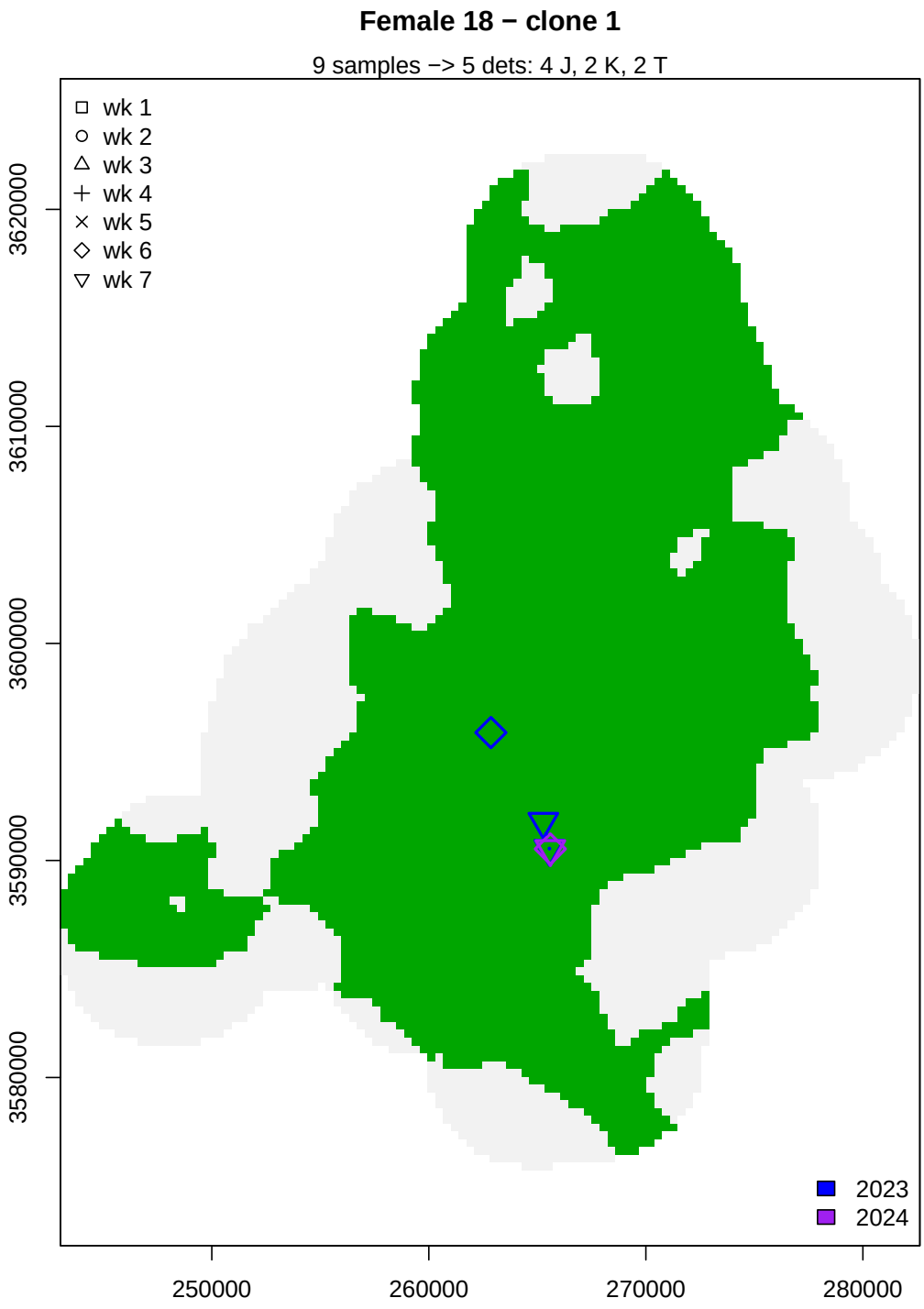
Female 18 − clone 1
9 samples −> 5 dets: 4 J, 2 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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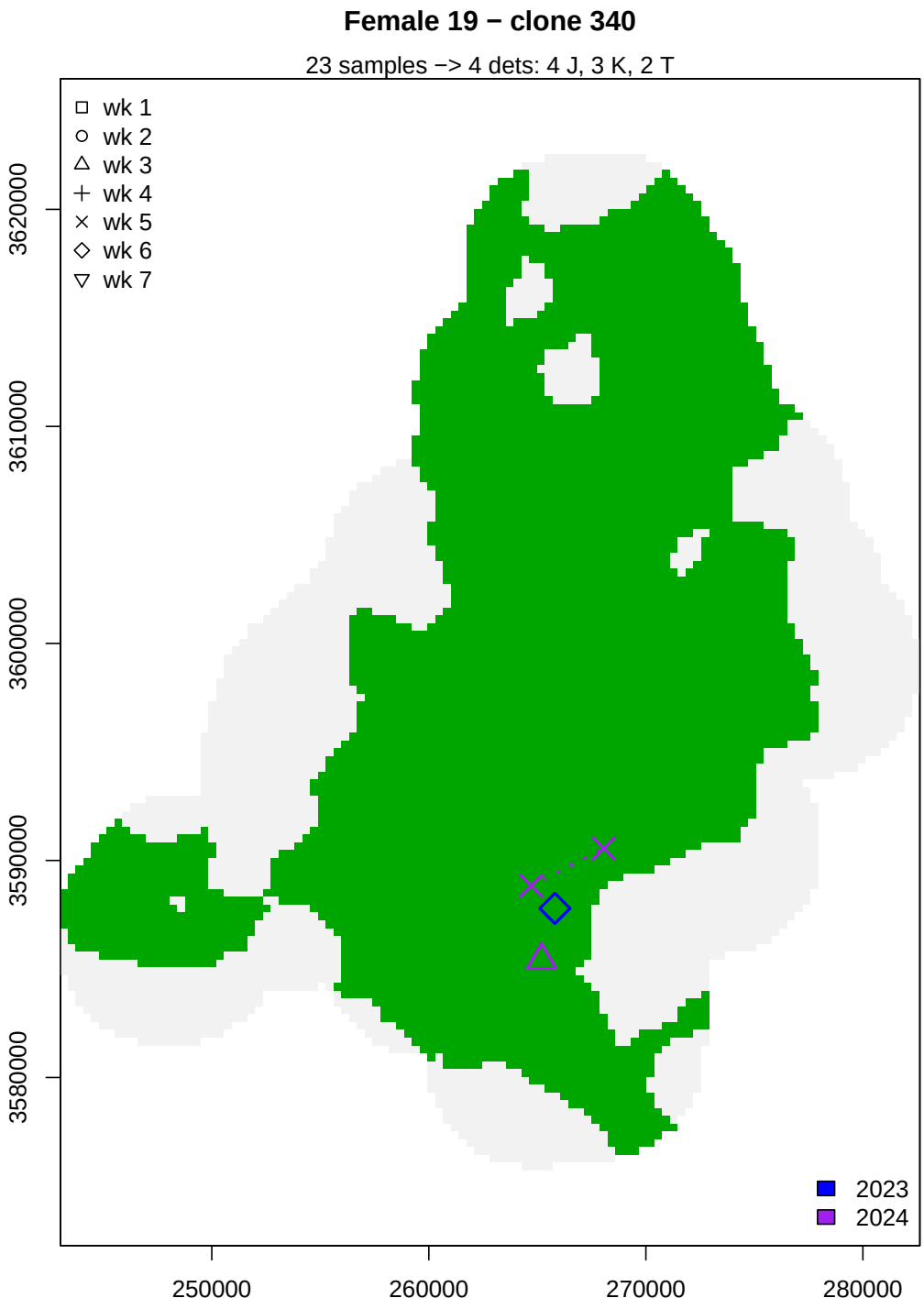
Female 19 − clone 340
23 samples −> 4 dets: 4 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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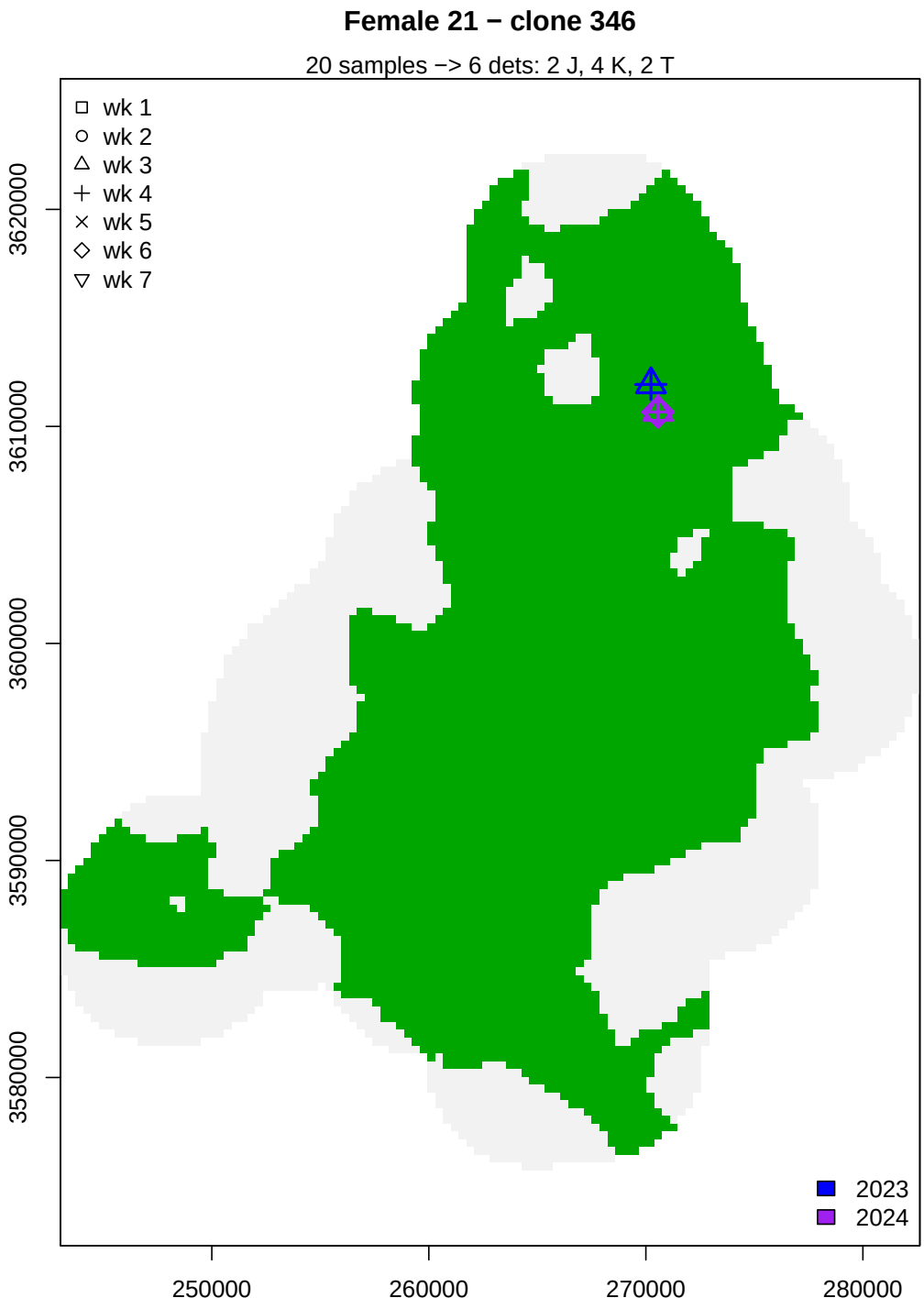
Female 21 − clone 346
20 samples −> 6 dets: 2 J, 4 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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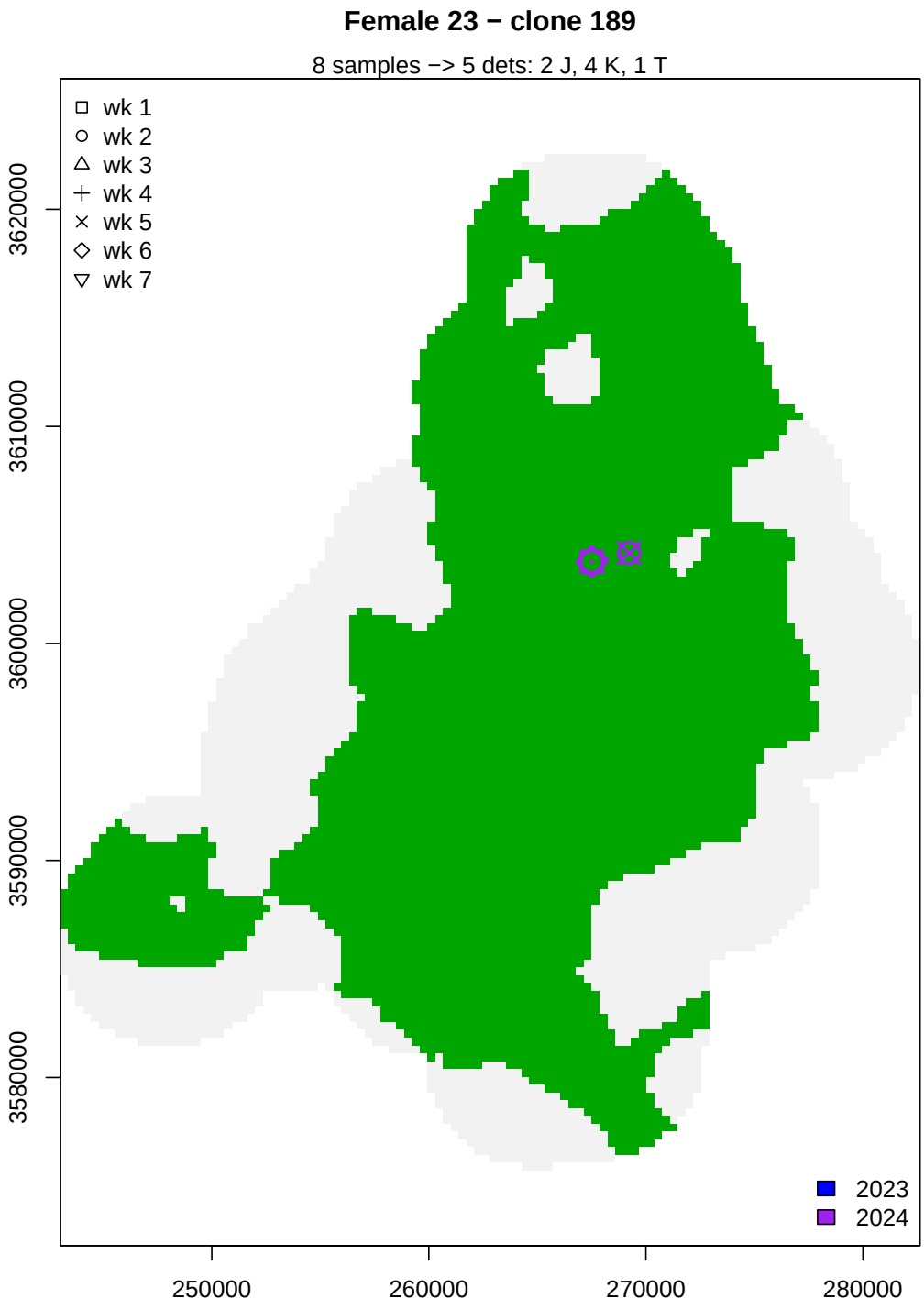
Female 23 − clone 189
8 samples −> 5 dets: 2 J, 4 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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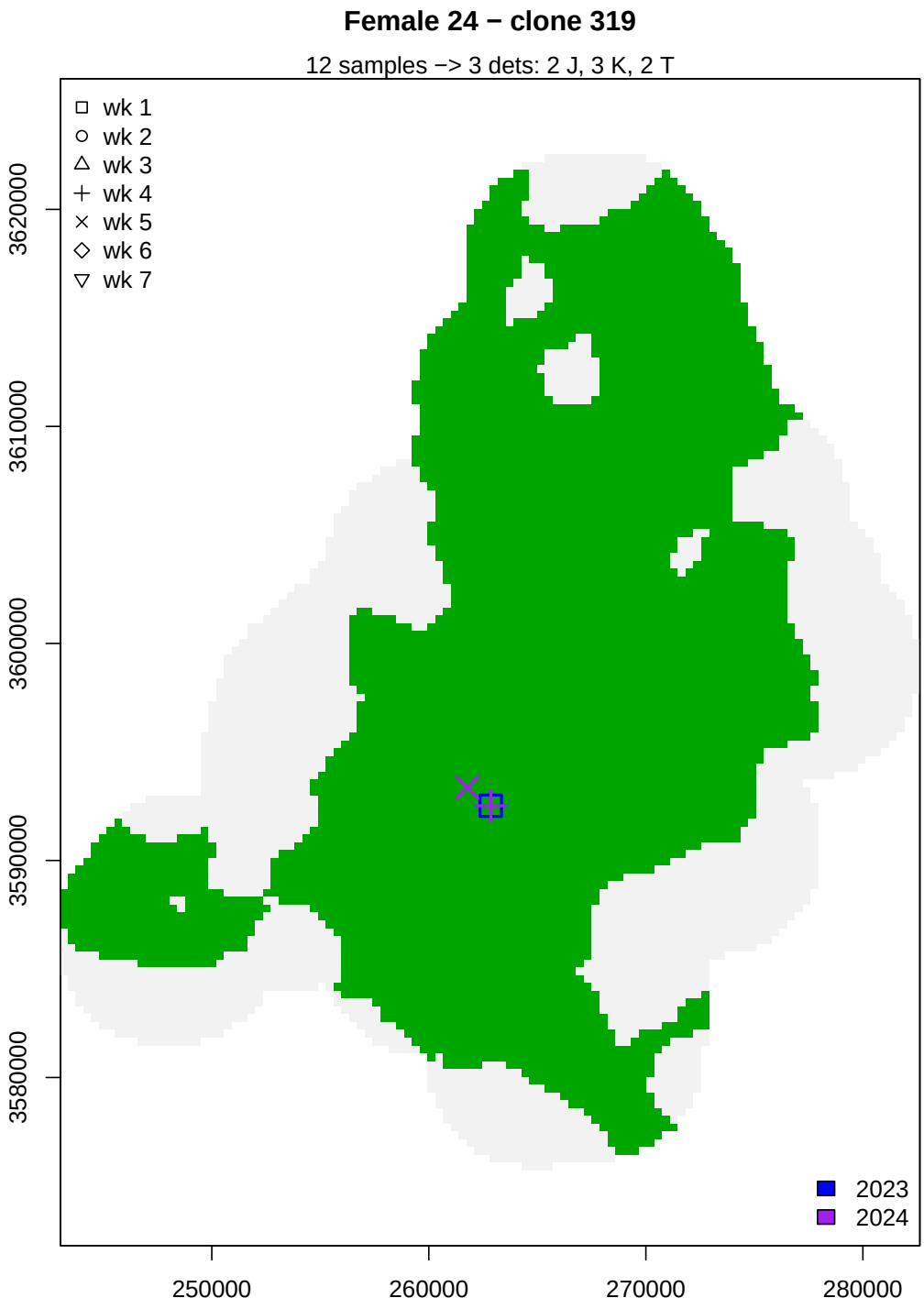
Female 24 − clone 319
12 samples −> 3 dets: 2 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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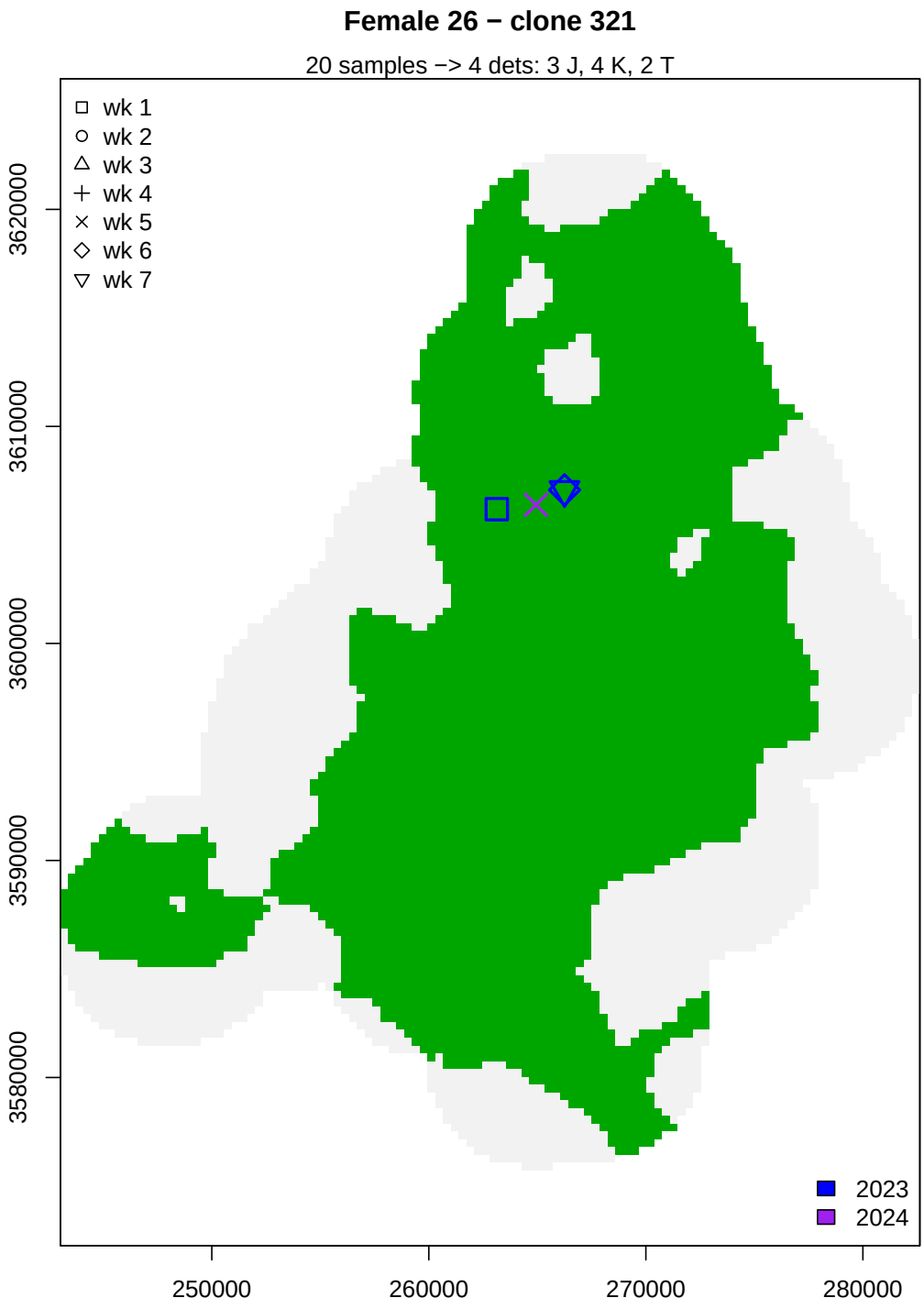
Female 26 − clone 321
20 samples −> 4 dets: 3 J, 4 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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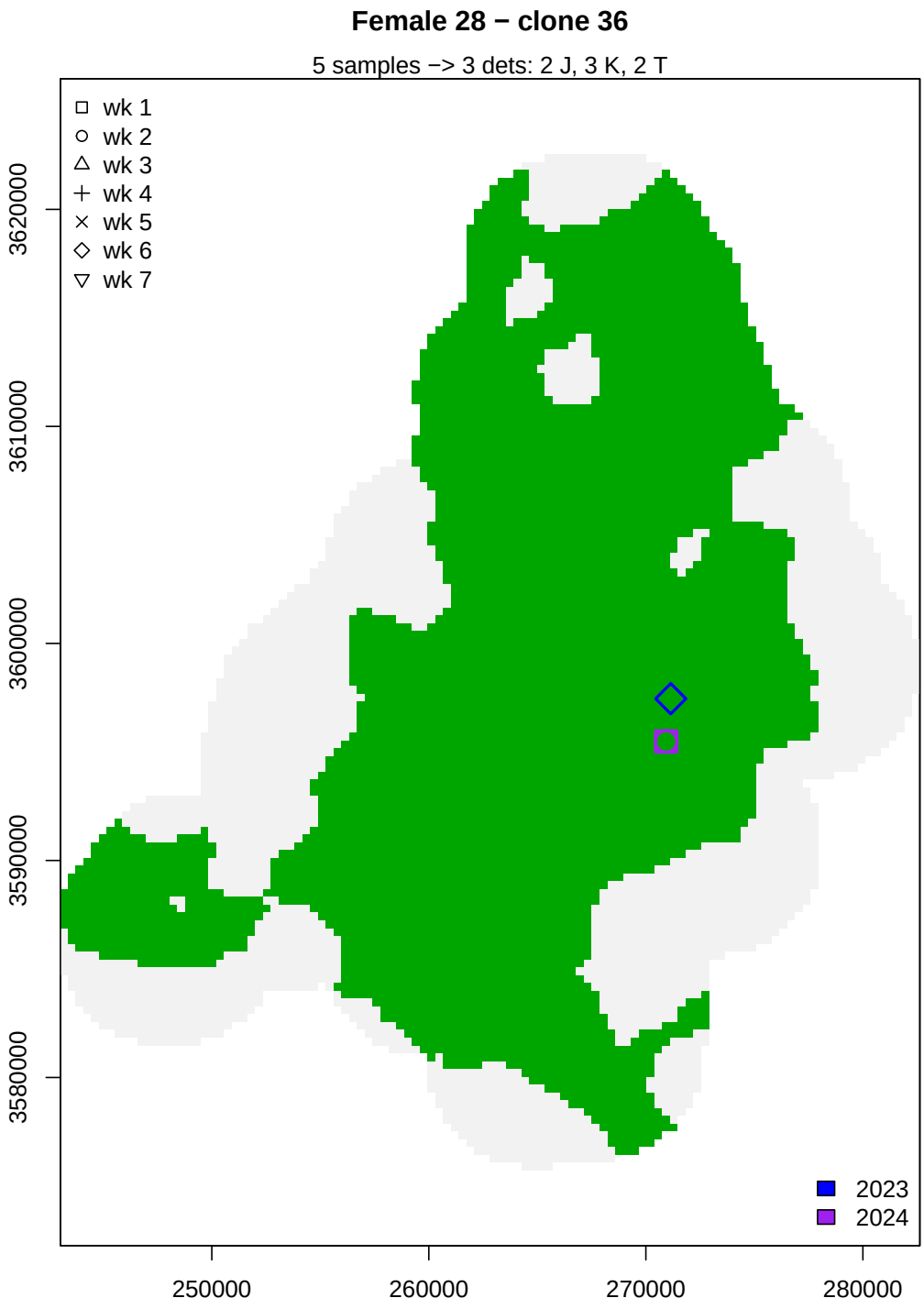
Female 28 − clone 36
5 samples −> 3 dets: 2 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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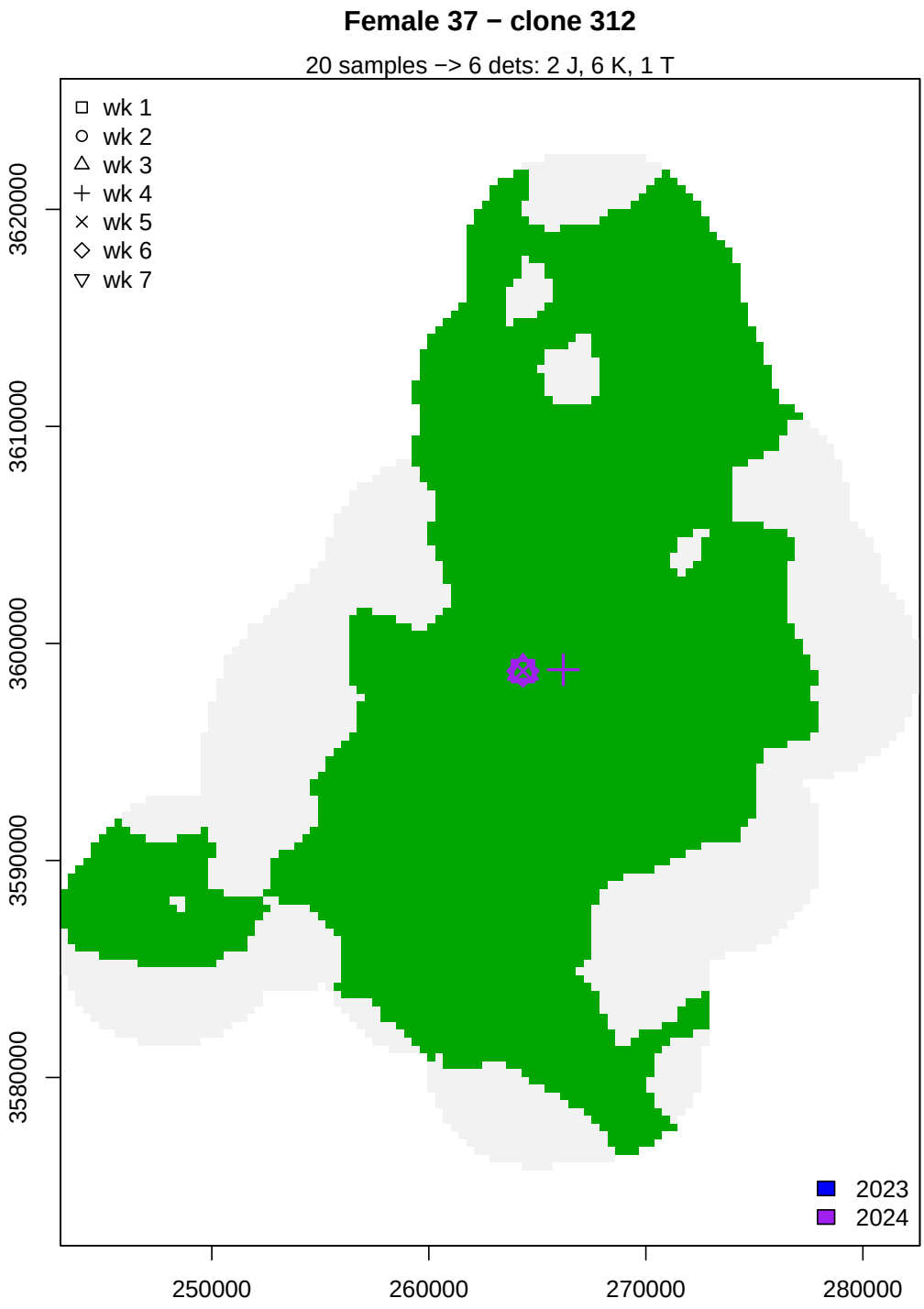
Female 37 − clone 312
20 samples −> 6 dets: 2 J, 6 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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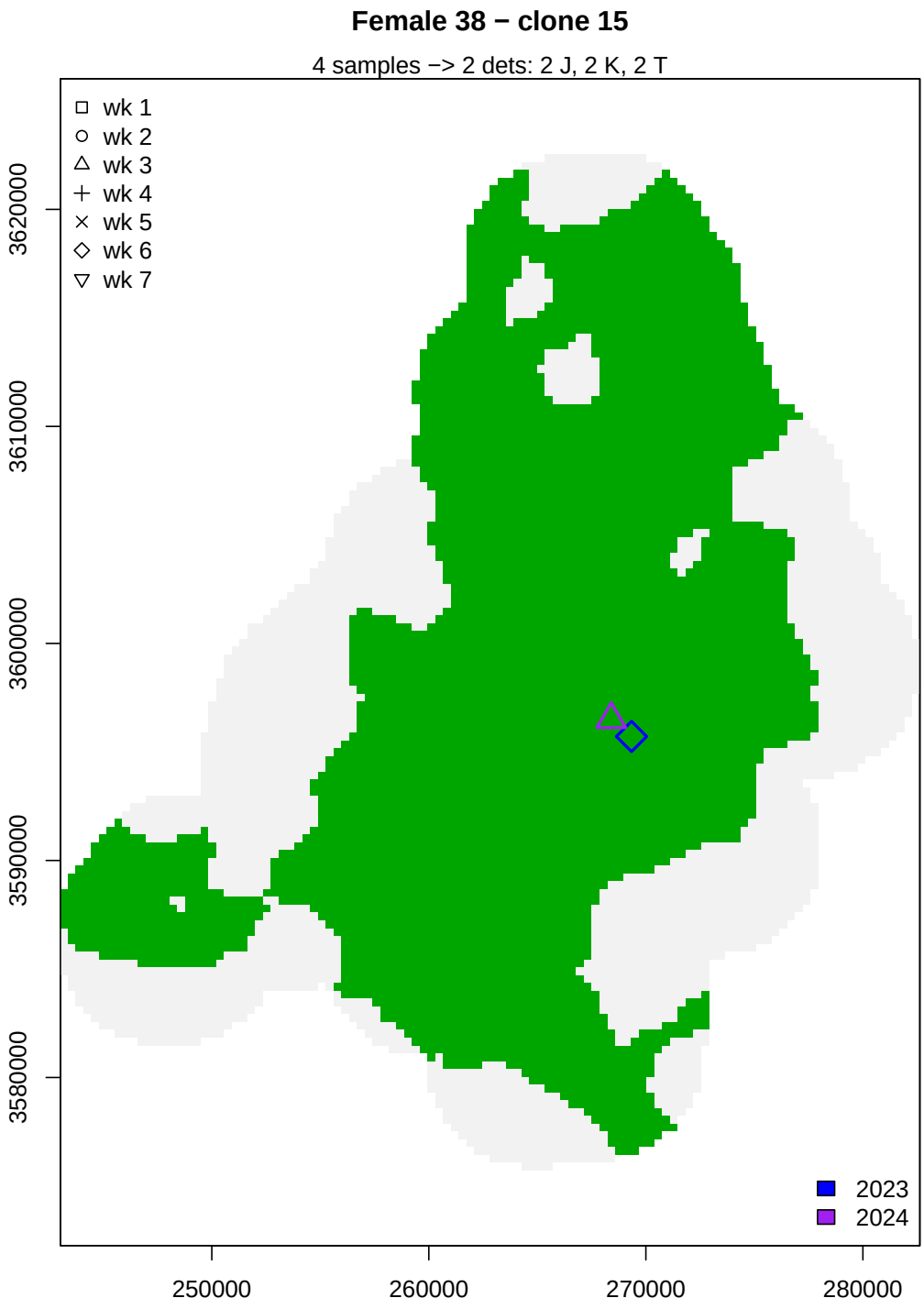
Female 38 − clone 15
4 samples −> 2 dets: 2 J, 2 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
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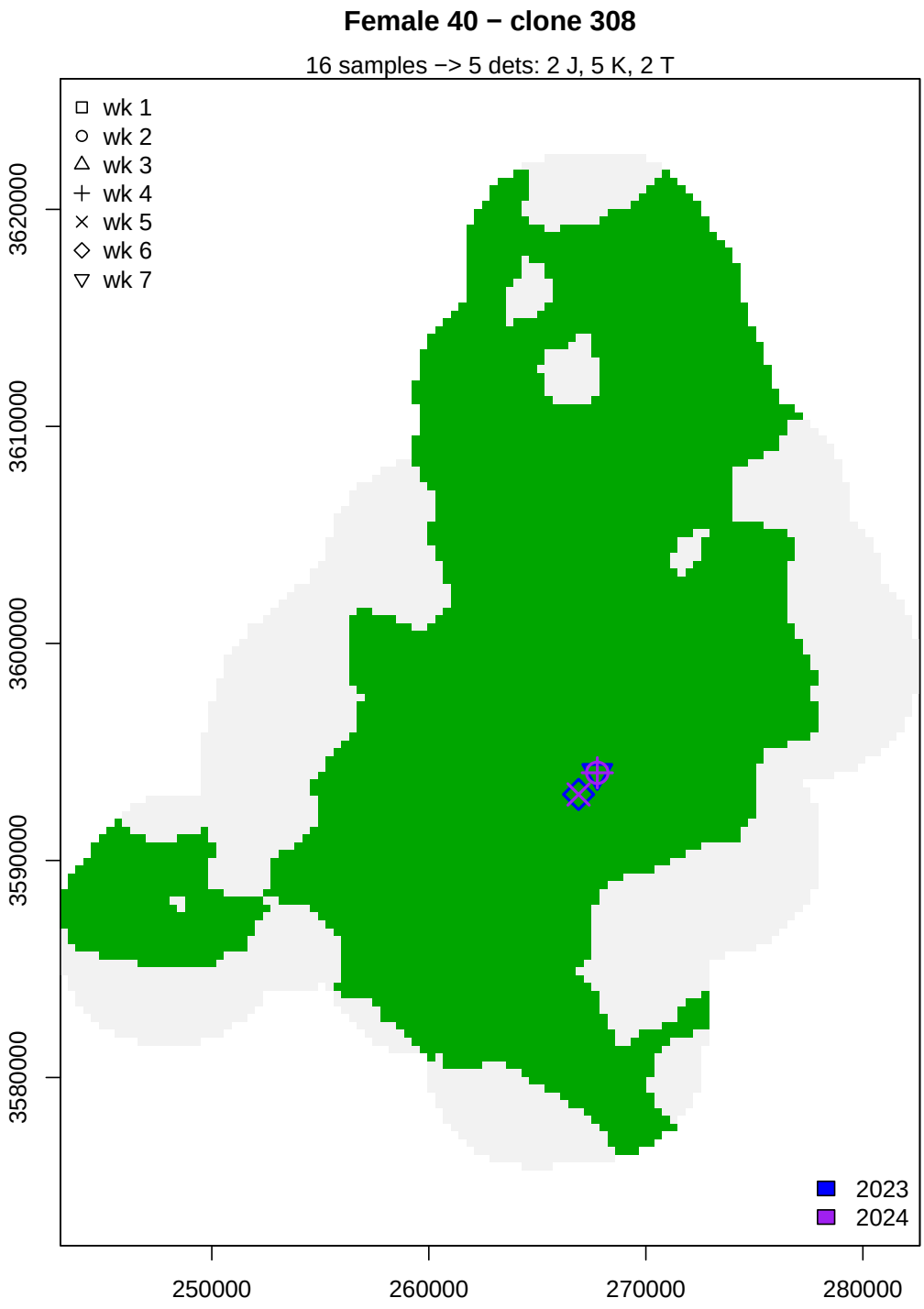
Female 40 − clone 308
16 samples −> 5 dets: 2 J, 5 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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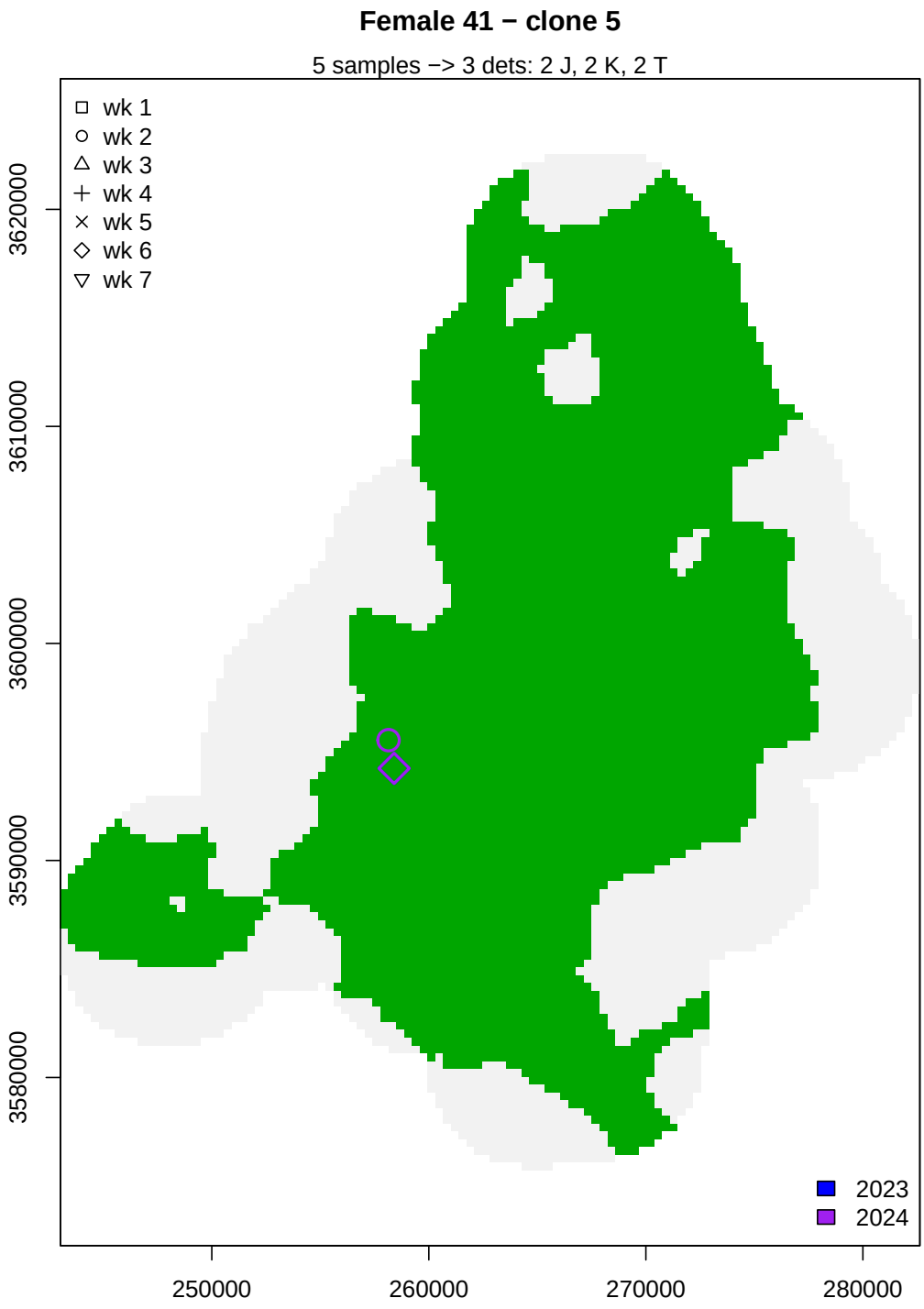
2023
2024


Female 41 − clone 5
5 samples −> 3 dets: 2 J, 2 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
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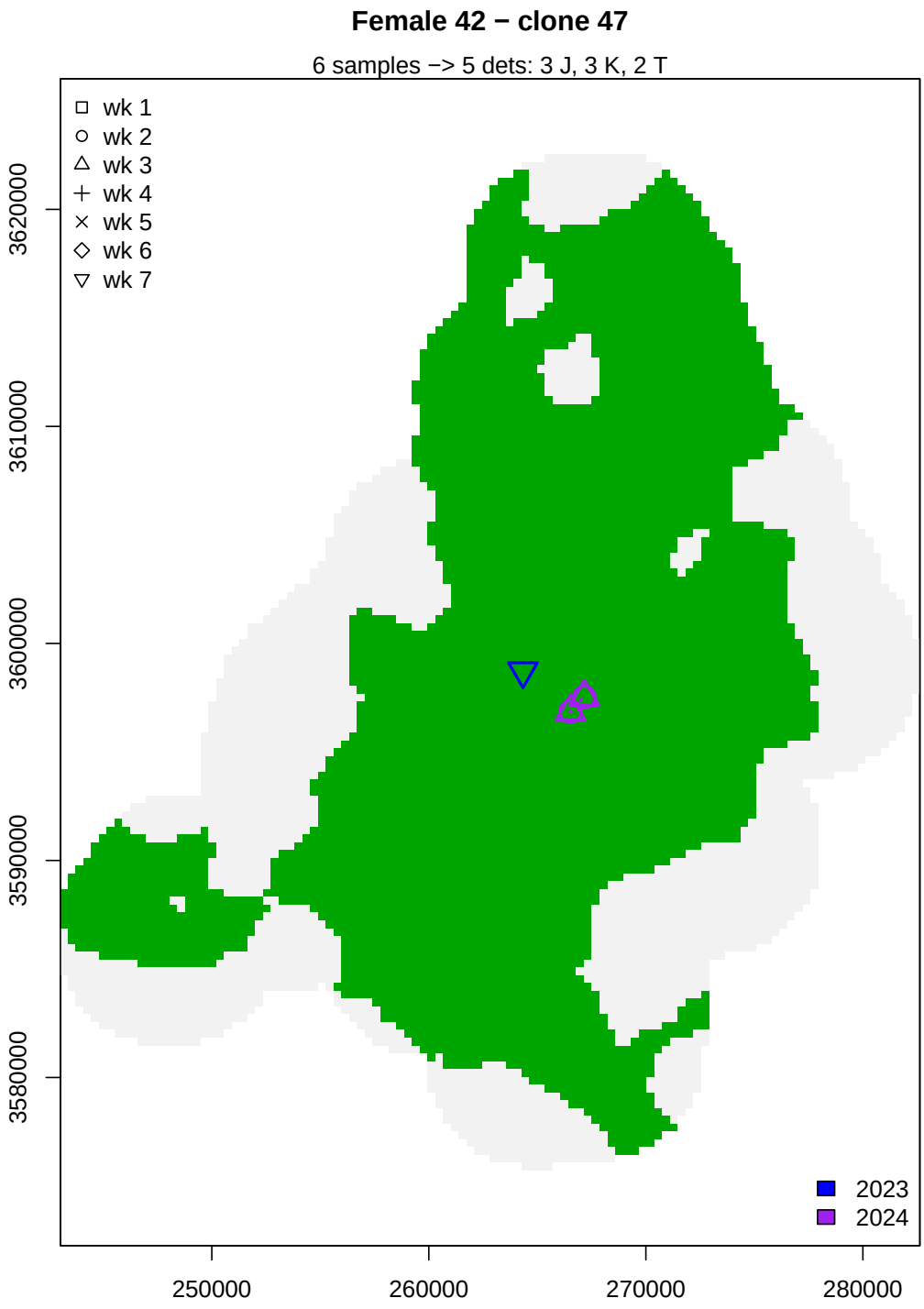
Female 42 − clone 47
6 samples −> 5 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
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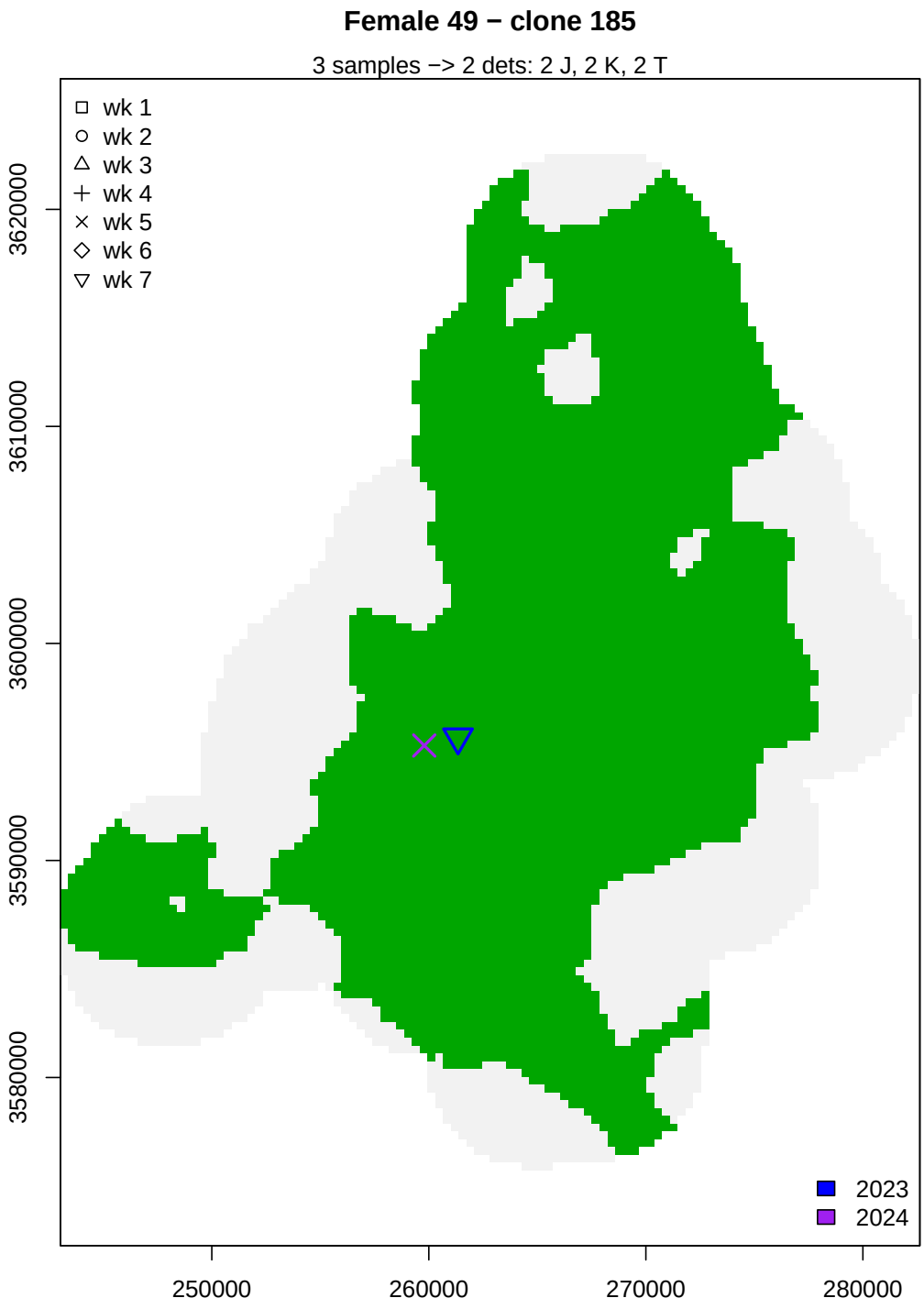
Female 49 − clone 185
3 samples −> 2 dets: 2 J, 2 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
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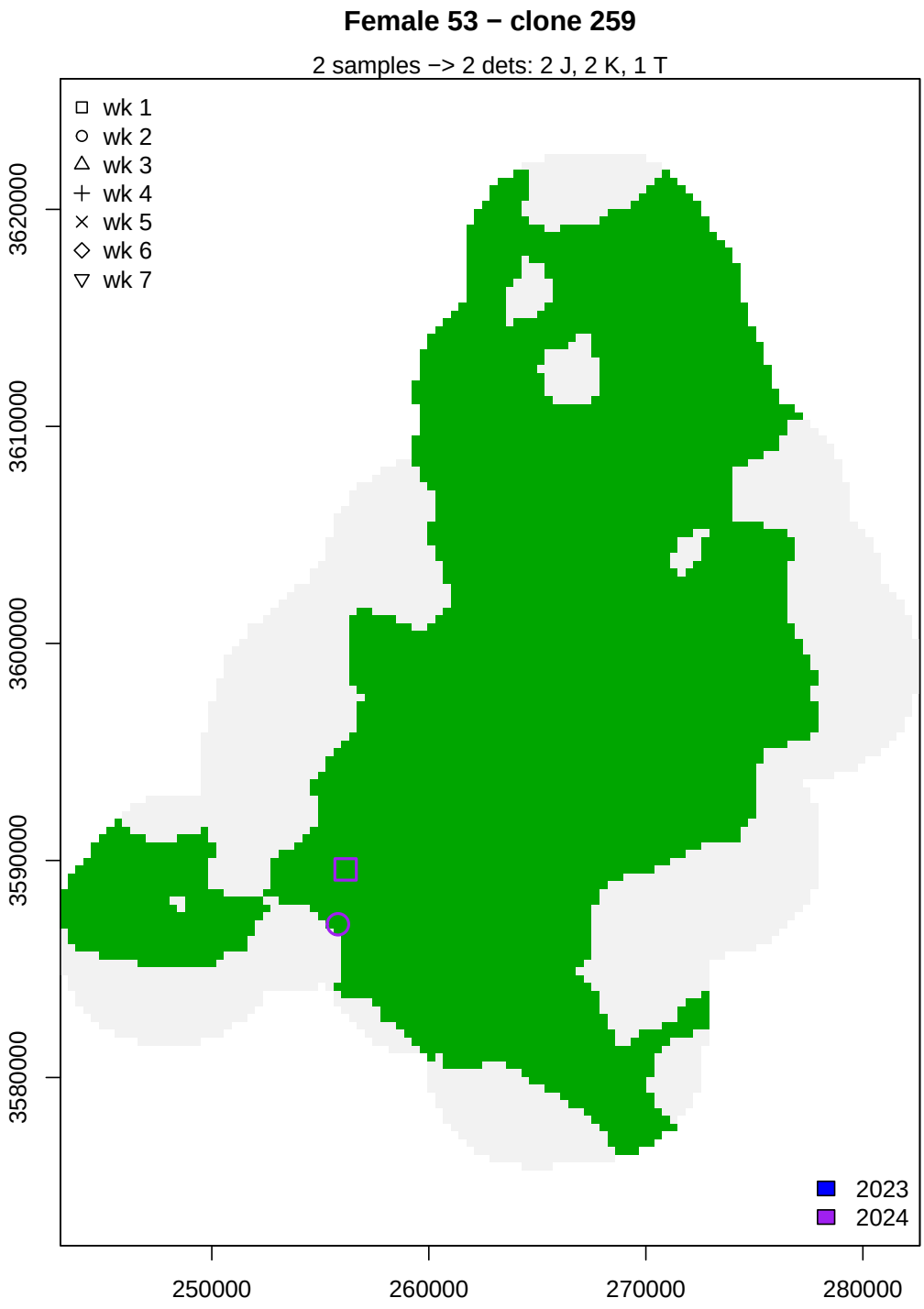
Female 53 − clone 259
2 samples −> 2 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
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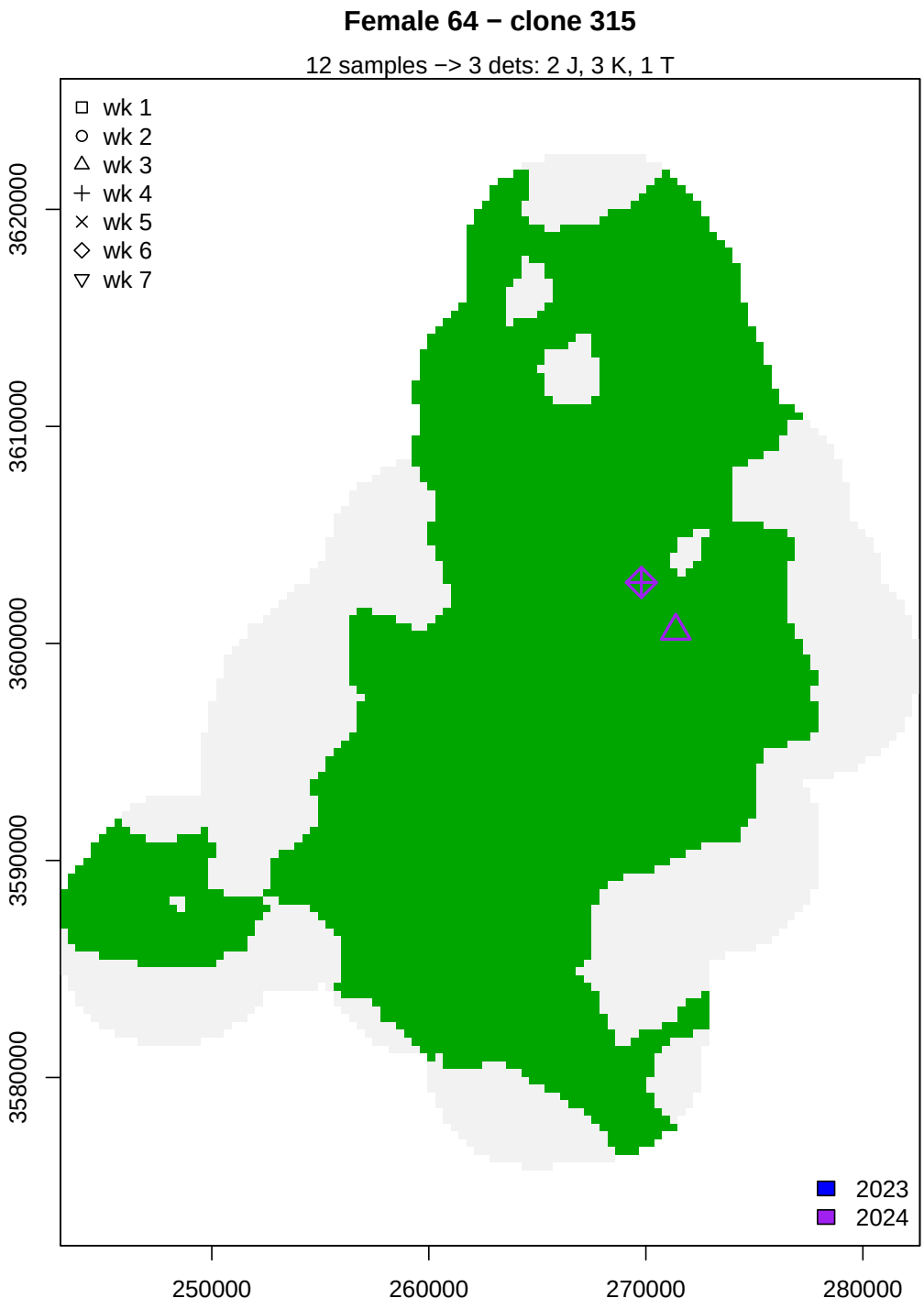
Female 64 − clone 315
12 samples −> 3 dets: 2 J, 3 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
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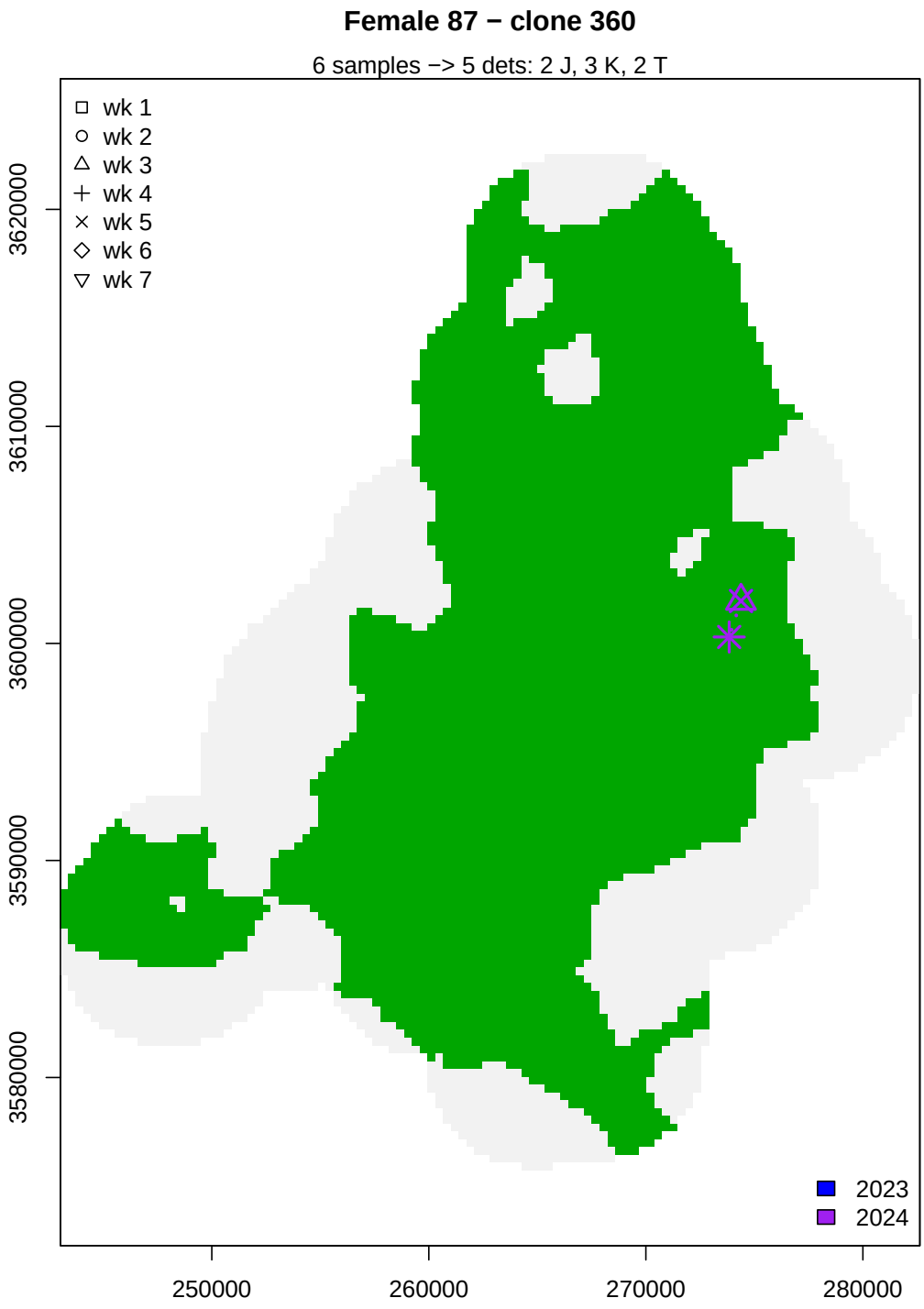
Female 87 − clone 360
6 samples −> 5 dets: 2 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
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3610000
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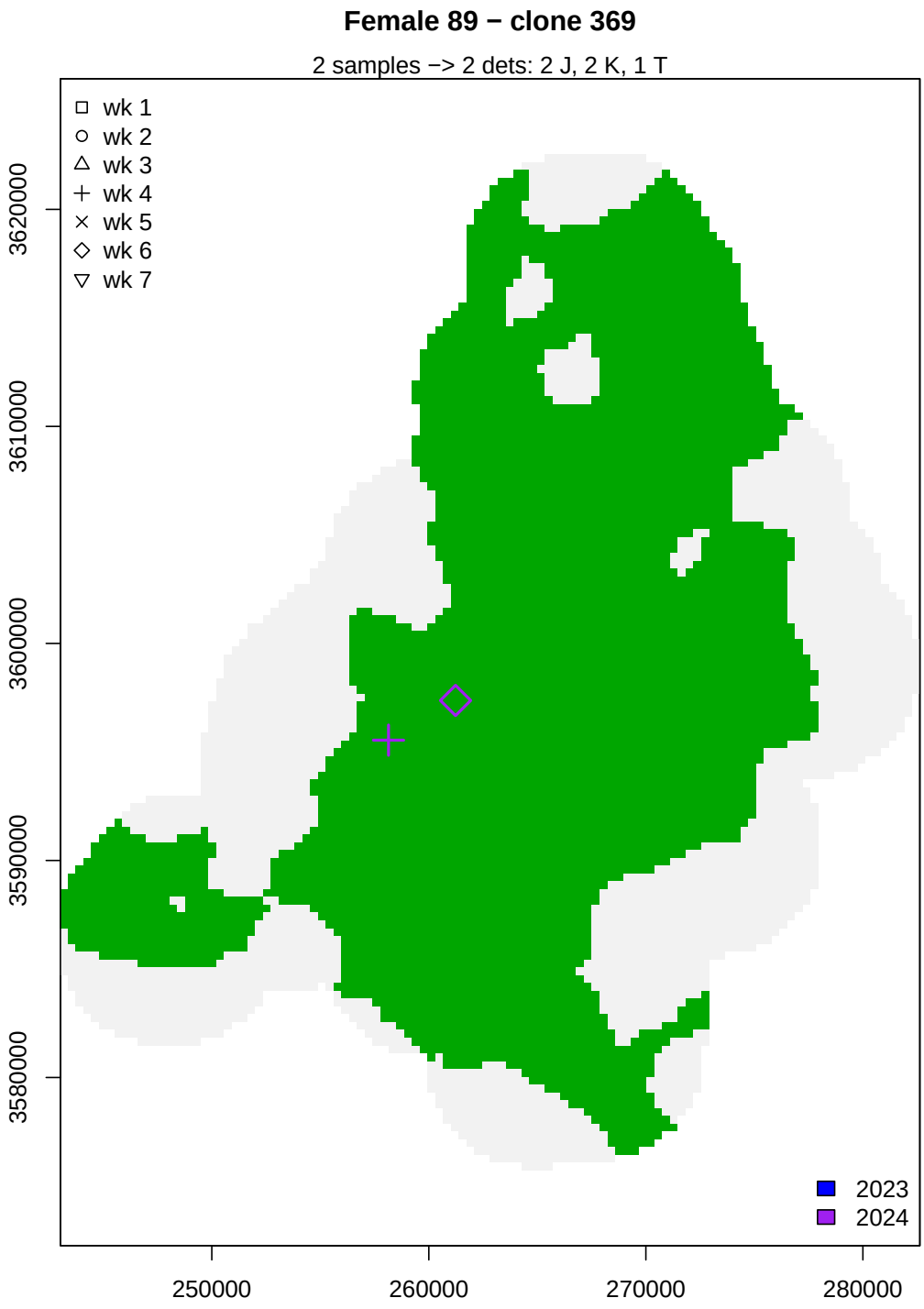
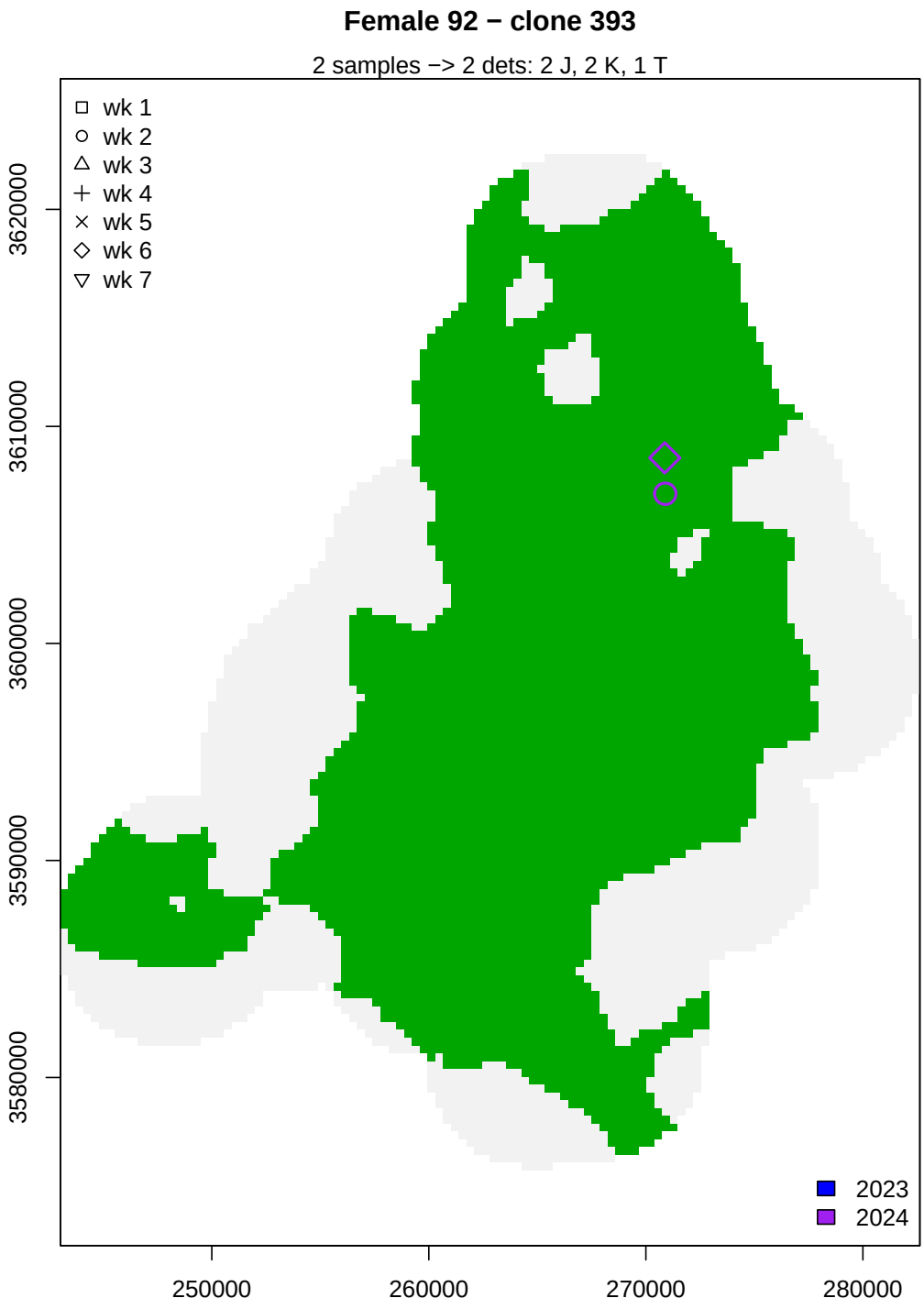
270000
280000


Female 89 − clone 369
2 samples −> 2 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



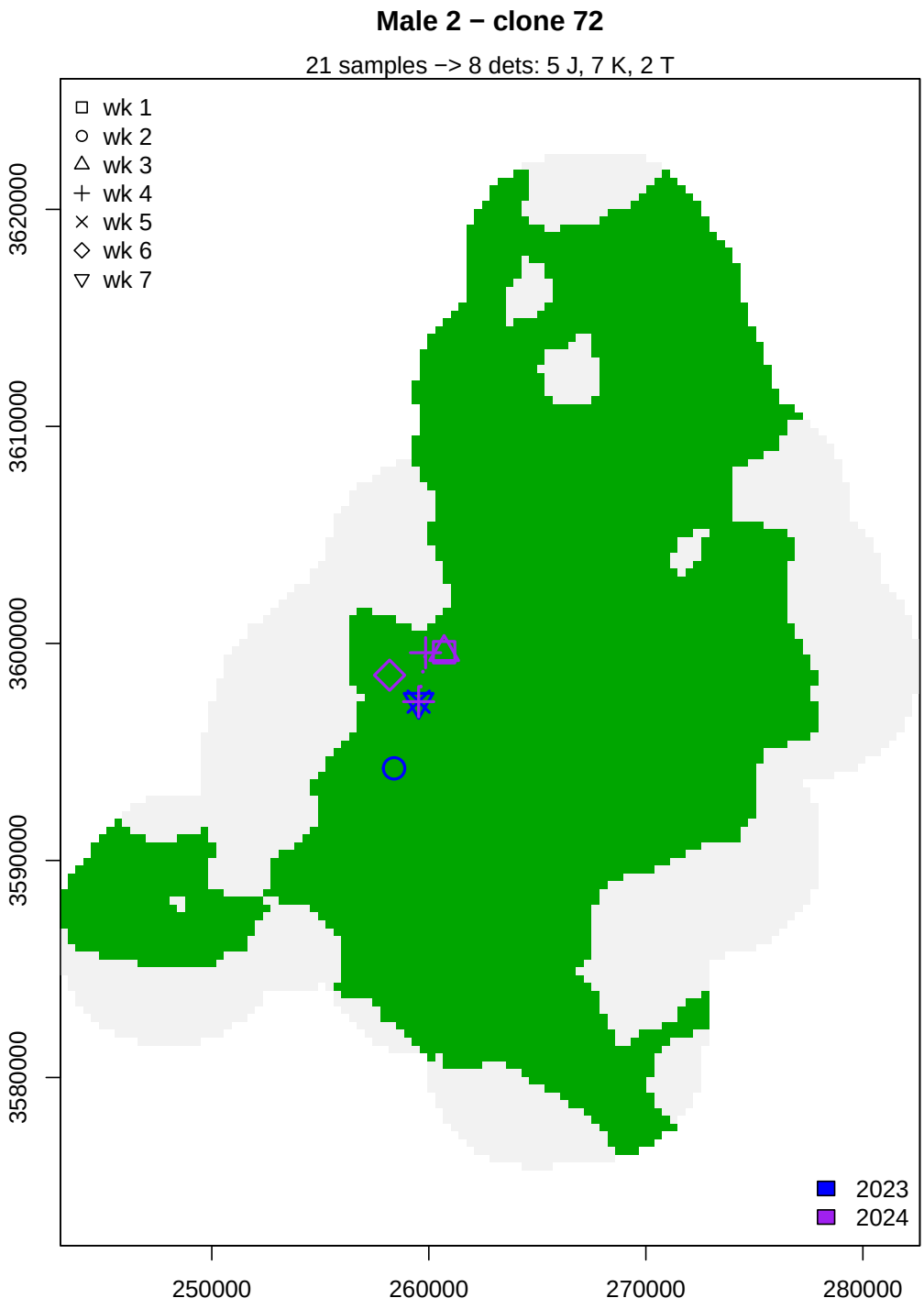
Males

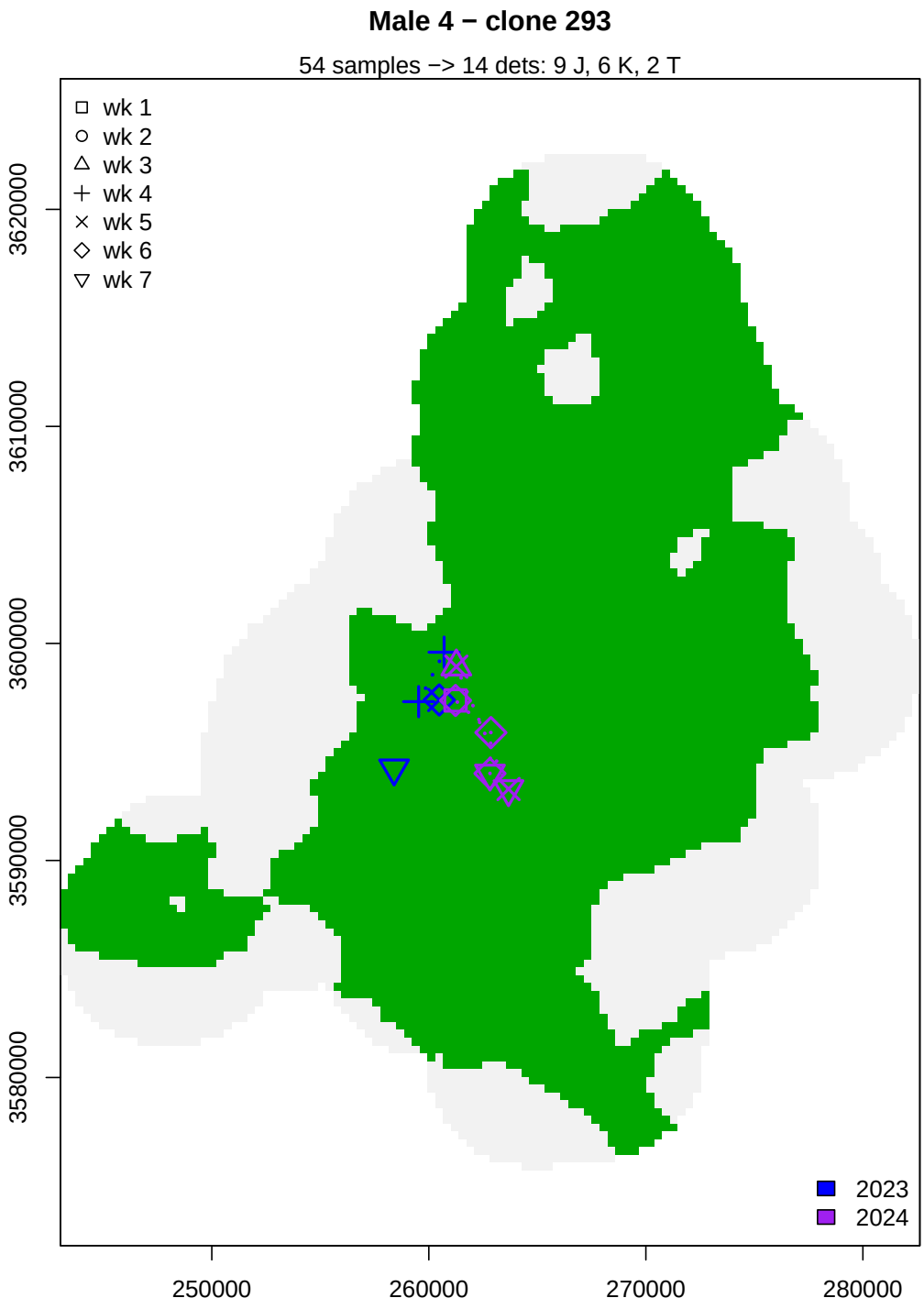
Then for males. . .

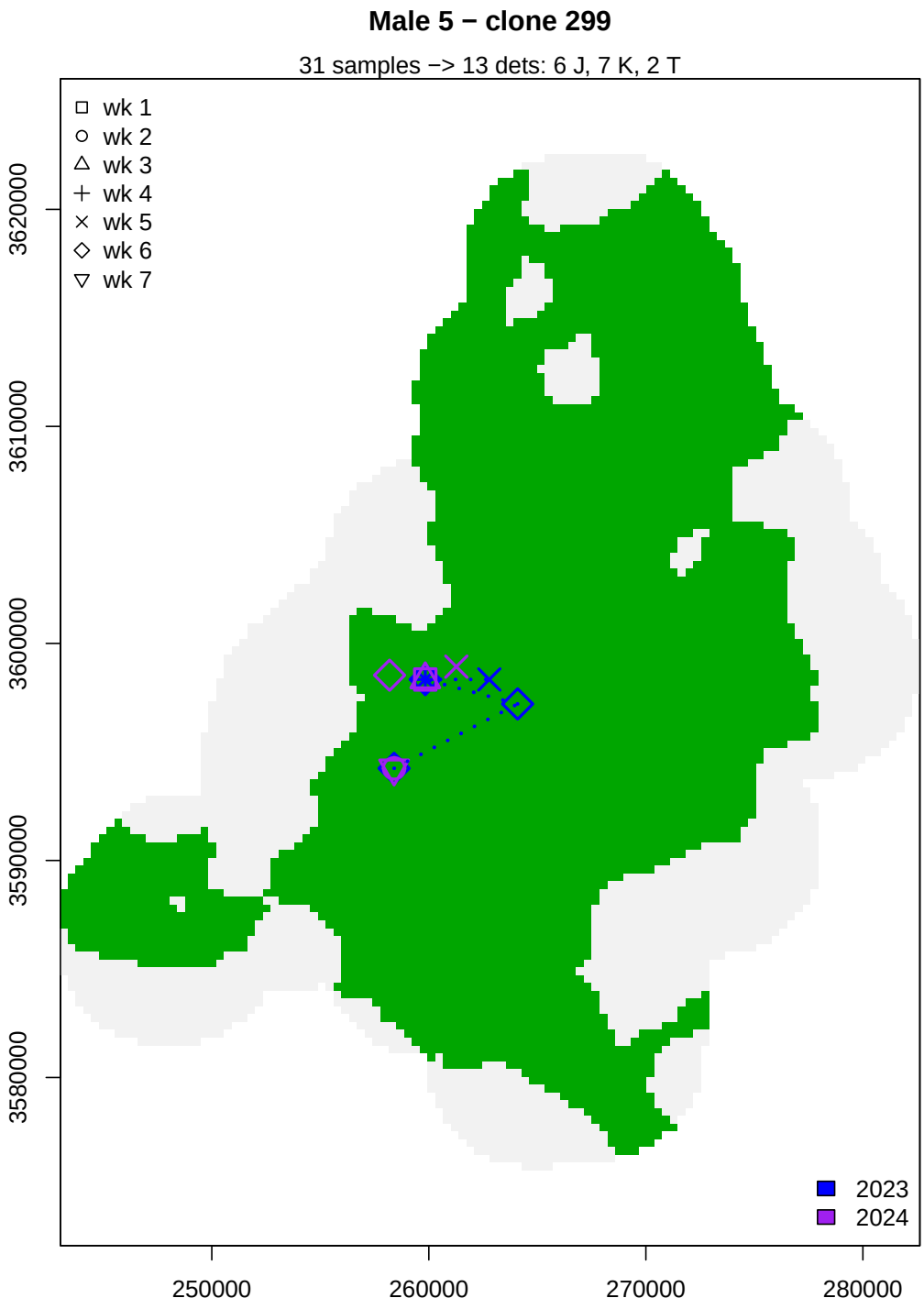
```r
#Males
caphist <- Mch$y
Mclones <- Mch$clone

sumstats_4d(caphist)
```

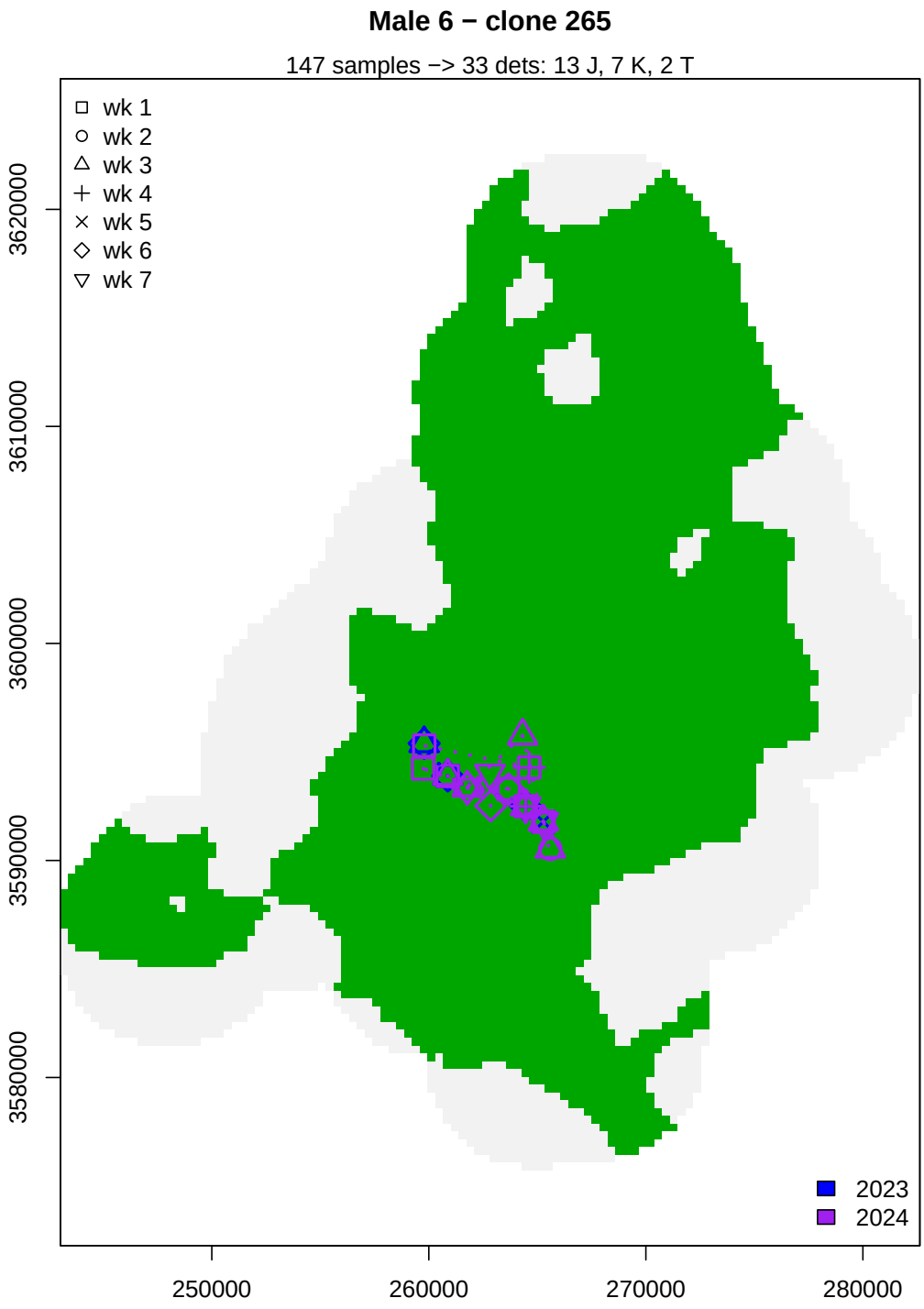
```
##        ss.totN     ss.totcaps     ss.avgcaps ss.indwsprecap    ss.avgtraps
##      90.000000     268.000000       2.977778      27.000000       2.077778
##      ss.avgyrs
##       1.166667
```

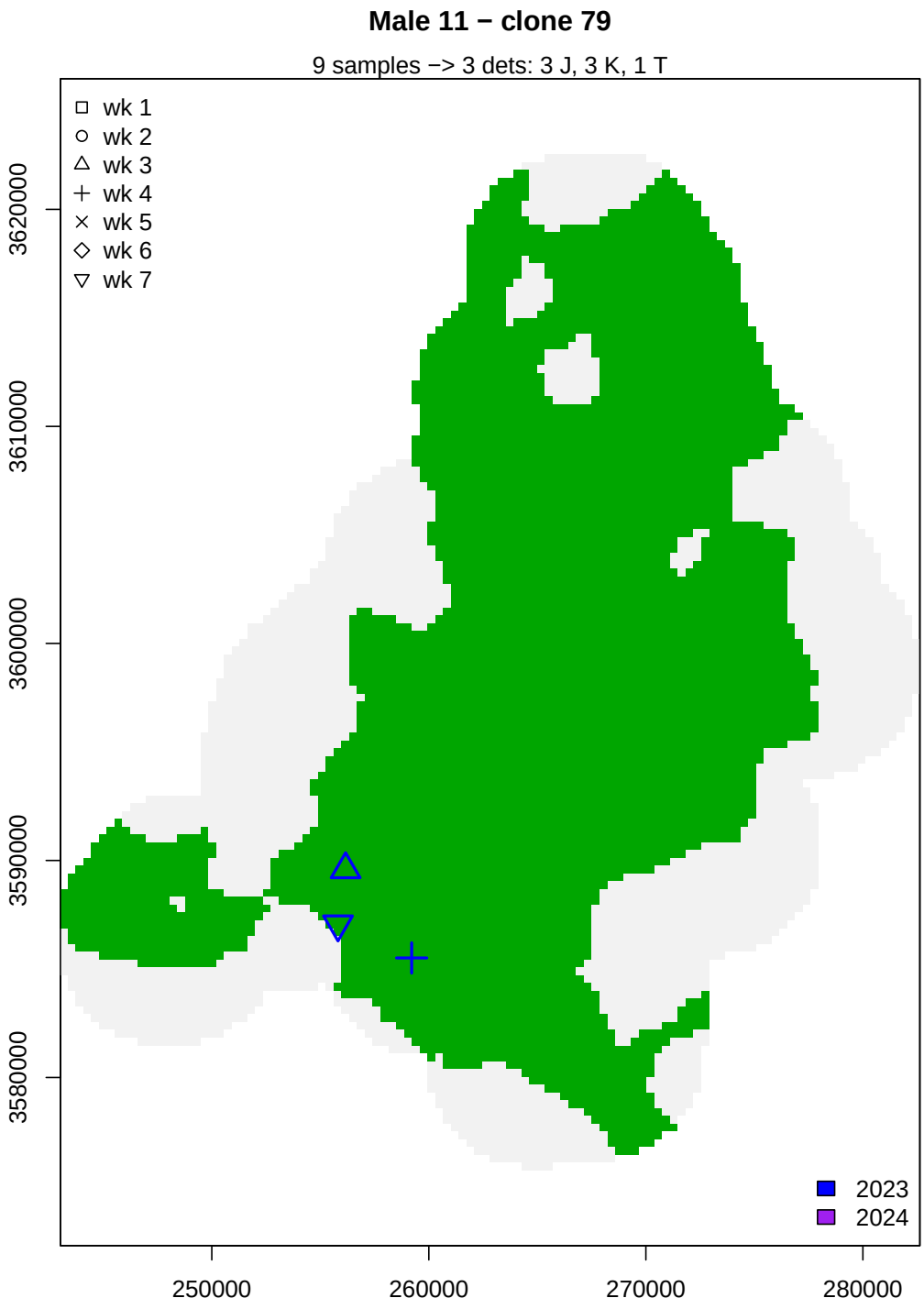
```r
nind <- dim(caphist)[1]

yr.col <- c("blue","purple")
wk.symbol <- 0:6

for(i in 1:nind) {
  i.ch <- caphist[i,,,]

  nTraps <- sum(rowSums(i.ch) > 0)
  if(nTraps > 1) {
    plot.me <- as.data.frame(which(i.ch == 1, arr.ind = TRUE))
    plot(S360, legend = FALSE, main = paste("Male",i,"- clone", Mclones[i]))
    mtext(paste(length(which(smd$finalClone == Mclones[i])), "samples ->",
                nrow(plot.me), "dets:",
                nTraps, "J,",
                length(unique(plot.me$dim2)), "K,",
                length(unique(plot.me$dim3)), "T"))
    points(x=locs[plot.me$dim1,1], locs[plot.me$dim1,2],
           pch = wk.symbol[plot.me$dim2], col = yr.col[plot.me$dim3],
           cex = 2, lwd = 2)

    for(r in 2:nrow(plot.me)) {
      if(plot.me$dim3[r]==plot.me$dim3[r-1] &
         plot.me$dim2[r]==plot.me$dim2[r-1]) {
        lines(x=c(locs[plot.me$dim1[r-1],1], locs[plot.me$dim1[r],1]),
              y=c(locs[plot.me$dim1[r-1],2], locs[plot.me$dim1[r],2]),
              lty="dotted", col = yr.col[plot.me$dim3[r]],
              lwd = 2.5)
      }
    }

    legend("topleft", bty = "n", pch = 0:6, legend = paste("wk",c(1:7)))
    legend("bottomright", bty = "n", fill = yr.col, legend = c("2023","2024"))

  }
}
```

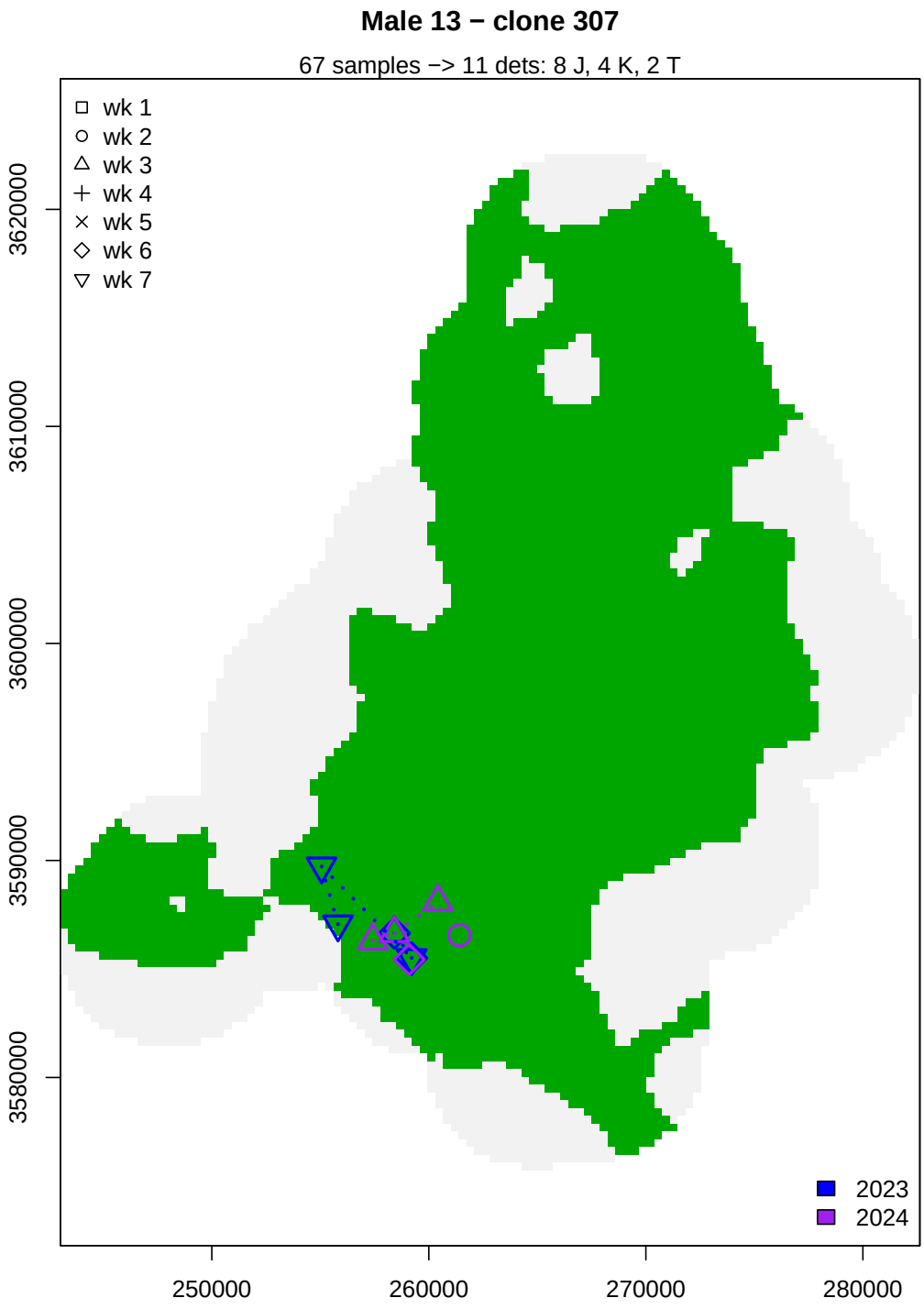

Male 2 − clone 72
21 samples −> 8 dets: 5 J, 7 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

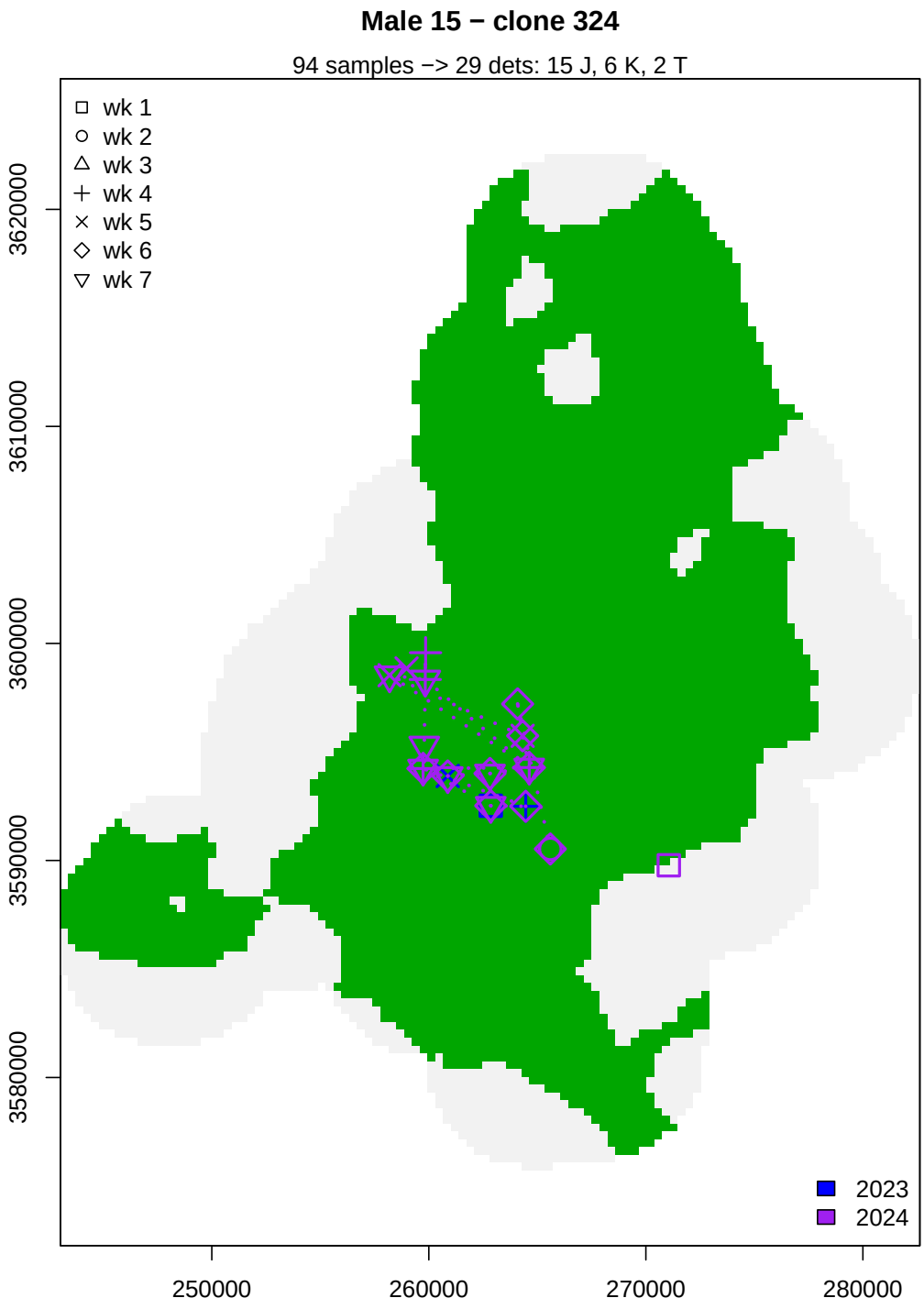

Male 4 − clone 293
54 samples −> 14 dets: 9 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

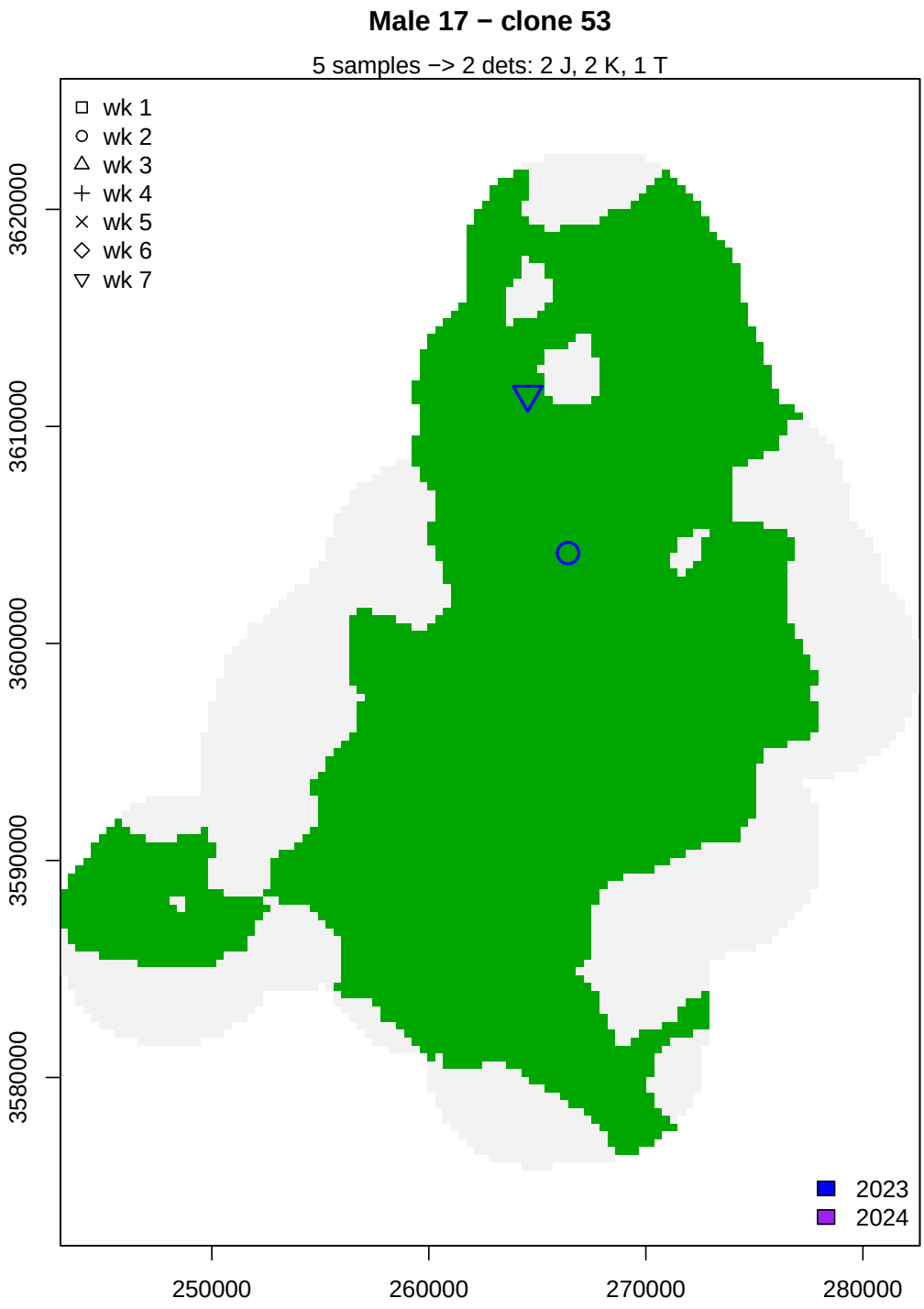

Male 5 − clone 299
31 samples −> 13 dets: 6 J, 7 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

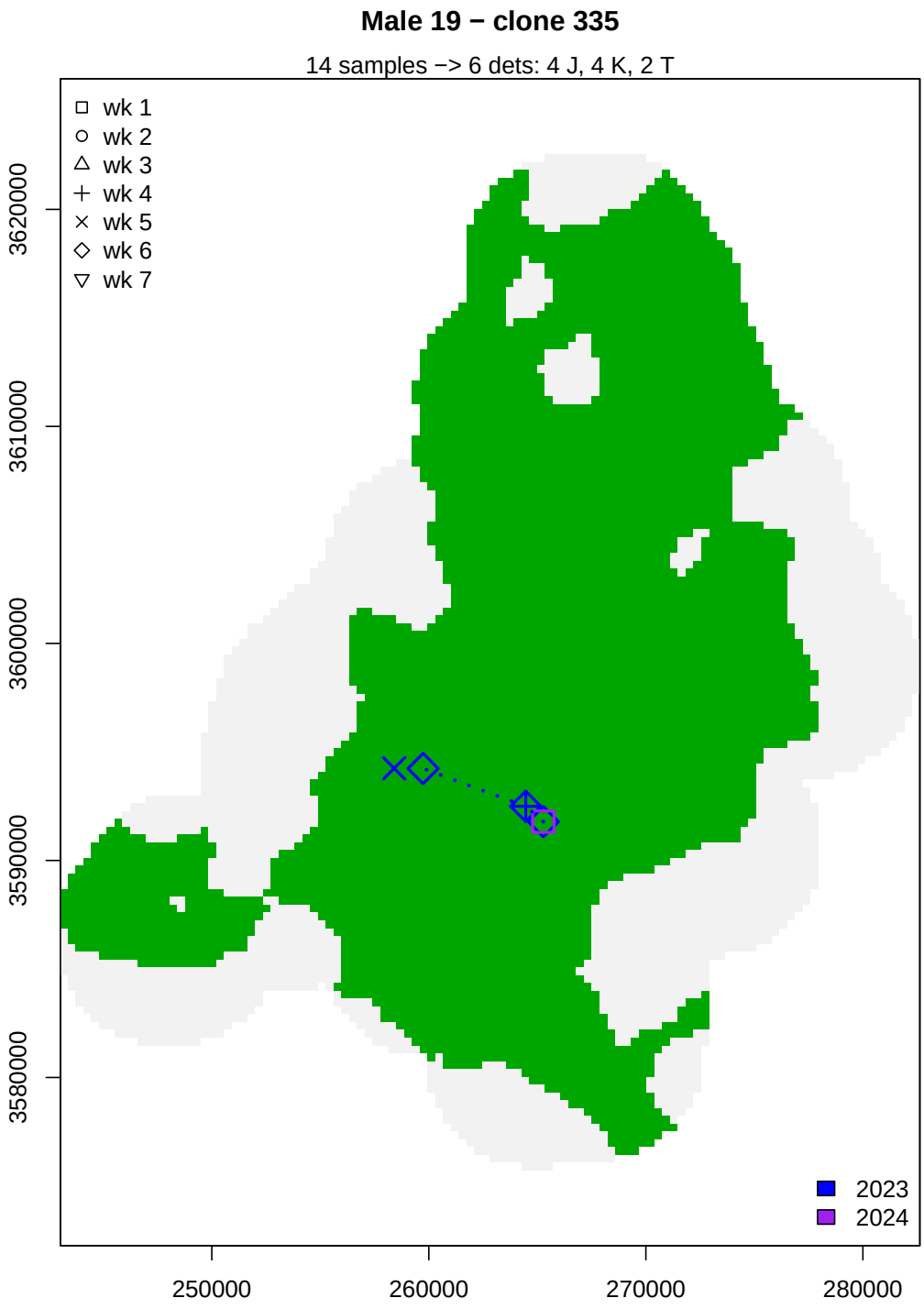

Male 6 − clone 265
147 samples −> 33 dets: 13 J, 7 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

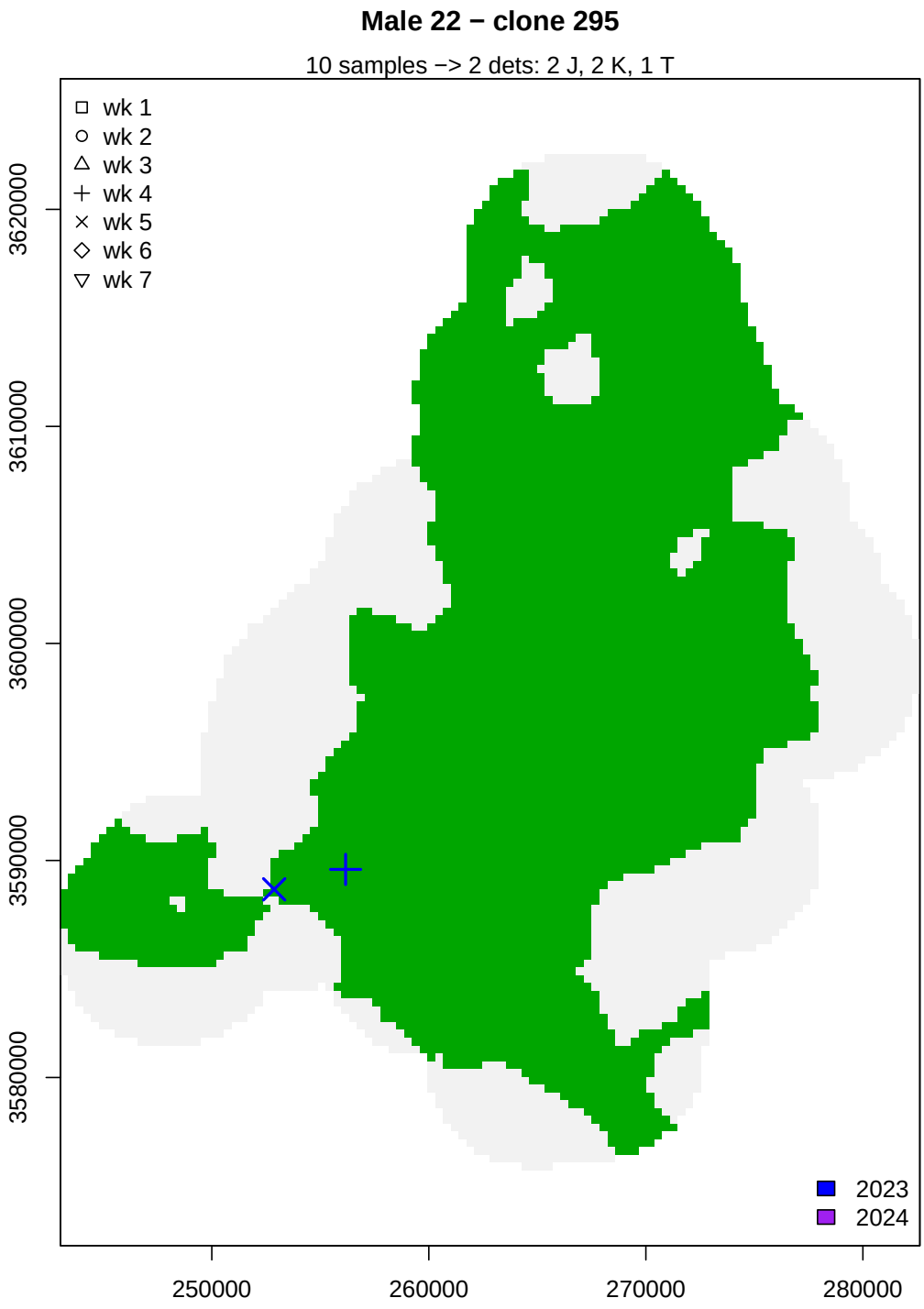

Male 11 − clone 79
9 samples −> 3 dets: 3 J, 3 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

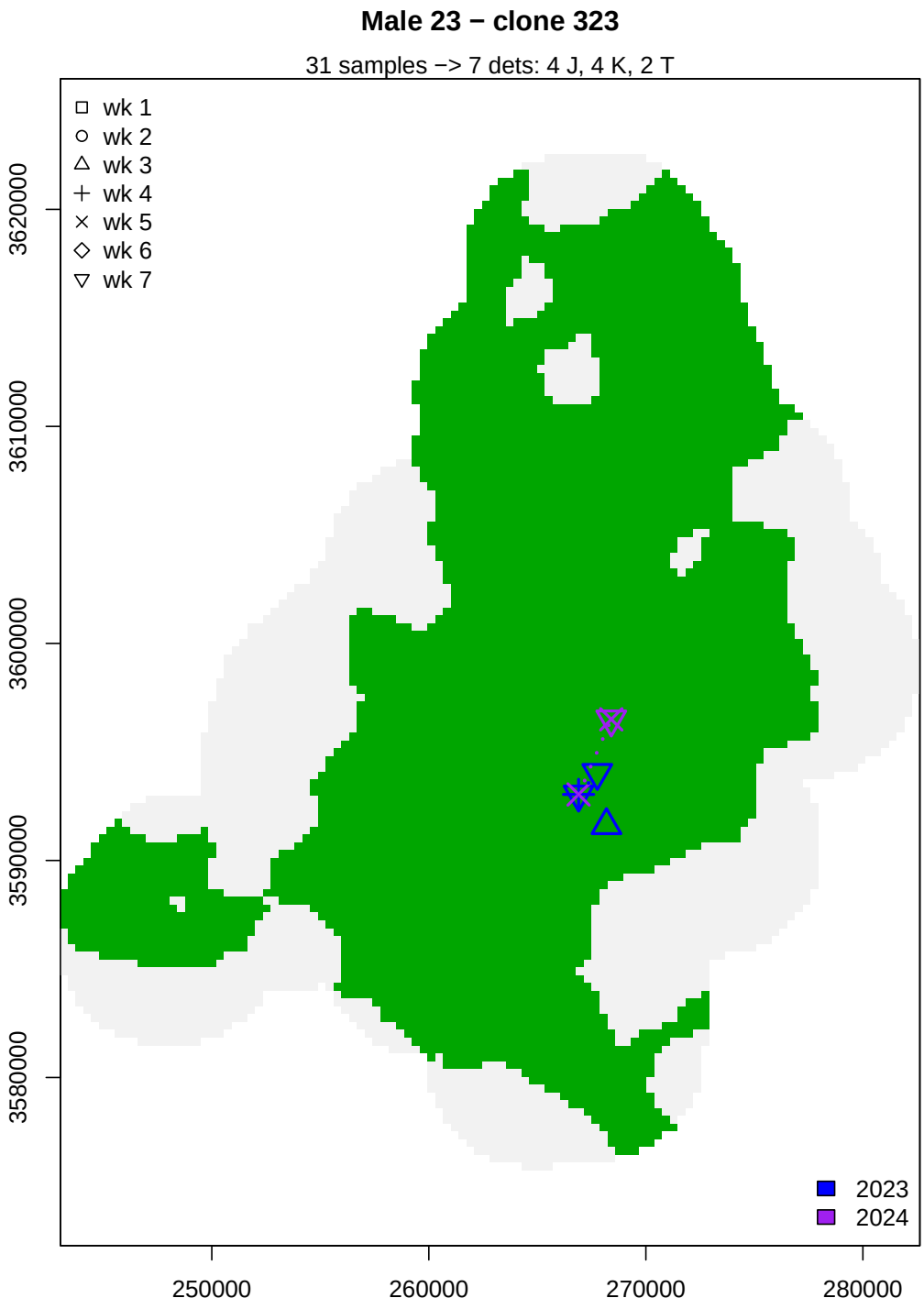

Male 13 − clone 307
67 samples −> 11 dets: 8 J, 4 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

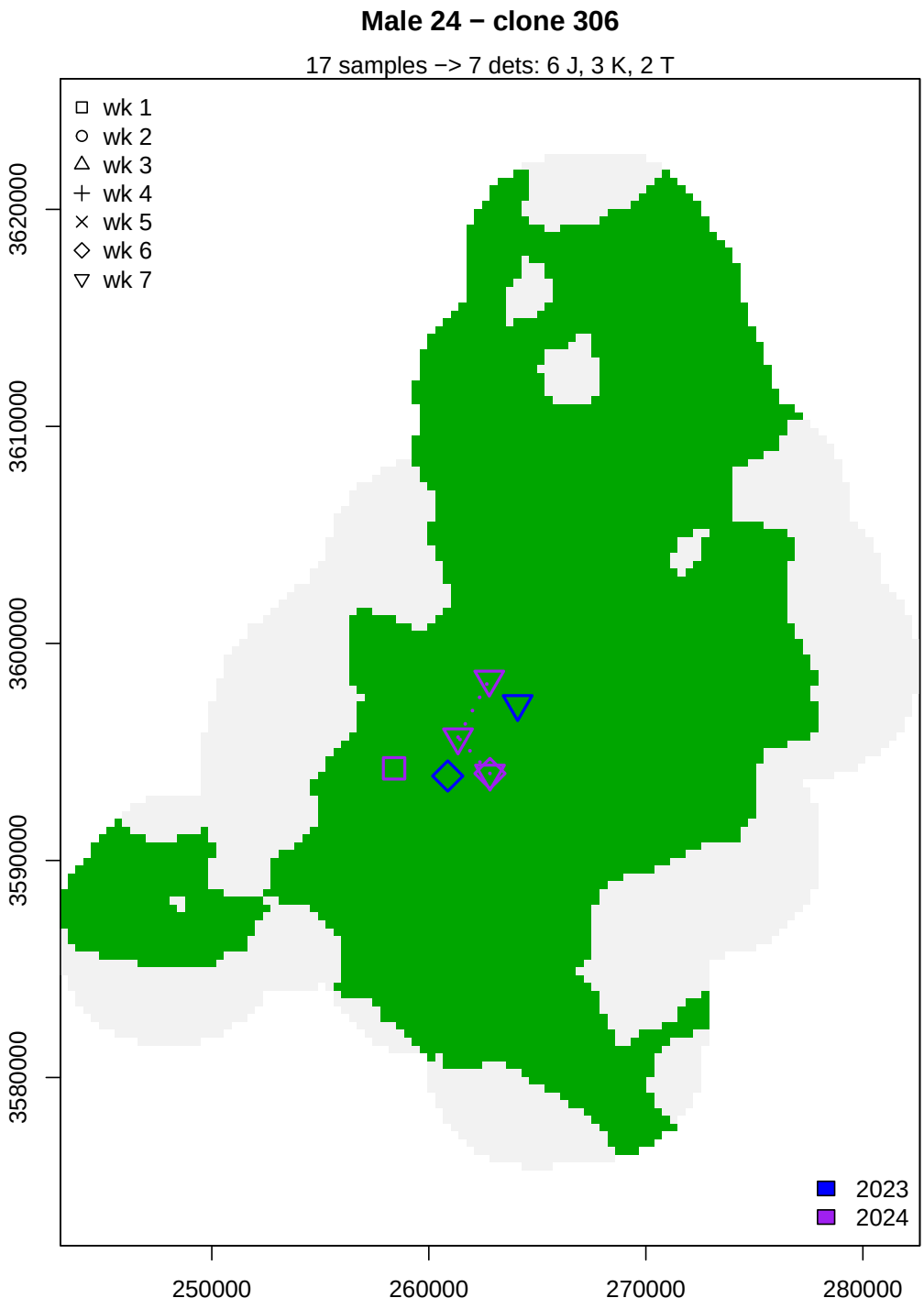

Male 15 − clone 324
94 samples −> 29 dets: 15 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

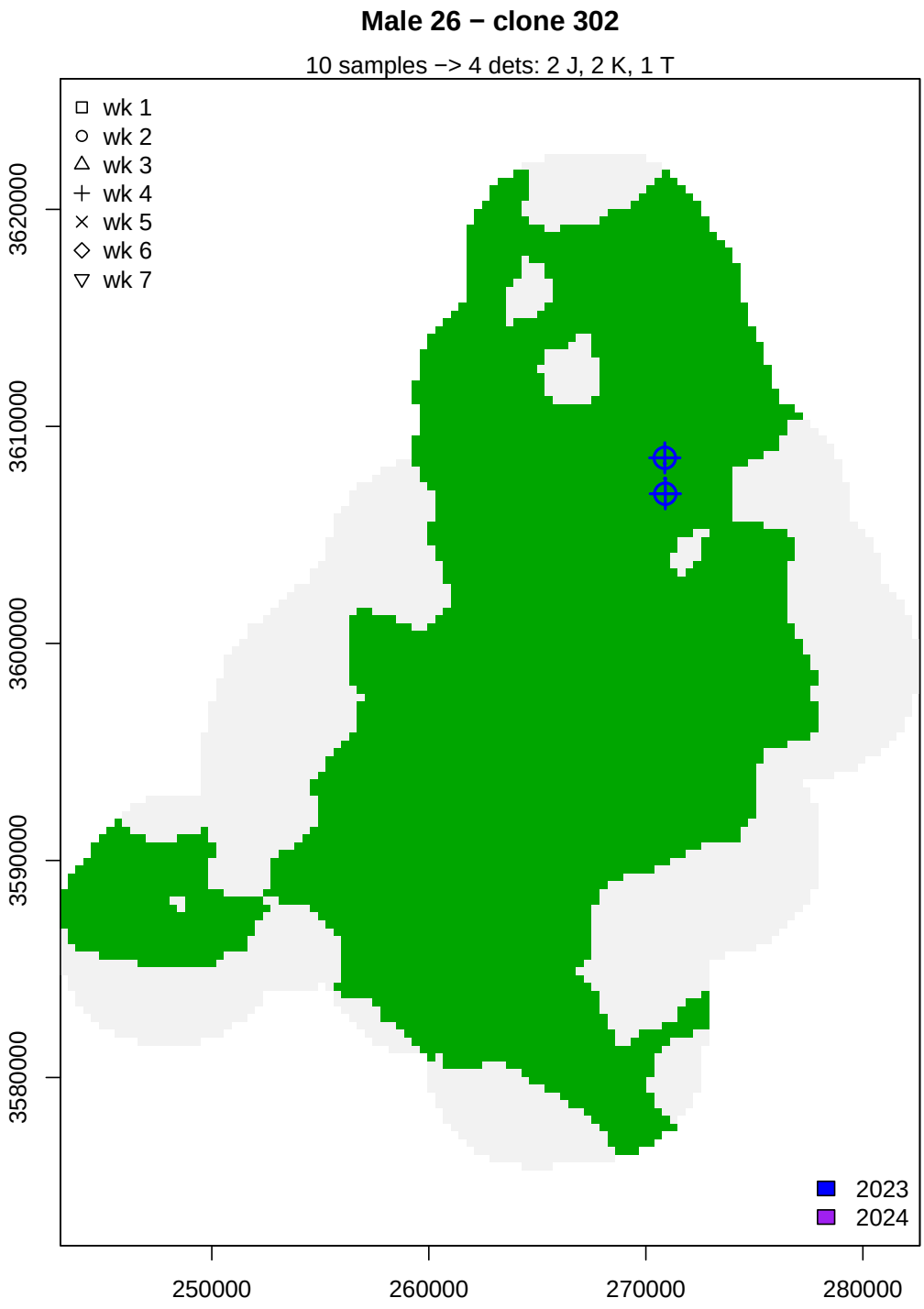

Male 17 − clone 53
5 samples −> 2 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

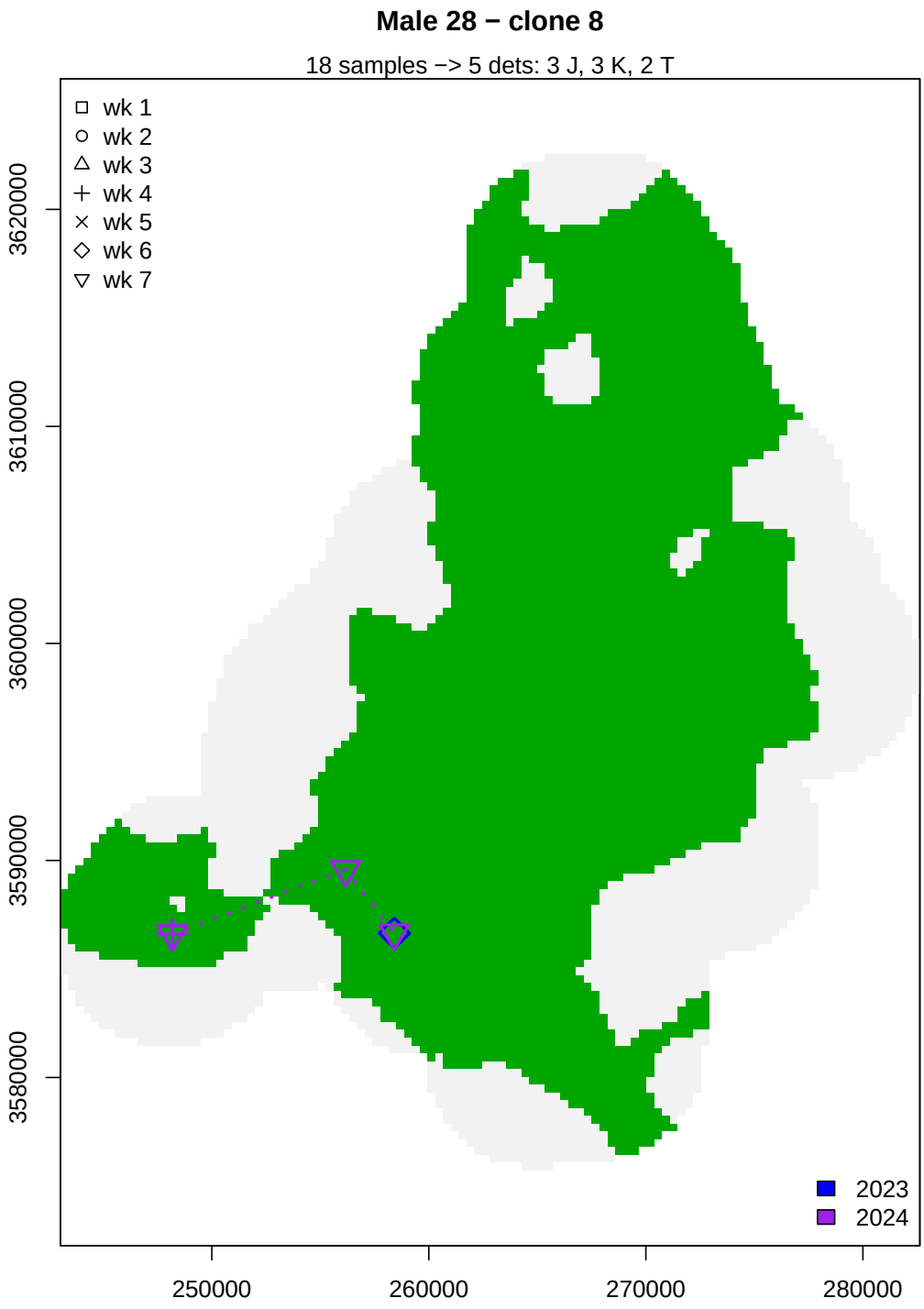

Male 19 − clone 335
14 samples −> 6 dets: 4 J, 4 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

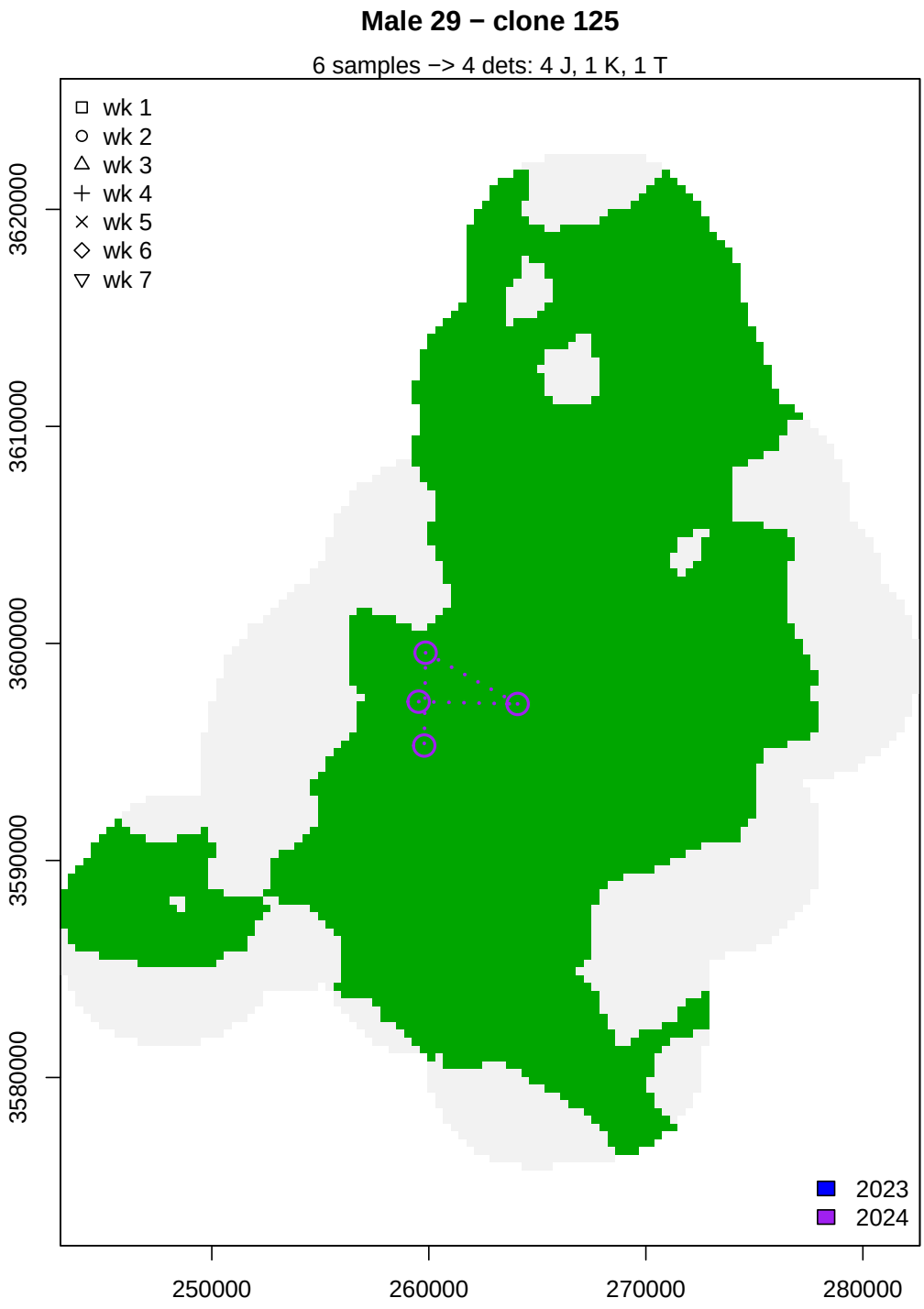

Male 22 − clone 295
10 samples −> 2 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

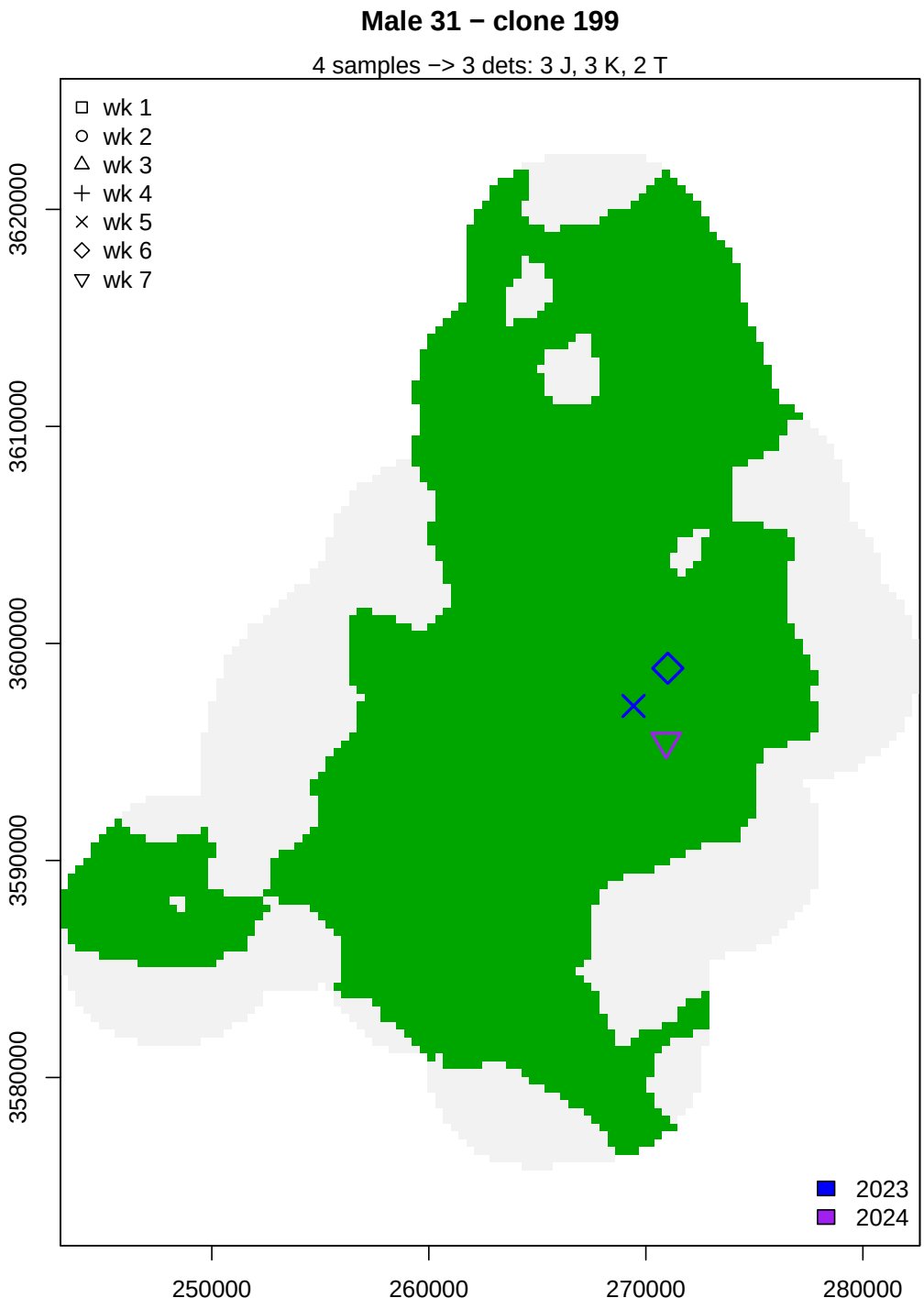

Male 23 − clone 323
31 samples −> 7 dets: 4 J, 4 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

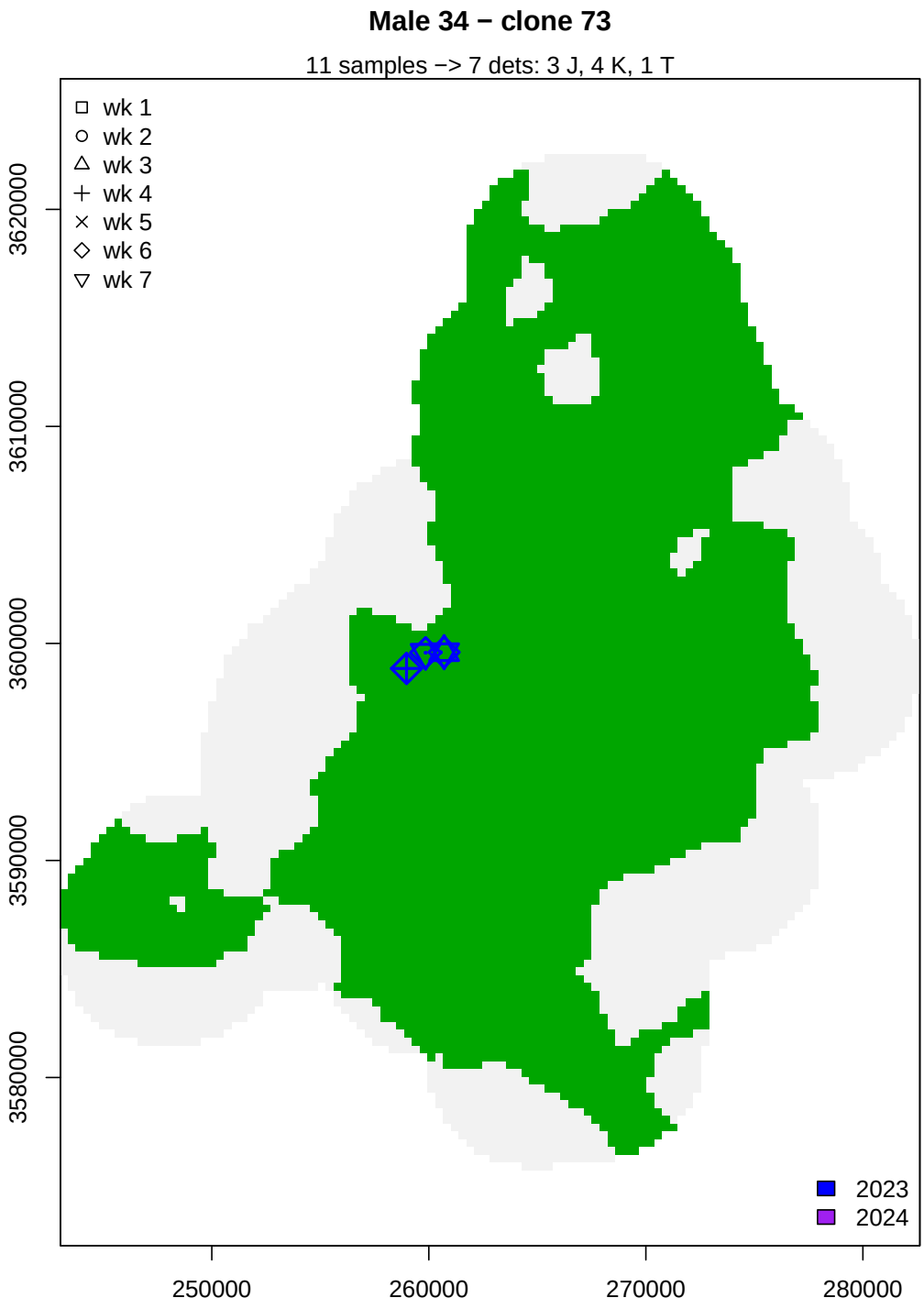

Male 24 − clone 306
17 samples −> 7 dets: 6 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

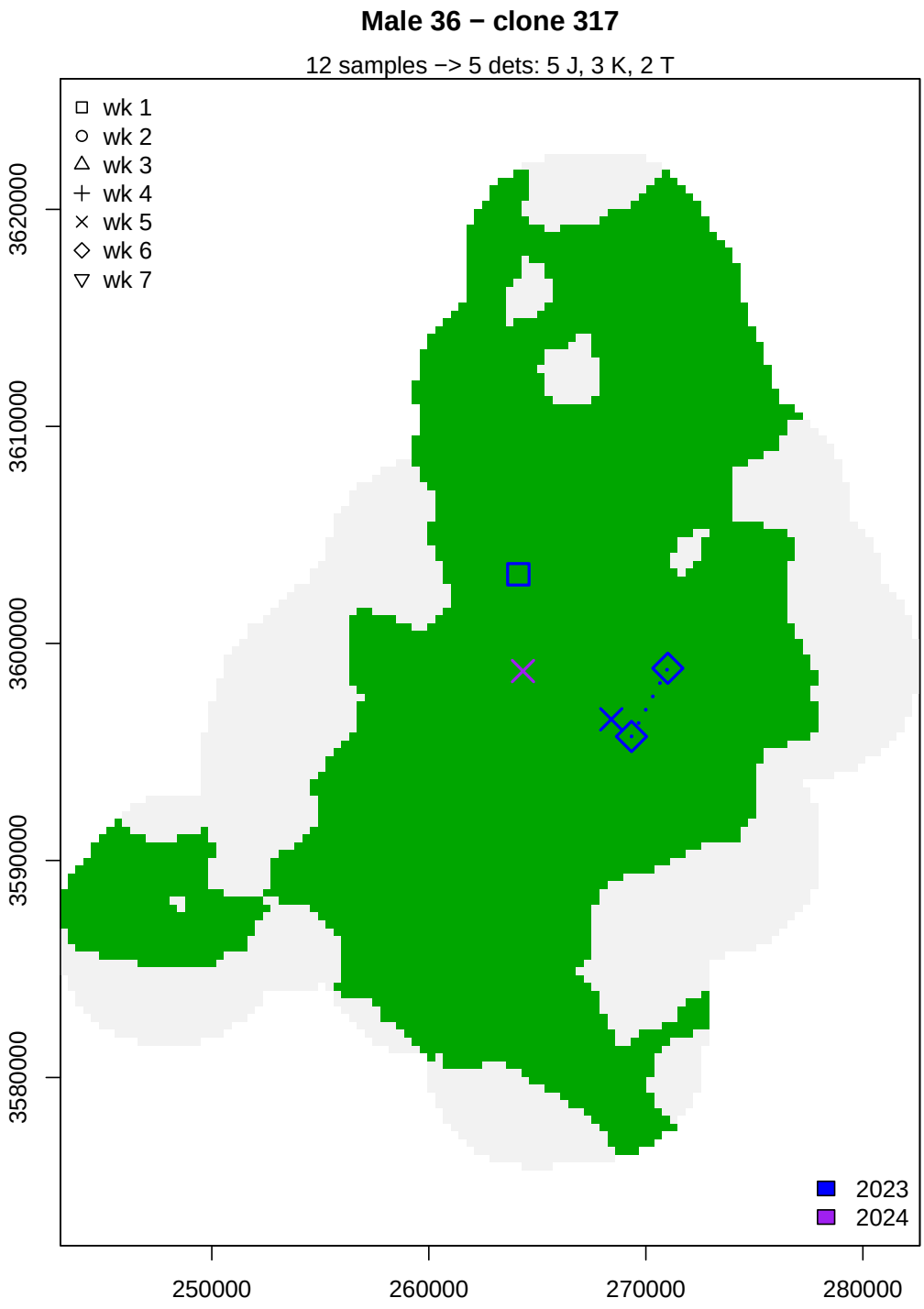

Male 26 − clone 302
10 samples −> 4 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

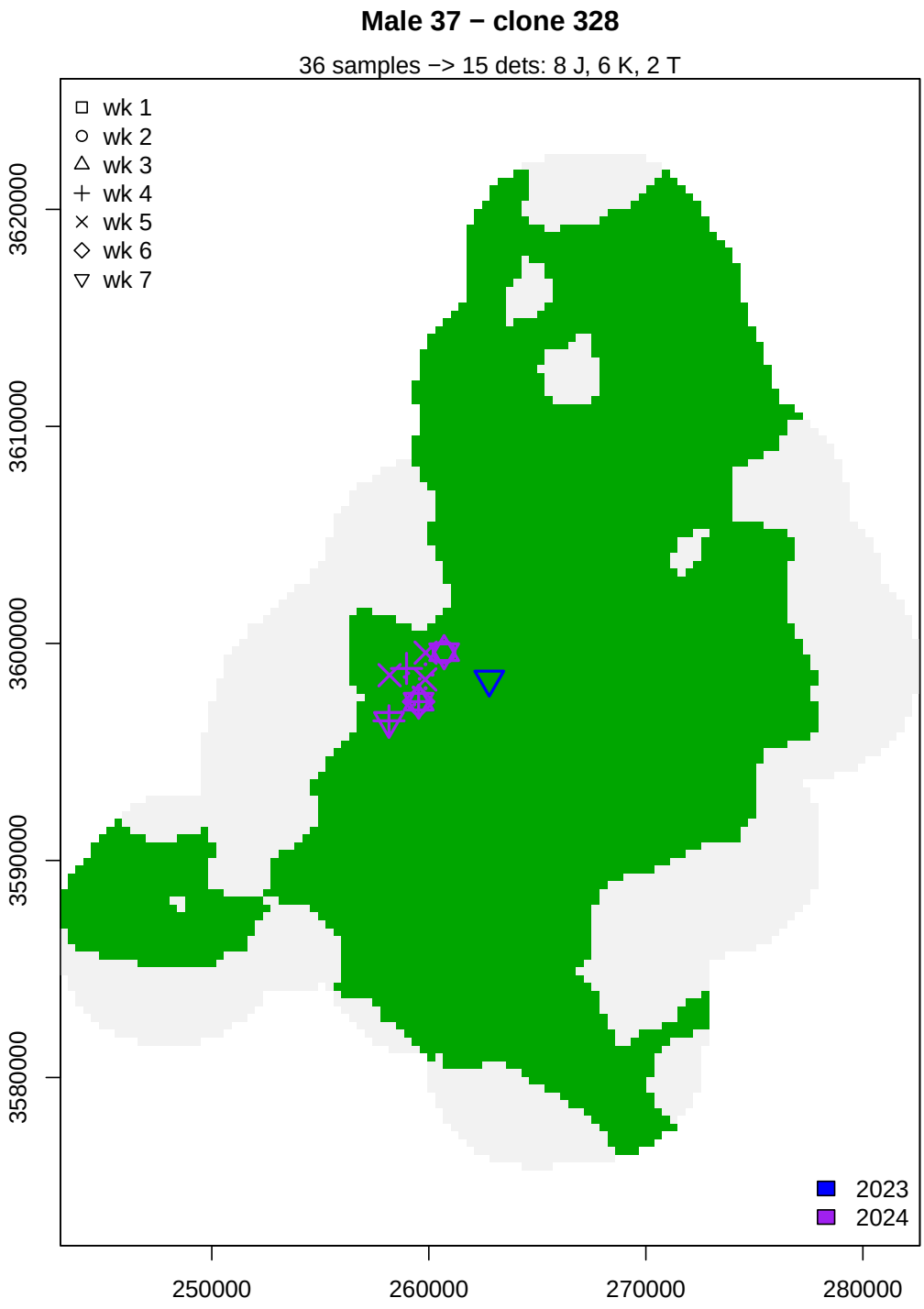

Male 28 − clone 8
18 samples −> 5 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

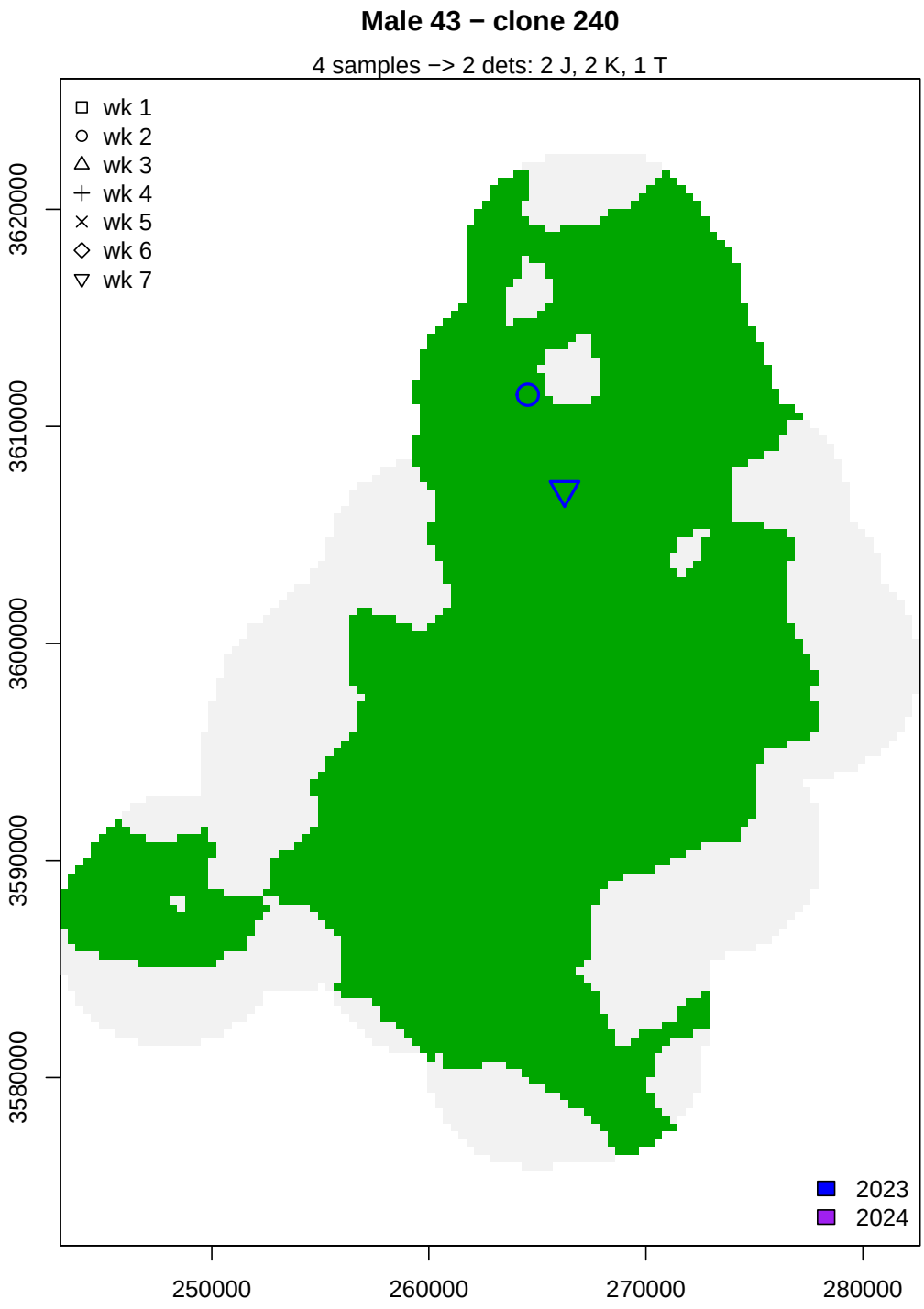

Male 29 − clone 125
6 samples −> 4 dets: 4 J, 1 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

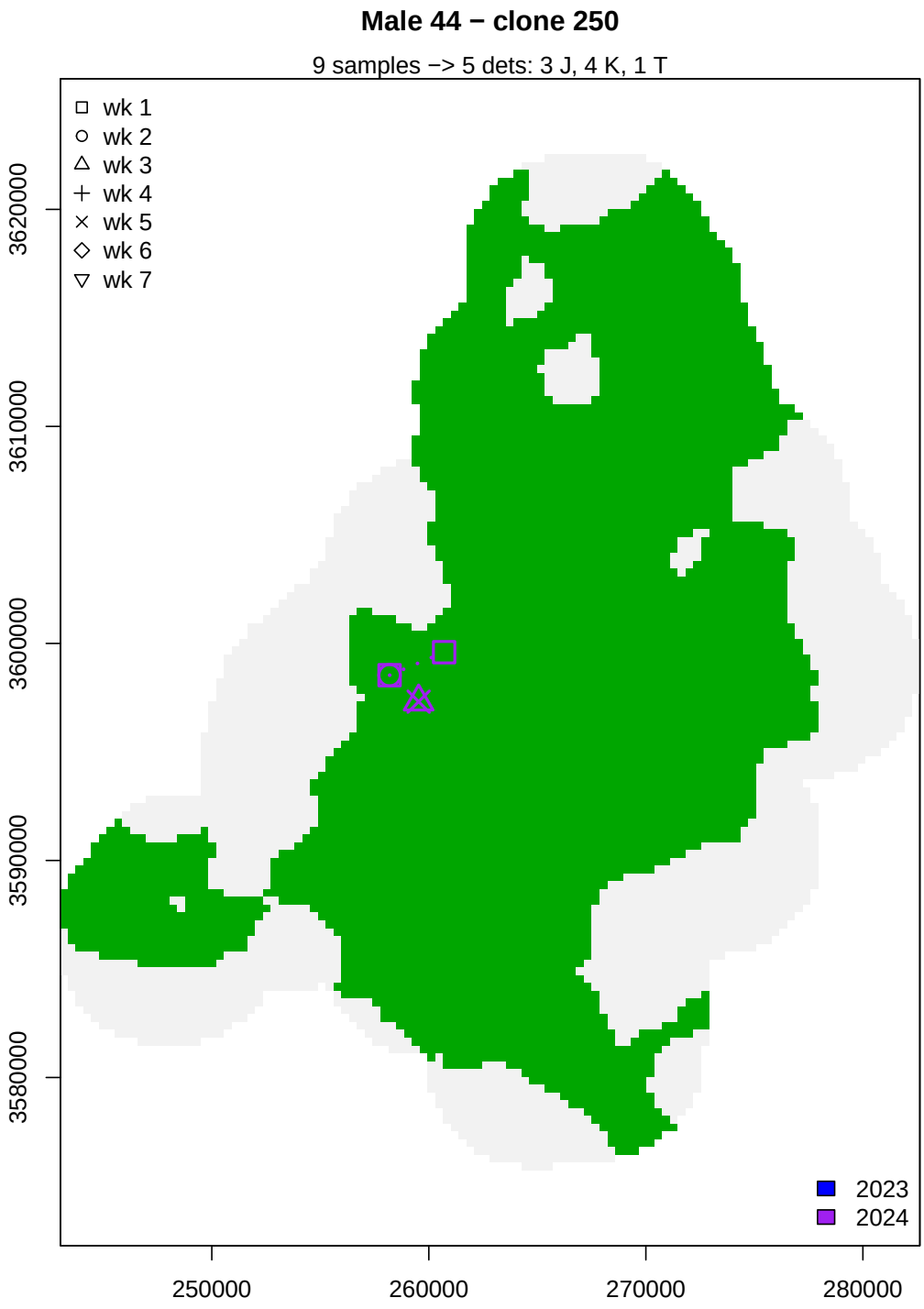

Male 31 − clone 199
4 samples −> 3 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

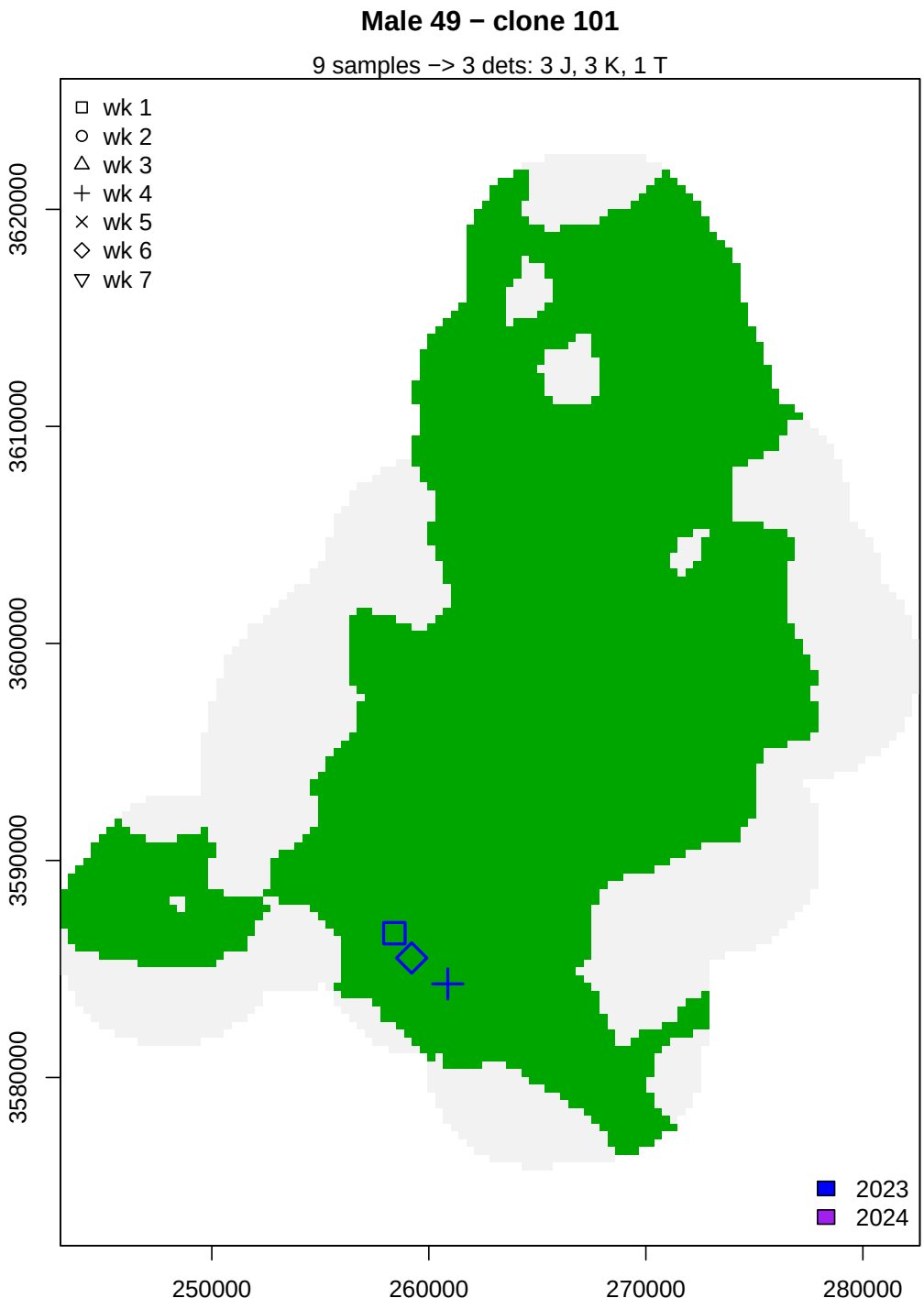

Male 34 − clone 73
11 samples −> 7 dets: 3 J, 4 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024

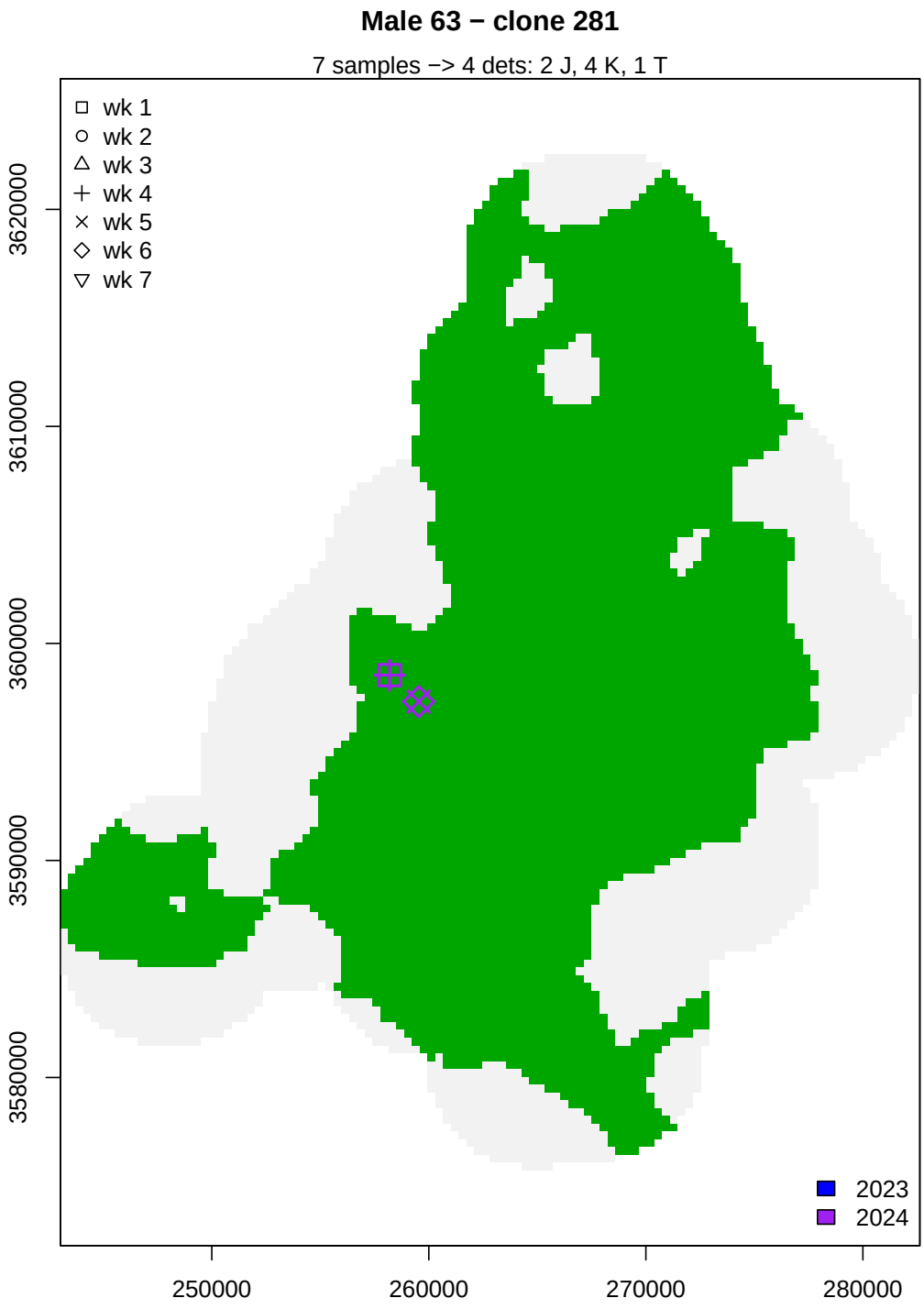

Male 36 − clone 317
12 samples −> 5 dets: 5 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

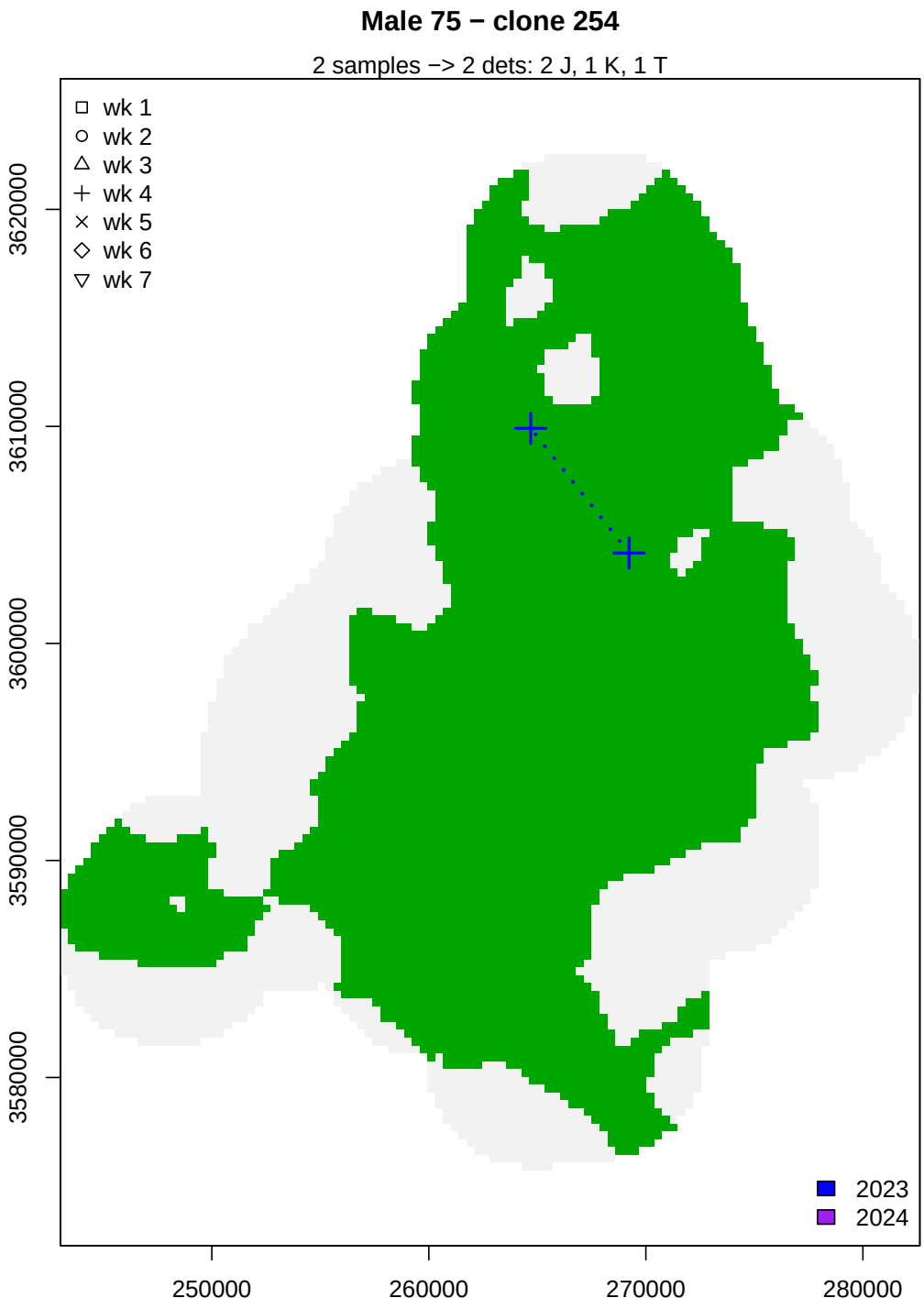

Male 37 − clone 328
36 samples −> 15 dets: 8 J, 6 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

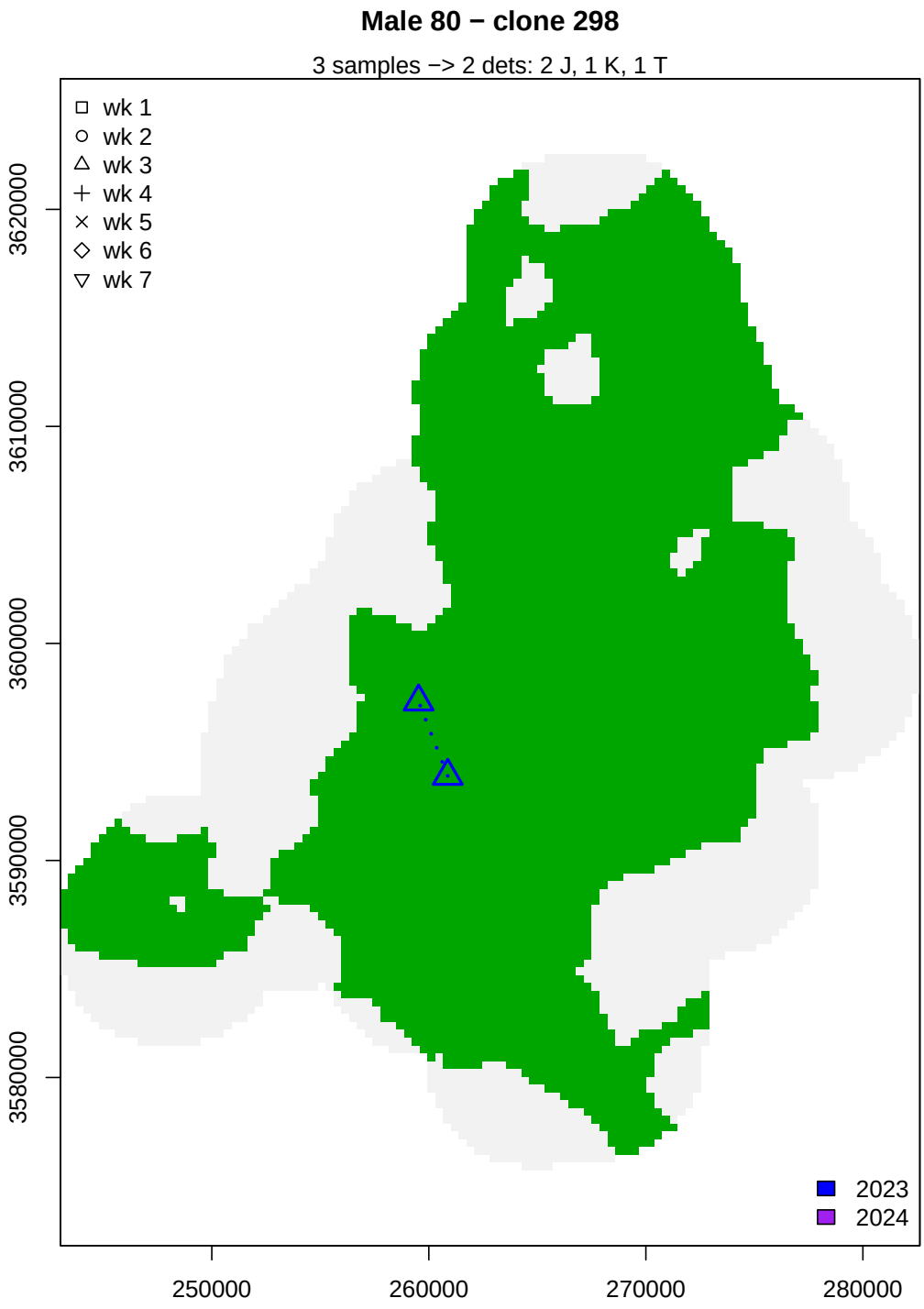

Male 43 − clone 240
4 samples −> 2 dets: 2 J, 2 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

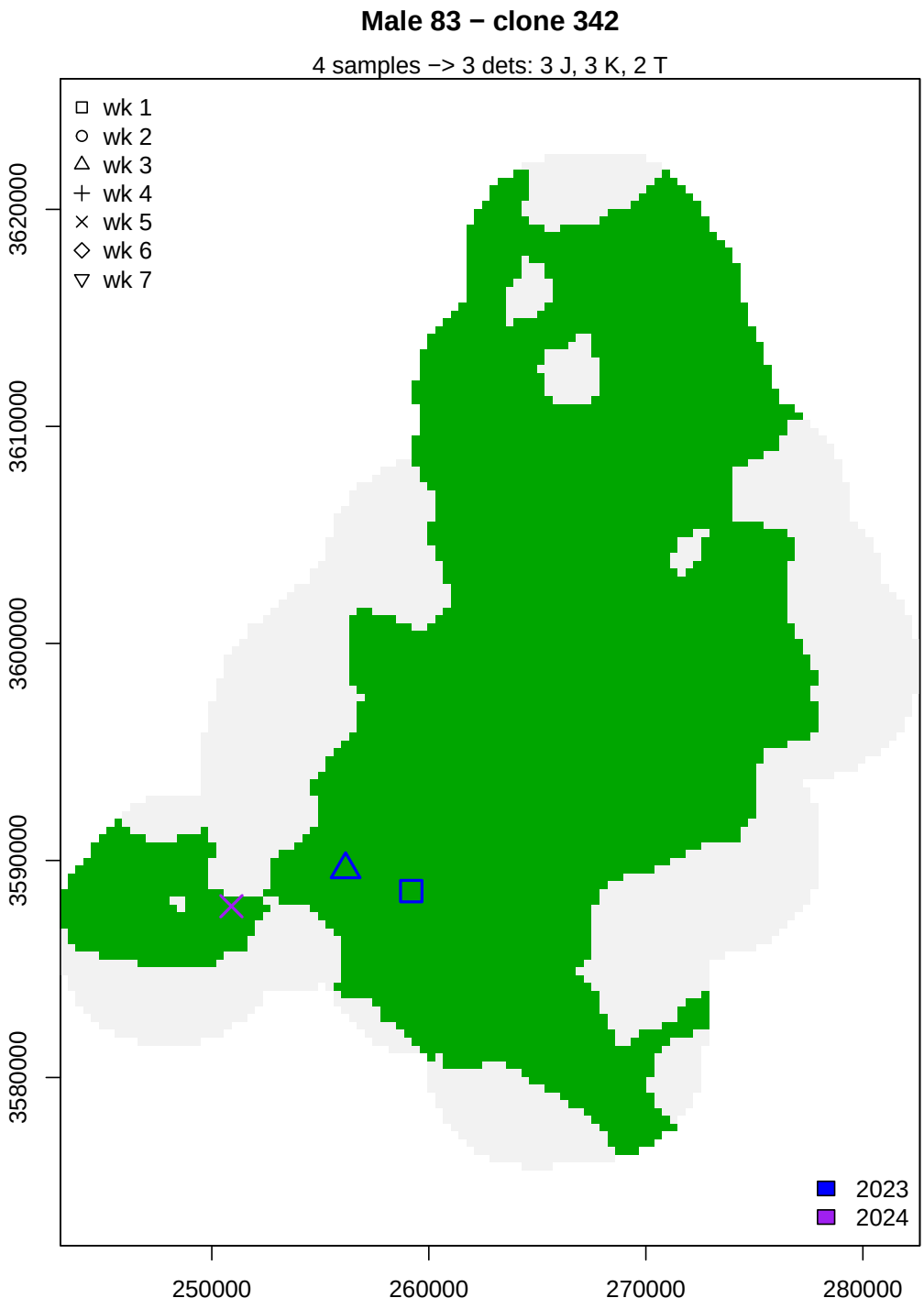

Male 44 − clone 250
9 samples −> 5 dets: 3 J, 4 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000

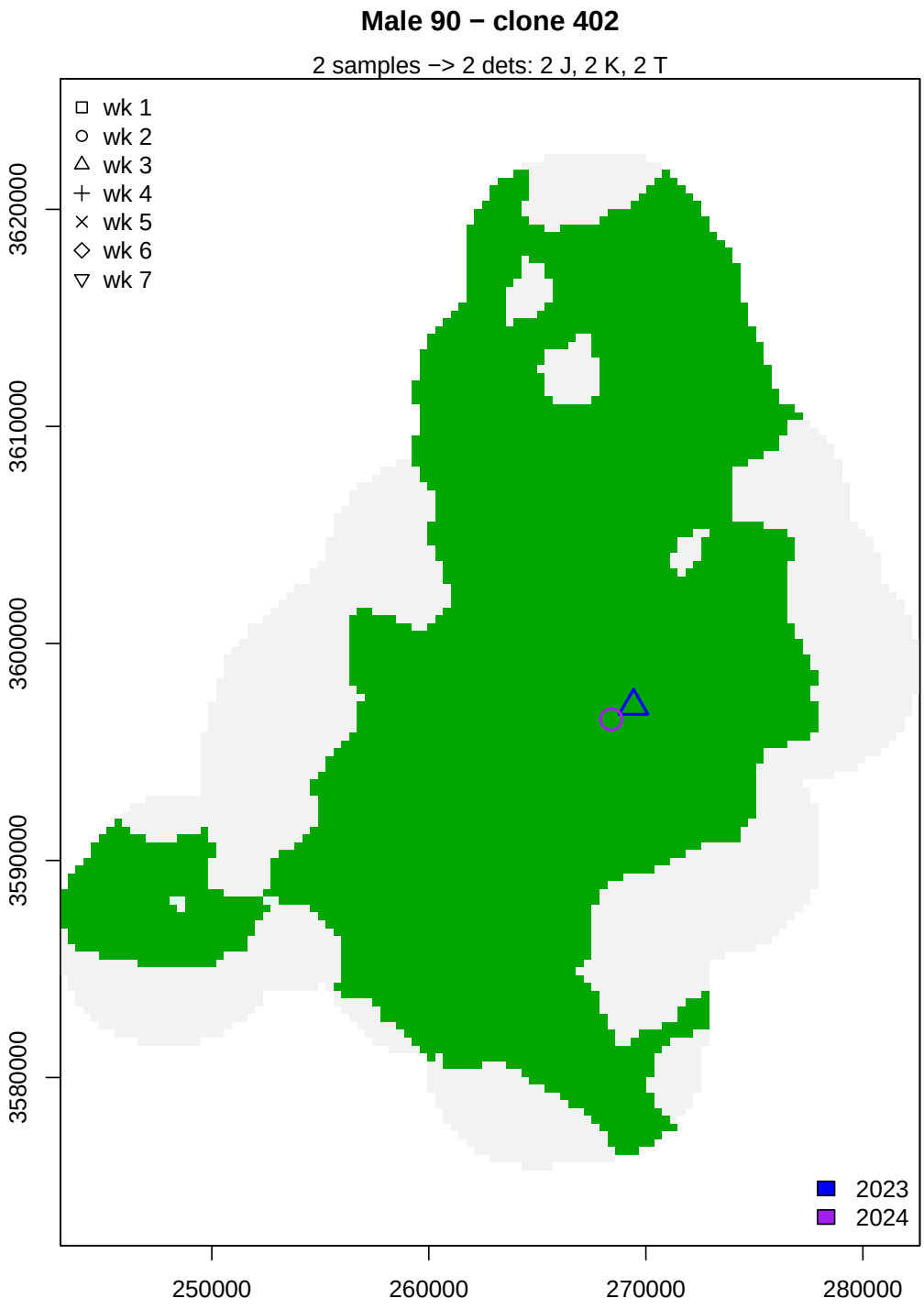

Male 49 − clone 101
9 samples −> 3 dets: 3 J, 3 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000


Male 63 − clone 281
7 samples −> 4 dets: 2 J, 4 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024


Male 75 − clone 254
2 samples −> 2 dets: 2 J, 1 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000
2023
2024


Male 80 − clone 298
3 samples −> 2 dets: 2 J, 1 K, 1 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000


Male 83 − clone 342
4 samples −> 3 dets: 3 J, 3 K, 2 T
wk 1
wk 2
wk 3
wk 4
wk 5
wk 6
wk 7
2023
2024
3620000
3610000
3600000
3590000
3580000
250000
260000
270000
280000



Overall, it seems we have little evidence for movement of activity centers between years. By far, the furthest movements recorded in our capture histories come in a single week!

Digging in…

Female clone 7

Looking more closely at the first female plotted (clone #7), which had a large within-week movement inferred…

```
#Recreating the object that was used to build the female capture history...

#Load sample metadata (smd)
smd <- read.csv("../sampleinfo/sampleMetaData_cloneID.csv", header = TRUE)

sampsALL <- smd[which(!is.na(smd$finalClone) &
                      smd$is.SCR == TRUE &
                      smd$year %in% c(2023,2024)),]

#Switch name of finalClone column for compatibility with makeCH function
colnames(sampsALL)[colnames(sampsALL) == "finalClone"] <- "cloneID"

Fsamps <- sampsALL[which(sampsALL$finalSEX == "F"),]
length(unique(Fsamps$cloneID[!is.na(Fsamps$cloneID)]))
```

```
## [1] 98
```

```
Fsamps[Fsamps$cloneID == 7,]
```

```
##                                    extID   ugaID is.SCR is.rep week year
## 40                   UGA_173101_P015_WD04  414-23   TRUE  FALSE    7 2023
## 43                   UGA_173101_P015_WD07  394-23   TRUE  FALSE    7 2023
## 102                  UGA_173101_P016_WA06  417-24   TRUE  FALSE    3 2024
## 113                  UGA_173101_P016_WB05  112-24   TRUE  FALSE    4 2024
## 136                  UGA_173101_P016_WD04   59-23   TRUE  FALSE    7 2023
## 179                  UGA_173101_P016_WG11  365-23   TRUE  FALSE    6 2023
## 250                  UGA_173101_P017_WE10  162-23   TRUE  FALSE    4 2023
## 265                  UGA_173101_P017_WG01  217-23   TRUE  FALSE    7 2023
## 273                  UGA_173101_P017_WG09  398-24   TRUE  FALSE    7 2024
## 282                  UGA_173101_P017_WH06   19-23   TRUE  FALSE    7 2023
## 326                  UGA_173101_P018_WD02  420-24   TRUE  FALSE    7 2024
## 349                  UGA_173101_P018_WF01   78-23   TRUE  FALSE    4 2023
## 355                  UGA_173101_P018_WF07  184-23   TRUE  FALSE    4 2023
## 369                  UGA_173101_P019_WA03   26-23   TRUE  FALSE    7 2023
## 393                  UGA_173101_P019_WC03  233-23   TRUE  FALSE    7 2023
## 484                  UGA_173101_P020_WB10   79-24   TRUE  FALSE    6 2024
## 537                  UGA_173101_P020_WG03  242-24   TRUE  FALSE    3 2024
## 539                  UGA_173101_P020_WG05  221-24   TRUE  FALSE    4 2024
## 547                  UGA_173101_P020_WH01  346-23   TRUE  FALSE    4 2023
## 573                  UGA_173101_P021_WB03  181-24   TRUE  FALSE    4 2024
## 635                  UGA_173101_P021_WG05  372-24   TRUE  FALSE    3 2024
## 718                  UGA_173101_P022_WF04  317-24   TRUE  FALSE    4 2024
## 721                  UGA_173101_P022_WF07  458-24   TRUE  FALSE    6 2024
## 748                  UGA_173101_P022_WH10  404-23   TRUE  FALSE    7 2023
## 755                  UGA_173101_P023_WA05  278-23   TRUE  FALSE    7 2023
## 758                  UGA_173101_P023_WA08  491-23   TRUE  FALSE    5 2023
```

```
## 770                    UGA_173101_P023_WB08  413-24   TRUE  FALSE    3 2024
## 771                    UGA_173101_P023_WB09  393-24   TRUE  FALSE    6 2024
## 785                    UGA_173101_P023_WC11  352-23   TRUE  FALSE    4 2023
## 788                    UGA_173101_P023_WD02  237-24   TRUE  FALSE    4 2024
## 813                    UGA_173101_P023_WF03  115-24   TRUE  FALSE    4 2024
## 878   GTseek_024_384i5_051_P098.002_767-23  767-23   TRUE  FALSE    7 2023
## 886   GTseek_024_384i5_059_P098.002_775-23  775-23   TRUE  FALSE    4 2023
## 918   GTseek_024_384i5_091_P098.002_785-23  785-23   TRUE  FALSE    2 2023
## 924   GTseek_024_384i5_097_P098.001_634-23  634-23   TRUE  FALSE    7 2023
## 981   GTseek_024_384i5_155_P098.002_806-23  806-23   TRUE  FALSE    7 2023
## 1012  GTseek_024_384i5_186_P098.004_648-24  648-24   TRUE  FALSE    7 2024
## 1085  GTseek_024_384i5_259_P098.001_695-23  695-23   TRUE  FALSE    7 2023
## 1109  GTseek_024_384i5_283_P098.002_867-23  867-23   TRUE  FALSE    4 2023
## 1110  GTseek_024_384i5_284_P098.004_683-24  683-24   TRUE  FALSE    4 2024
## 1152  GTseek_024_384i5_326_P098.003_537-24  537-24   TRUE  FALSE    3 2024
## 1186  GTseek_024_384i5_360_P098.003_554-24  554-24   TRUE  FALSE    4 2024
## 1237  GTseek_025_384i5_027_P098.006_996-24  996-24   TRUE  FALSE    6 2024
## 1439  GTseek_025_384i5_229_P098.005_871-24  871-24   TRUE  FALSE    4 2024
## 1525 GTseek_025_384i5_315_P098.006_3009-24 3009-24   TRUE  FALSE    7 2024
## 1543  GTseek_025_384i5_333_P098.005_954-24  954-24   TRUE  FALSE    4 2024
## 1559 GTseek_025_384i5_349_P098.006_3025-24 3025-24   TRUE  FALSE    7 2024
## 1638 GTseek_026_384i5_055_P098.010_2768-24 2768-24   TRUE  FALSE    4 2024
## 1649 GTseek_026_384i5_071_P098.009_1600-24 1600-24   TRUE  FALSE    4 2024
## 1713 GTseek_026_384i5_159_P098.010_2821-24 2821-24   TRUE  FALSE    4 2024
## 1776 GTseek_026_384i5_285_P098.010_2880-24 2880-24   TRUE  FALSE    4 2024
## 1856 GTseek_028_384i5_031_P098.UGA2_351-23  351-23   TRUE  FALSE    7 2023
## 1890  GTseek_028_384i5_065_P098.UGA1_79-23   79-23   TRUE  FALSE    6 2023
## 1949 GTseek_028_384i5_124_P098.UGA4_400-24  400-24   TRUE  FALSE    4 2024
## 2121 GTseek_028_384i5_296_P098.UGA3_276-24  276-24   TRUE  FALSE    6 2024
##       snare       utmSnare
## 40   HS 042 259530x3597317
## 43   HS 045 258191x3598537
## 102  HS 045 258191x3598537
## 113  HS 042 259530x3597317
## 136  HS 042 259530x3597317
## 179  HS 042 259530x3597317
## 250  HS 042 259530x3597317
## 265  HS 042 259530x3597317
## 273  HS 066 268064x3590555
## 282  HS 042 259530x3597317
## 326  HS 042 259530x3597317
## 349  HS 042 259530x3597317
## 355  HS 042 259530x3597317
## 369  HS 042 259530x3597317
## 393  HS 042 259530x3597317
## 484  HS 042 259530x3597317
## 537  HS 045 258191x3598537
## 539  HS 042 259530x3597317
## 547  HS 042 259530x3597317
## 573  HS 042 259530x3597317
## 635  HS 045 258191x3598537
## 718  HS 041 258115x3597270
## 721  HS 042 259530x3597317
## 748  HS 045 258191x3598537
```

```
## 755  HS 045 258191x3598537
## 758  HS 042 259530x3597317
## 770  HS 045 258191x3598537
## 771  HS 042 259530x3597317
## 785  HS 042 259530x3597317
## 788  HS 041 258115x3597270
## 813  HS 041 258115x3597270
## 878  HS 045 258191x3598537
## 886  HS 042 259530x3597317
## 918  HS 042 259530x3597317
## 924  HS 045 258191x3598537
## 981  HS 045 258191x3598537
## 1012 HS 045 258191x3598537
## 1085 HS 042 259530x3597317
## 1109 HS 042 259530x3597317
## 1110 HS 041 258115x3597270
## 1152 HS 045 258191x3598537
## 1186 HS 042 259530x3597317
## 1237 HS 042 259530x3597317
## 1439 HS 042 259530x3597317
## 1525 HS 042 259530x3597317
## 1543 HS 041 258115x3597270
## 1559 HS 042 259530x3597317
## 1638 HS 042 259530x3597317
## 1649 HS 041 258115x3597270
## 1713 HS 042 259530x3597317
## 1776 HS 042 259530x3597317
## 1856 HS 045 258191x3598537
## 1890 HS 042 259530x3597317
## 1949 HS 042 259530x3597317
## 2121 HS 042 259530x3597317
##                                                                                  fqfile
## 40    RAPiD-Genomics_F534_UGA_173101_P015_WD04_i5-502_i7-91_S14778_L003_R1_001.fastq
## 43    RAPiD-Genomics_F534_UGA_173101_P015_WD07_i5-502_i7-86_S14781_L003_R1_001.fastq
## 102  RAPiD-Genomics_F534_UGA_173101_P016_WA06_i5-504_i7-102_S14840_L003_R1_001.fastq
## 113  RAPiD-Genomics_F534_UGA_173101_P016_WB05_i5-504_i7-113_S14851_L003_R1_001.fastq
## 136  RAPiD-Genomics_F534_UGA_173101_P016_WD04_i5-504_i7-136_S14874_L003_R1_001.fastq
## 179  RAPiD-Genomics_F534_UGA_173101_P016_WG11_i5-504_i7-179_S14916_L003_R1_001.fastq
## 250  RAPiD-Genomics_F534_UGA_173101_P017_WE10_i5-512_i7-154_S14987_L003_R1_001.fastq
## 265  RAPiD-Genomics_F534_UGA_173101_P017_WG01_i5-512_i7-169_S15002_L003_R1_001.fastq
## 273  RAPiD-Genomics_F534_UGA_173101_P017_WG09_i5-512_i7-177_S15010_L003_R1_001.fastq
## 282  RAPiD-Genomics_F534_UGA_173101_P017_WH06_i5-512_i7-186_S15019_L003_R1_001.fastq
## 326  RAPiD-Genomics_F534_UGA_173101_P018_WD02_i5-537_i7-134_S15063_L003_R1_001.fastq
## 349  RAPiD-Genomics_F534_UGA_173101_P018_WF01_i5-537_i7-157_S15088_L003_R1_001.fastq
## 355  RAPiD-Genomics_F534_UGA_173101_P018_WF07_i5-537_i7-163_S15094_L003_R1_001.fastq
## 369  RAPiD-Genomics_F534_UGA_173101_P019_WA03_i5-1102_i7-82_S15108_L003_R1_001.fastq
## 393  RAPiD-Genomics_F534_UGA_173101_P019_WC03_i5-1102_i7-40_S15132_L003_R1_001.fastq
## 484    RAPiD-Genomics_F566_UGA_173101_P020_WB10_i5-505_i7-55_S3912_L001_R1_001.fastq
## 537    RAPiD-Genomics_F566_UGA_173101_P020_WG03_i5-505_i7-22_S3965_L001_R1_001.fastq
## 539    RAPiD-Genomics_F566_UGA_173101_P020_WG05_i5-505_i7-69_S3967_L001_R1_001.fastq
## 547    RAPiD-Genomics_F566_UGA_173101_P020_WH01_i5-505_i7-11_S3975_L001_R1_001.fastq
## 573       RAPiD-Genomics_F566_UGA_173101_P021_WB03_i5-501_i7-2_S15_L003_R1_001.fastq
## 635      RAPiD-Genomics_F566_UGA_173101_P021_WG05_i5-501_i7-69_S77_L003_R1_001.fastq
## 718   RAPiD-Genomics_F566_UGA_173101_P022_WF04_i5-1098_i7-45_S7646_L003_R1_001.fastq
```

```
## 721   RAPiD-Genomics_F566_UGA_173101_P022_WF07_i5-1098_i7-33_S7649_L003_R1_001.fastq
## 748   RAPiD-Genomics_F566_UGA_173101_P022_WH10_i5-1098_i7-34_S7676_L003_R1_001.fastq
## 755   RAPiD-Genomics_F566_UGA_173101_P023_WA05_i5-1099_i7-38_S7683_L003_R1_001.fastq
## 758   RAPiD-Genomics_F566_UGA_173101_P023_WA08_i5-1099_i7-16_S7686_L003_R1_001.fastq
## 770   RAPiD-Genomics_F566_UGA_173101_P023_WB08_i5-1099_i7-57_S7698_L003_R1_001.fastq
## 771   RAPiD-Genomics_F566_UGA_173101_P023_WB09_i5-1099_i7-63_S7699_L003_R1_001.fastq
## 785   RAPiD-Genomics_F566_UGA_173101_P023_WC11_i5-1099_i7-70_S7713_L003_R1_001.fastq
## 788   RAPiD-Genomics_F566_UGA_173101_P023_WD02_i5-1099_i7-72_S7716_L003_R1_001.fastq
## 813   RAPiD-Genomics_F566_UGA_173101_P023_WF03_i5-1099_i7-79_S7741_L003_R1_001.fastq
## 878                                                                            <NA>
## 886                                                                            <NA>
## 918                                                                            <NA>
## 924                                                                            <NA>
## 981                                                                            <NA>
## 1012                                                                           <NA>
## 1085                                                                           <NA>
## 1109                                                                           <NA>
## 1110                                                                           <NA>
## 1152                                                                           <NA>
## 1186                                                                           <NA>
## 1237                                                                           <NA>
## 1439                                                                           <NA>
## 1525                                                                           <NA>
## 1543                                                                           <NA>
## 1559                                                                           <NA>
## 1638                                                                           <NA>
## 1649                                                                           <NA>
## 1713                                                                           <NA>
## 1776                                                                           <NA>
## 1856                                                                           <NA>
## 1890                                                                           <NA>
## 1949                                                                           <NA>
## 2121                                                                           <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 40             Hair Snare    CGA   1063 UGA_173101_P015_WD04 FALSE       LGC
## 43             Hair Snare    CGA    308 UGA_173101_P015_WD07 FALSE       LGC
## 102            Hair Snare    CGA   1123 UGA_173101_P016_WA06 FALSE       LGC
## 113            Hair Snare    CGA   1239 UGA_173101_P016_WB05 FALSE       LGC
## 136            Hair Snare    CGA    857 UGA_173101_P016_WD04 FALSE       LGC
## 179            Hair Snare    CGA    783 UGA_173101_P016_WG11 FALSE       LGC
## 250            Hair Snare    CGA    755 UGA_173101_P017_WE10 FALSE       LGC
## 265            Hair Snare    CGA    651 UGA_173101_P017_WG01 FALSE       LGC
## 273            Hair Snare    CGA    462 UGA_173101_P017_WG09 FALSE       LGC
## 282            Hair Snare    CGA   1187 UGA_173101_P017_WH06 FALSE       LGC
## 326            Hair Snare    CGA   1222 UGA_173101_P018_WD02 FALSE       LGC
## 349            Hair Snare    CGA   1210 UGA_173101_P018_WF01 FALSE       LGC
## 355            Hair Snare    CGA   1178 UGA_173101_P018_WF07 FALSE       LGC
## 369            Hair Snare    CGA   1289 UGA_173101_P019_WA03 FALSE       LGC
## 393            Hair Snare    CGA   1283 UGA_173101_P019_WC03 FALSE       LGC
## 484            Hair Snare    CGA   1265 UGA_173101_P020_WB10 FALSE       LGC
## 537            Hair Snare    CGA   1246 UGA_173101_P020_WG03 FALSE       LGC
## 539            Hair Snare    CGA   1158 UGA_173101_P020_WG05 FALSE       LGC
## 547            Hair Snare    CGA   1301 UGA_173101_P020_WH01 FALSE       LGC
## 573            Hair Snare    CGA   1276 UGA_173101_P021_WB03 FALSE       LGC
```

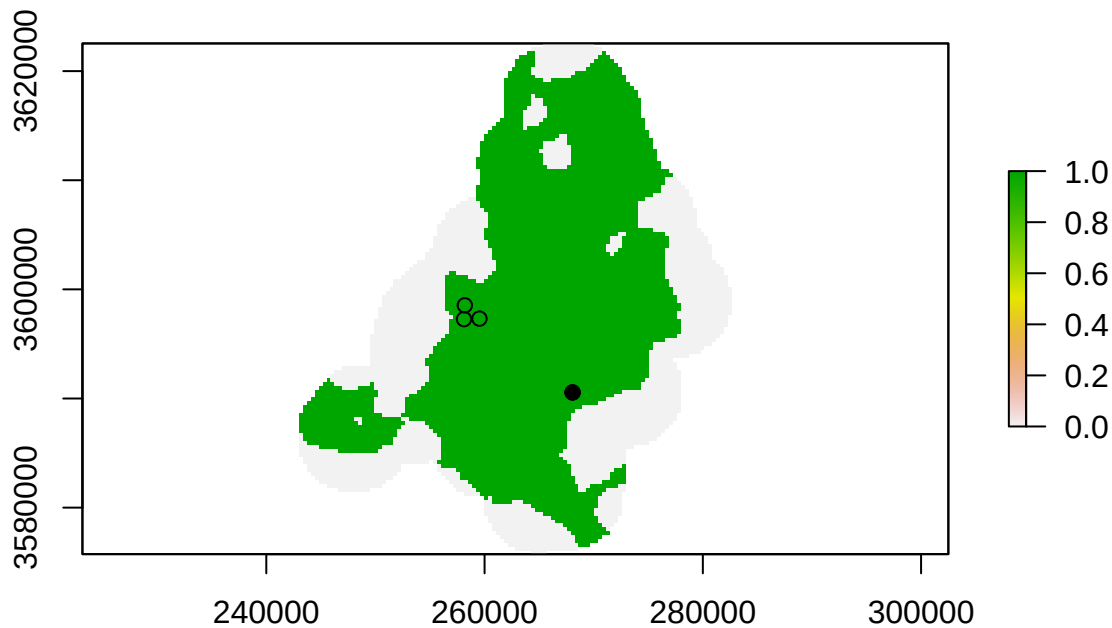
```
## 635            Hair Snare    CGA   1246 UGA_173101_P021_WG05 FALSE       LGC
## 718            Hair Snare    CGA   1227 UGA_173101_P022_WF04 FALSE       LGC
## 721            Hair Snare    CGA   1040 UGA_173101_P022_WF07 FALSE       LGC
## 748            Hair Snare    CGA   1293 UGA_173101_P022_WH10 FALSE       LGC
## 755            Hair Snare    CGA   1249 UGA_173101_P023_WA05 FALSE       LGC
## 758            Hair Snare    CGA   1255 UGA_173101_P023_WA08 FALSE       LGC
## 770            Hair Snare    CGA   1266 UGA_173101_P023_WB08 FALSE       LGC
## 771            Hair Snare    CGA   1256 UGA_173101_P023_WB09 FALSE       LGC
## 785            Hair Snare    CGA   1143 UGA_173101_P023_WC11 FALSE       LGC
## 788            Hair Snare    CGA   1054 UGA_173101_P023_WD02 FALSE       LGC
## 813            Hair Snare    CGA   1240 UGA_173101_P023_WF03 FALSE       LGC
## 878           Hair Sample    CGA    123           GTS_767-23 FALSE    GTSeek
## 886           Hair Sample    CGA    123           GTS_775-23 FALSE    GTSeek
## 918           Hair Sample    CGA    123           GTS_785-23 FALSE    GTSeek
## 924           Hair Sample    CGA    115           GTS_634-23 FALSE    GTSeek
## 981           Hair Sample    CGA    123           GTS_806-23 FALSE    GTSeek
## 1012          Hair Sample    CGA    120           GTS_648-24 FALSE    GTSeek
## 1085          Hair Sample    CGA    117           GTS_695-23 FALSE    GTSeek
## 1109          Hair Sample    CGA    124           GTS_867-23 FALSE    GTSeek
## 1110          Hair Sample    CGA    123           GTS_683-24 FALSE    GTSeek
## 1152          Hair Sample    CGA    123           GTS_537-24 FALSE    GTSeek
## 1186          Hair Sample    CGA    122           GTS_554-24 FALSE    GTSeek
## 1237          Hair Sample    CGA    125           GTS_996-24 FALSE    GTSeek
## 1439          Hair Sample    CGA    125           GTS_871-24 FALSE    GTSeek
## 1525          Hair Sample    CGA    125          GTS_3009-24 FALSE    GTSeek
## 1543          Hair Sample    CGA    124           GTS_954-24 FALSE    GTSeek
## 1559          Hair Sample    CGA    126          GTS_3025-24 FALSE    GTSeek
## 1638          Hair Sample    CGA    123          GTS_2768-24 FALSE    GTSeek
## 1649          Hair Sample    CGA    124          GTS_1600-24 FALSE    GTSeek
## 1713          Hair Sample    CGA    124          GTS_2821-24 FALSE    GTSeek
## 1776          Hair Sample    CGA    125          GTS_2880-24 FALSE    GTSeek
## 1856          Hair Sample    CGA    123           GTS_351-23 FALSE    GTSeek
## 1890          Hair Sample    CGA    123            GTS_79-23 FALSE    GTSeek
## 1949          Hair Sample    CGA    125           GTS_400-24 FALSE    GTSeek
## 2121          Hair Sample    CGA    124           GTS_276-24 FALSE    GTSeek
##      sexGT COLclone cloneID finalSEX
## 40       F        7       7        F
## 43       F        7       7        F
## 102      F        7       7        F
## 113      F        7       7        F
## 136      F        7       7        F
## 179      F        7       7        F
## 250      F        7       7        F
## 265      F        7       7        F
## 273      F        7       7        F
## 282      F        7       7        F
## 326      F        7       7        F
## 349      F        7       7        F
## 355      F        7       7        F
## 369      F        7       7        F
## 393      F        7       7        F
## 484      F        7       7        F
## 537      F        7       7        F
## 539      F        7       7        F
```

```
## 547      F        7       7        F
## 573      F        7       7        F
## 635      F        7       7        F
## 718      F        7       7        F
## 721  undet        7       7        F
## 748      F        7       7        F
## 755      F        7       7        F
## 758      F        7       7        F
## 770      F        7       7        F
## 771      F        7       7        F
## 785      F        7       7        F
## 788      F        7       7        F
## 813      F        7       7        F
## 878      F      309       7        F
## 886      F      311       7        F
## 918      F      311       7        F
## 924      F      309       7        F
## 981      F      311       7        F
## 1012     F      311       7        F
## 1085     F      309       7        F
## 1109     F      311       7        F
## 1110     F      311       7        F
## 1152     F      311       7        F
## 1186     F      311       7        F
## 1237     F      309       7        F
## 1439     F      311       7        F
## 1525     F      311       7        F
## 1543     F      309       7        F
## 1559     F      311       7        F
## 1638     F      309       7        F
## 1649     F      309       7        F
## 1713     F      309       7        F
## 1776     F      309       7        F
## 1856     F      311       7        F
## 1890     F      311       7        F
## 1949     F      311       7        F
## 2121     F      311       7        F
```

```r
table(Fsamps$snare[which(Fsamps$cloneID == 7)])
```

```
##
## HS 041 HS 042 HS 045 HS 066
##      6     35     13      1
```

```r
plot(S360)
points(locs[rownames(locs) %in% Fsamps$utmSnare[Fsamps$cloneID == 7],])
points(locs[rownames(locs) %in% Fsamps$utmSnare[which(Fsamps$cloneID == 7 & Fsamps$snare == "HS 066")],
```




```
Fsamps[Fsamps$cloneID == 7 & Fsamps$snare == "HS 066",]
```

```
##                    extID  ugaID is.SCR is.rep week year  snare       utmSnare
## 273 UGA_173101_P017_WG09 398-24   TRUE  FALSE    7 2024 HS 066 268064x3590555
##                                                                        fqfile
## 273 RAPiD-Genomics_F534_UGA_173101_P017_WG09_i5-512_i7-177_S15010_L003_R1_001.fastq
##     knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 273          Hair Snare    CGA    462 UGA_173101_P017_WG09 FALSE       LGC
##     sexGT COLclone cloneID finalSEX
## 273     F        7       7        F
```

So the recapture inference in this case is from `COLONY`, not `PLINK` only, but the sample that is distant from the rest of the set did have fewer SNPs genotyped. Still well enough to make a reliable inference here, especially given the number of samples identified as part of this clone. It seems unlikely that this is a false-positive recapture.

Female clone 45

Clone #45 (the 9th female plotted) also had a large within-week movement (week 4 of 2023)...

```
#Load sample metadata (smd)
smd <- read.csv("../sampleinfo/sampleMetaData_cloneID.csv", header = TRUE)

sampsALL <- smd[which(!is.na(smd$finalClone) &
```

```r
                        smd$is.SCR == TRUE &
                        smd$year %in% c(2023,2024)),]

#Switch name of finalClone column for compatibility with makeCH function
colnames(sampsALL)[colnames(sampsALL) == "finalClone"] <- "cloneID"

Fsamps <- sampsALL[which(sampsALL$finalSEX == "F"),]
length(unique(Fsamps$cloneID[!is.na(Fsamps$cloneID)]))
```

```
## [1] 98
```

```r
Fsamps[Fsamps$cloneID == 45,]
```

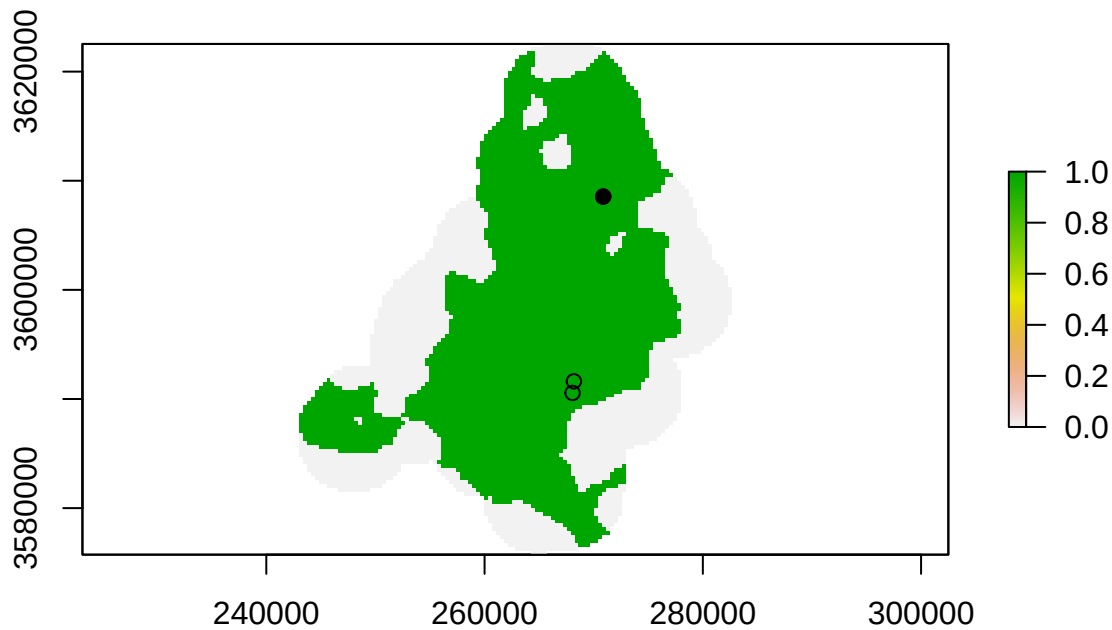
```
##                                        extID  ugaID is.SCR is.rep week year
## 107                   UGA_173101_P016_WA11 372-23   TRUE  FALSE    3 2023
## 168                   UGA_173101_P016_WF12 370-23   TRUE  FALSE    2 2023
## 196                   UGA_173101_P017_WA04 339-23   TRUE  FALSE    2 2023
## 222                   UGA_173101_P017_WC06  21-23   TRUE   TRUE    2 2023
## 337                   UGA_173101_P018_WE01 103-23   TRUE  FALSE    6 2023
## 359                   UGA_173101_P018_WF11 476-23   TRUE  FALSE    6 2023
## 404                   UGA_173101_P019_WD02 299-23   TRUE  FALSE    4 2023
## 753                   UGA_173101_P023_WA03  21-23   TRUE   TRUE    2 2023
## 798                   UGA_173101_P023_WD12 197-23   TRUE  FALSE    6 2023
## 996   GTseek_024_384i5_170_P098.003_996-23 996-23   TRUE  FALSE    2 2023
## 1031  GTseek_024_384i5_205_P098.001_678-23 678-23   TRUE  FALSE    2 2023
## 1165  GTseek_024_384i5_339_P098.002_895-23 895-23   TRUE  FALSE    6 2023
## 1342  GTseek_025_384i5_132_P098.007_886-23 886-23   TRUE  FALSE    6 2023
## 1404  GTseek_025_384i5_194_P098.007_987-23 987-23   TRUE  FALSE    6 2023
## 1924 GTseek_028_384i5_099_P098.UGA1_107-23 107-23   TRUE  FALSE    4 2023
## 1958 GTseek_028_384i5_133_P098.UGA1_143-23 143-23   TRUE  FALSE    3 2023
## 2014 GTseek_028_384i5_189_P098.UGA2_459-23 459-23   TRUE  FALSE    2 2023
##       snare       utmSnare
## 107  HS 068 268186x3591620
## 168  HS 068 268186x3591620
## 196  HS 068 268186x3591620
## 222  HS 068 268186x3591620
## 337  HS 068 268186x3591620
## 359  HS 066 268064x3590555
## 404  HS 068 268186x3591620
## 753  HS 068 268186x3591620
## 798  HS 066 268064x3590555
## 996  HS 068 268186x3591620
## 1031 HS 068 268186x3591620
## 1165 HS 066 268064x3590555
## 1342 HS 066 268064x3590555
## 1404 HS 066 268064x3590555
## 1924 HS 126 270872x3608550
## 1958 HS 068 268186x3591620
## 2014 HS 068 268186x3591620
##                                                                      fqfile
## 107  RAPiD-Genomics_F534_UGA_173101_P016_WA11_i5-504_i7-107_S14845_L003_R1_001.fastq
## 168  RAPiD-Genomics_F534_UGA_173101_P016_WF12_i5-504_i7-168_S14905_L003_R1_001.fastq
```

```
## 196  RAPiD-Genomics_F534_UGA_173101_P017_WA04_i5-512_i7-100_S14933_L003_R1_001.fastq
## 222  RAPiD-Genomics_F534_UGA_173101_P017_WC06_i5-512_i7-126_S14959_L003_R1_001.fastq
## 337  RAPiD-Genomics_F534_UGA_173101_P018_WE01_i5-537_i7-145_S15074_L003_R1_001.fastq
## 359  RAPiD-Genomics_F534_UGA_173101_P018_WF11_i5-537_i7-167_S15098_L003_R1_001.fastq
## 404  RAPiD-Genomics_F534_UGA_173101_P019_WD02_i5-1102_i7-72_S15143_L003_R1_001.fastq
## 753   RAPiD-Genomics_F566_UGA_173101_P023_WA03_i5-1099_i7-82_S7681_L003_R1_001.fastq
## 798    RAPiD-Genomics_F566_UGA_173101_P023_WD12_i5-1099_i7-9_S7726_L003_R1_001.fastq
## 996                                                                            <NA>
## 1031                                                                           <NA>
## 1165                                                                           <NA>
## 1342                                                                           <NA>
## 1404                                                                           <NA>
## 1924                                                                           <NA>
## 1958                                                                           <NA>
## 2014                                                                           <NA>
##      knownSEX  sampSOURCE region nSNPgt               vcfID  FILT GTservice
## 107           Hair Snare    CGA   1286 UGA_173101_P016_WA11 FALSE       LGC
## 168           Hair Snare    CGA   1161 UGA_173101_P016_WF12 FALSE       LGC
## 196           Hair Snare    CGA   1288 UGA_173101_P017_WA04 FALSE       LGC
## 222           Hair Snare    CGA    875 UGA_173101_P017_WC06 FALSE       LGC
## 337           Hair Snare    CGA   1283 UGA_173101_P018_WE01 FALSE       LGC
## 359           Hair Snare    CGA   1270 UGA_173101_P018_WF11 FALSE       LGC
## 404           Hair Snare    CGA    825 UGA_173101_P019_WD02 FALSE       LGC
## 753           Hair Snare    CGA    928 UGA_173101_P023_WA03 FALSE       LGC
## 798           Hair Snare    CGA    649 UGA_173101_P023_WD12 FALSE       LGC
## 996          Hair Sample    CGA    121           GTS_996-23 FALSE    GTSeek
## 1031         Hair Sample    CGA    120           GTS_678-23 FALSE    GTSeek
## 1165         Hair Sample    CGA    121           GTS_895-23 FALSE    GTSeek
## 1342         Hair Sample    CGA    121           GTS_886-23 FALSE    GTSeek
## 1404         Hair Sample    CGA    122           GTS_987-23 FALSE    GTSeek
## 1924         Hair Sample    CGA    123           GTS_107-23 FALSE    GTSeek
## 1958         Hair Sample    CGA    123           GTS_143-23 FALSE    GTSeek
## 2014         Hair Sample    CGA    123           GTS_459-23 FALSE    GTSeek
##      sexGT COLclone cloneID finalSEX
## 107      F       45      45        F
## 168      F       45      45        F
## 196      F       45      45        F
## 222      F       45      45        F
## 337      F       45      45        F
## 359      F       45      45        F
## 404      F       45      45        F
## 753      F       45      45        F
## 798      F       45      45        F
## 996      F      332      45        F
## 1031     F      332      45        F
## 1165     F      332      45        F
## 1342     F      332      45        F
## 1404     F      332      45        F
## 1924     F      332      45        F
## 1958     F      332      45        F
## 2014     F      332      45        F
```

```
table(Fsamps$snare[which(Fsamps$cloneID == 45)])
```

```
##
## HS 066 HS 068 HS 126
##      5     11      1
```

```r
plot(S360)
points(locs[rownames(locs) %in% Fsamps$utmSnare[Fsamps$cloneID == 45],])
points(locs[rownames(locs) %in%
              Fsamps$utmSnare[which(Fsamps$cloneID == 45 &
                                    Fsamps$snare == "HS 126")],], pch = 19)
```




```r
Fsamps[Fsamps$cloneID == 45 & Fsamps$snare == "HS 126",]
```

```
##                                    extID  ugaID is.SCR is.rep week year
## 1924 GTseek_028_384i5_099_P098.UGA1_107-23 107-23   TRUE  FALSE    4 2023
##       snare       utmSnare fqfile knownSEX  sampSOURCE region nSNPgt      vcfID
## 1924 HS 126 270872x3608550   <NA>          Hair Sample    CGA    123 GTS_107-23
##       FILT GTservice sexGT COLclone cloneID finalSEX
## 1924 FALSE    GTSeek     F      332      45        F
```

```r
#This was a GTSeek sample originally assigned to clone ID 332...

#Look closer at the original COLONY clone
Fsamps[Fsamps$COLclone == 332,]
```

```
##                                     extID  ugaID is.SCR is.rep week year
## 996   GTseek_024_384i5_170_P098.003_996-23 996-23   TRUE  FALSE    2 2023
## 1031  GTseek_024_384i5_205_P098.001_678-23 678-23   TRUE  FALSE    2 2023
## 1165  GTseek_024_384i5_339_P098.002_895-23 895-23   TRUE  FALSE    6 2023
## 1342  GTseek_025_384i5_132_P098.007_886-23 886-23   TRUE  FALSE    6 2023
## 1404  GTseek_025_384i5_194_P098.007_987-23 987-23   TRUE  FALSE    6 2023
## 1924 GTseek_028_384i5_099_P098.UGA1_107-23 107-23   TRUE  FALSE    4 2023
## 1958 GTseek_028_384i5_133_P098.UGA1_143-23 143-23   TRUE  FALSE    3 2023
## 2014 GTseek_028_384i5_189_P098.UGA2_459-23 459-23   TRUE  FALSE    2 2023
##       snare       utmSnare fqfile knownSEX  sampSOURCE region nSNPgt      vcfID
## 996  HS 068 268186x3591620   <NA>          Hair Sample    CGA    121 GTS_996-23
## 1031 HS 068 268186x3591620   <NA>          Hair Sample    CGA    120 GTS_678-23
## 1165 HS 066 268064x3590555   <NA>          Hair Sample    CGA    121 GTS_895-23
## 1342 HS 066 268064x3590555   <NA>          Hair Sample    CGA    121 GTS_886-23
## 1404 HS 066 268064x3590555   <NA>          Hair Sample    CGA    122 GTS_987-23
## 1924 HS 126 270872x3608550   <NA>          Hair Sample    CGA    123 GTS_107-23
## 1958 HS 068 268186x3591620   <NA>          Hair Sample    CGA    123 GTS_143-23
## 2014 HS 068 268186x3591620   <NA>          Hair Sample    CGA    123 GTS_459-23
##       FILT GTservice sexGT COLclone cloneID finalSEX
## 996  FALSE    GTSeek     F      332      45        F
## 1031 FALSE    GTSeek     F      332      45        F
## 1165 FALSE    GTSeek     F      332      45        F
## 1342 FALSE    GTSeek     F      332      45        F
## 1404 FALSE    GTSeek     F      332      45        F
## 1924 FALSE    GTSeek     F      332      45        F
## 1958 FALSE    GTSeek     F      332      45        F
## 2014 FALSE    GTSeek     F      332      45        F
```

```
#This includes samples from HS 66, 68, and 126!

#Check kinship of this original COLONY clone
kingrel <- read.table("recapPLINK.kin0", header = FALSE)
colnames(kingrel) <- c("id1","id2","nsnp","hehet","ibs0","kinship")
kingrel$kinship[which(kingrel$id1 %in% Fsamps$vcfID[which(Fsamps$COLclone == 332)] & kingrel$id2 %in% Fs
```

```
##  [1] 0.488636 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000
##  [9] 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000
## [17] 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000
## [25] 0.500000 0.500000 0.500000 0.500000
```

```
#And kinship of the sample in question to other members of the final clone
kingrel$kinship[which(kingrel$id1 == "GTS_107-23" & kingrel$id2 %in% Fsamps$vcfID[which(Fsamps$cloneID =
```

```
##  [1] 0.489583 0.489130 0.489583 0.468750 0.500000 0.478261 0.486111 0.480769
##  [9] 0.454545 0.500000 0.500000 0.500000 0.500000 0.500000
```

```
kingrel$kinship[which(kingrel$id1 %in% Fsamps$vcfID[which(Fsamps$cloneID ==45)] & kingrel$id2 == "GTS_1
```

```
## [1] 0.5 0.5
```

```
#Kinship is high to all other members of this final clone!
```

Given the high kinship coefficients with all members of `finalClone 45` and the presence of GTSeek samples from `HS 066`, `HS 068`, and `HS 126` in `COLONY` clone `332`, this movement also appears to be real!